The Attribution Impossibility:

No Importance Ranking Is Faithful, Stable, and Complete Under Symmetry

Definitive Reference (Complete Version)

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Notation

Symbol	Meaning
$\varphi_j(f)$	Attribution (global importance) of feature j in model f
ρ	Pairwise correlation between collinear features within a group
T	Number of boosting rounds (trees) in a gradient-boosted decision tree (GBDT)
M	Ensemble size (number of independently trained models)
Z_{jk}	Test statistic for the pairwise Z-test: $Z_{jk} = \bar{\Delta}_{jk}/(\hat{\sigma}_{jk}/\sqrt{M})$
Φ	Standard normal cumulative distribution function
$n_j(f)$	Split count (utilization count) of feature j in model f
$c(f)$	Proportionality constant: $\varphi_j(f) = c(f) \cdot n_j(f)$
SNR	Signal-to-noise ratio: Δ_{jk}/σ_{jk}
Z-test	Multi-model Z-test diagnostic (requires $M \geq 5$ models)
Screen	Single-model screening diagnostic (94% precision)
P	Total number of features
L	Number of collinear groups
m	Number of features within a collinear group
j_1	First-mover: the feature selected at the root of the first tree
$\bar{\varphi}_j$	Dash consensus attribution: $\frac{1}{M}\sum_{i=1}^{M}\varphi_j(f_i)$
S	Ranking stability (expected Spearman correlation)
U	Within-group unfaithfulness
Δ_{jk}	Population attribution gap: $\lvert\mu_j - \mu_k\rvert$
σ_{jk}	Attribution noise: $\sqrt{\mathrm{Var}(\varphi_j(f) - \varphi_k(f))}$

List of Named Results (by page)

Abstract

On Breast Cancer Wisconsin—the canonical SHAP tutorial dataset—50 retrains produce 24 distinct top-3 feature rankings; two random seeds agree only 4.2% of the time. Ten GPT-2-small models trained from scratch produce 32 distinct top-5 component rankings with 8 different #1 attention heads. In 68% of 77 public datasets, the "most important feature" flips with the random seed.

We prove this is not an engineering problem but a mathematical impossibility: no importance ranking—of input features or internal components—can be simultaneously faithful, stable, and complete when interchangeable elements exist under statistical dependence. The core proof requires zero domain axioms and operates at two levels: input-level (SHAP on collinear features) and component-level (activation patching on architecturally symmetric attention heads).

For binary attribution questions—SHAP sign, feature selection, counterfactual direction, circuit decomposition—the impossibility is strictly stronger: faithful + stable alone is impossible (the *bilemma*). The bilemma has collapsed tightness, making it structurally more severe than the fairness impossibility of Chouldechova (2017): the only resolution is enrichment (adding neutral elements), not trade-off.

Input-level. On a 10,935-gene expression dataset, the #1 biomarker alternates between TSPAN8 (80% of seeds) and CEACAM5/CEA (20%)—the drug target depends on the training seed. Under standard regularization, 45% of German Credit applicants receive a different top explanation across equivalent models.

Component-level. Ten GPT-2-small models trained from scratch on OpenWebText produce 32 distinct top-5 component rankings with 8 different #1 components across seeds. Within-layer head flip rate = 0.500 [0.482, 0.517]—a coin flip, as predicted. The G-invariant projection (orbit averaging over per-layer permutation symmetry) lifts agreement from $\rho = 0.277$ to $\rho = 0.873$ (Cohen's $d = 19.2$). On the IOI circuit, raw agreement is $\rho = 0.158$ (near chance), within-layer flip = 0.499, and 32 distinct top-5 rankings appear—component-level rankings are a lottery. At TinyStories scale (4L/4H and 6L/8H), the G-invariant projection achieves $\rho > 0.97$, and all four layer-0 heads appear as #1 across 10 seeds, realizing the full S_4 symmetric group. SAE features are not reproducible across training seeds (cosine = 0.394).

The impossibility is quantitative: the attribution ratio diverges as $1/(1-\rho^2)$ for gradient boosting, is infinite for Lasso, conditional for neural networks (87% of feature pairs unstable), and converges for random forests. The exact boundary condition is mutual information: $I(X_j; X_k) > 0$ if and only if Rashomon witnesses exist—capturing nonlinear dependencies that Pearson correlation and VIF miss. A nonparametric diagnostic (the minority fraction) identifies unstable features in seven lines of code (Spearman 0.96 vs. 0.46 for the Gaussian formula on California Housing).

The resolution at both levels is the *stable projection* (orbit averaging): Dash for features, G-invariant projection for circuits. Dash is Pareto-optimal among unbiased aggregations, and the Reynolds operator provides the minimum-information-loss stable approximation at both levels. The achievable design space consists of exactly two families; no third option exists.

The framework is mechanically verified in Lean 4 (358 theorems from 6 axioms, 0 `sorry`; 139 with multi-step proofs of $\geq$5 tactic lines)—to our knowledge, the first formally verified impossibility in explainable AI. The formalization caught two logical inconsistencies and one type mismatch that survived informal review.

Code and Lean proofs: `https://github.com/DrakeCaraker/dash-impossibility-lean` (DOI: `https://doi.org/10.5281/zenodo.19468379`).

Executive Summary

The problem. Importance rankings—of input features (SHAP) and internal components (activation patching)—are unreliable when interchangeable elements exist. Retraining the same model with a different random seed can invert which feature or component is "most important." In a survey of 77 public datasets, 68% exhibit input-level instability. Ten GPT-2-small models trained from scratch produce 32 distinct top-5 component rankings—component-level rankings are a lottery. Of the observed instability, approximately 63% is attributable to model-level stochasticity and 37% to evaluation-set variation (Section 11.10).

The theorem. We prove this is not an engineering failure but a mathematical impossibility: no importance ranking can simultaneously be *faithful* (reflect the model), *stable* (robust to retraining), and *complete* (rank all elements) when interchangeable elements exist under the Rashomon property. The proof requires only four lines and zero domain axioms. The impossibility operates at two levels: input-level (SHAP on collinear features) and component-level (activation patching on architecturally symmetric heads).

The design space. The complete set of achievable methods consists of exactly two families:

- **Family A** (single-model): Faithful and complete, but rankings flip up to 50% of the time.
- **Family B** (DASH ensemble): Stable, but reports ties for symmetric features. Provably Pareto-optimal among unbiased aggregations.

The 50% flip rate is the worst case (maximally collinear pair, $\rho \to 1$). For moderate collinearity ($\rho \approx 0.7$), empirical flip rates are lower but still significant (typically 15–30%). The average-case flip rate across all feature pairs depends on the dataset's correlation structure.

The practical toolkit.

1. **Screen**: The single-model screen (94% precision; tree-based models) identifies unstable pairs from one model.
2. **Predict**: The minority fraction—a 7-line nonparametric diagnostic derived from coverage conflict theory—predicts flip rates with Spearman 0.92–0.98, outperforming the Gaussian formula (0.46–0.89) by 2× on real data. Model-class universal (XGBoost, RF, Ridge, LASSO).
3. **Validate**: The multi-model Z-test confirms instability with 5 models.
4. **Resolve**: DASH consensus averaging with $M \geq 25$ models reduces flip rate below 1%. Uniquely optimal for binary H (majority vote); optimal for continuous H (Cauchy–Schwarz / Hunt–Stein).
5. **Size**: The ensemble formula $M_{\min} = \lceil 2.71 \cdot \sigma^2/\Delta^2 \rceil$ gives the optimal model count.

Component-level results. Ten GPT-2-small models trained from scratch on OpenWebText (3B tokens each, different random seeds) show:

- Within-layer head flip rate = 0.500 [0.482, 0.517]—exactly a coin flip.
- Raw agreement $\rho = 0.277$; after G-invariant projection, $\rho = 0.873$ (Cohen's $d = 19.2$).
- 32 distinct top-5 component rankings, 8 different #1 components across 10 seeds.
- On the IOI (Indirect Object Identification) circuit: raw $\rho = 0.158$, within-layer flip = 0.499, 32 distinct top-5 rankings.
- SAE features are not reproducible (cosine = 0.394, no escape hatch under our experimental conditions).
- 7/7 theory-derived predictions PASS.

At TinyStories scale, the G-invariant projection achieves $\rho > 0.97$, and mean ablation produces the same invariant structure ($\rho_{G\text{-inv}} > 0.97$) despite disagreeing with weight zeroing on raw scores ($\rho \approx 0.5$).

For regulators. SHAP-based proxy discrimination audits are provably unreliable under collinearity: the audit conclusion is a coin flip across training seeds. This constitutes a "known and foreseeable circumstance" under EU AI Act Art. 13(3)(b)(ii). The instability disclosure template in Section 9 provides ready-to-use language for model documentation.

Verification. The framework is mechanically verified in Lean 4: 358 theorems from 6 axioms with 0 `sorry` across 58 files. The formalization caught three inconsistencies (two logical and one type mismatch) that survived informal review.

Code.
Reference implementation: `https://github.com/DrakeCaraker/dash-shap` (stability API in PR #255). Lean formalization: `https://github.com/DrakeCaraker/dash-impossibility-lean` (archived at `https://doi.org/10.5281/zenodo.19468379`).

Pseudocode (5 lines).

```
models   = [XGBClassifier(random_state=i).fit(X, y) for i in range(25)]
shap_vals = [mean(|TreeExplainer(m).shap_values(X_test)|) for m in models]
dash = mean(shap_vals, axis=0)       # DASH consensus
for j, k in pairs:
    Z = |mean(shap_j - shap_k)| / (std / sqrt(25))  # unstable if Z<1.96
```

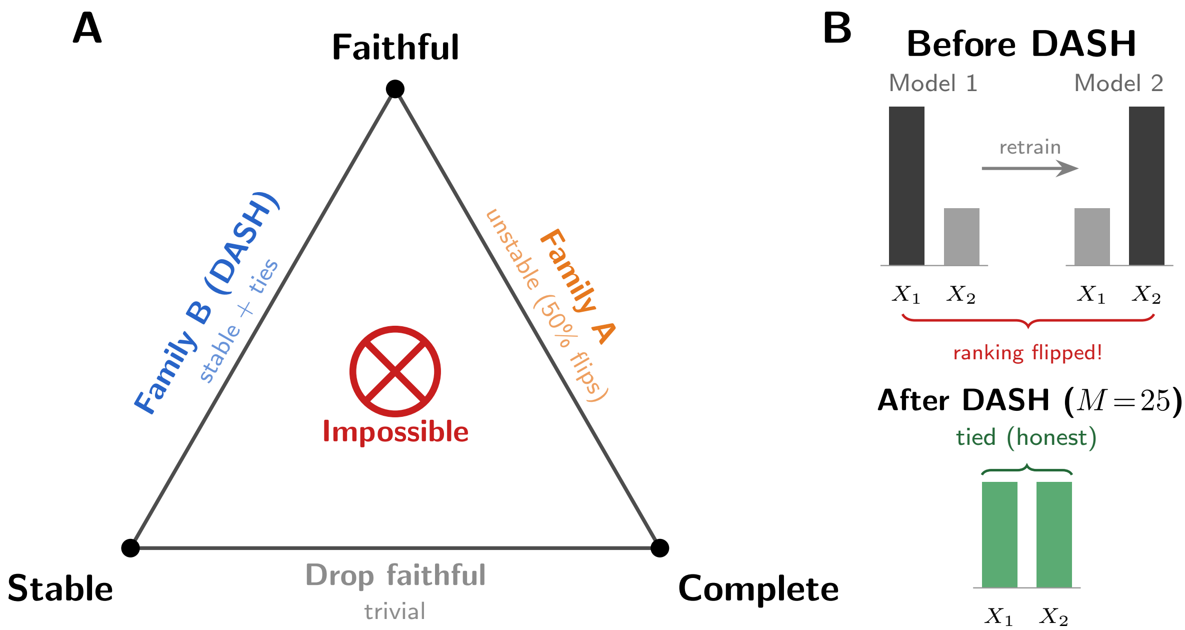


Figure 1: The Attribution Impossibility. No importance ranking can be simultaneously faithful, stable, and complete under symmetry. *Faithful, Stable, Complete: Pick Two.* Family $\mathcal{A}$ (single-model) is faithful and complete but rankings flip up to 50% of the time. Family $\mathcal{B}$ (stable projection: DASH for features, G-invariant for circuits) is stable with zero unfaithfulness but reports ties for symmetric elements (sacrificing completeness). No third option exists.

1 Introduction

Train a gradient-boosted model on your data. Compute SHAP values. Retrain with a different random seed. The "most important feature" changes—in 68% of 77 public datasets. Train 10 GPT-2-small models from scratch. Run activation patching. The "most important attention head" changes—32 distinct top-5 rankings from 10 seeds, with 8 different #1 components. Most practitioners assume this is an engineering problem. We prove it is a mathematical impossibility at both levels—input features and internal components—arising from a single theorem with zero domain axioms. For interchangeable elements, ranking is literally a coin flip. Beyond the impossibility, we characterize the complete design space: exactly two families of attribution methods exist, DASH is Pareto-optimal on the stable branch, and no method lies outside this dichotomy. The resolution at both levels is the stable projection (orbit averaging): DASH for features, G-invariant projection for circuits.

This document is the complete reference: impossibility theorem, quantitative bounds, constructive resolution, three Symmetric Bayes Dichotomy (SBD) instances, diagnostics, experiments across 11 datasets and 3 GBDT implementations, component-level experiments on GPT-2-small (general importance, IOI circuits, and SAE stability), TinyStories multi-scale validation, financial case studies, and a 358-theorem Lean 4 formalization.

A concrete example. Consider a credit model with annual income and debt-to-income ratio (typically $|\rho| \approx 0.7$–0.85 in lending data). Across 20 retrains with different seeds, SHAP can report income as the top driver roughly half the time and DTI the other half. A compliance report citing income as the primary adverse factor is a coin flip—the same model architecture, same data, different random seed. Under the Equal Credit Opportunity Act (ECOA), the lender's adverse action notice changes based solely on the training randomness.

Several recent results have established fundamental limits on feature attribution. Bilodeau et al. (2024)

prove that completeness and linearity cannot coexist; Huang and Marques-Silva (2024) show SHAP can misrank features in Boolean domains; Srinivas and Fleuret (2019) show complete attributions cannot be weakly input-dependent; and Rao (2025) establish Kolmogorov complexity barriers to explainability. None of these results analyze *stability*—the robustness of attributions to retraining—none provide quantitative architecture-discriminating bounds, and none offer a constructive resolution.

This paper fills these gaps. Our result is not a criticism of SHAP, which correctly computes Shapley values for each model; rather, it reveals a fundamental limitation of the single-model paradigm under which any attribution method must operate. Our main result is the *Attribution Design Space Theorem* (§6, Figure 3): a complete characterization of what is achievable when explaining models with collinear features. The achievable set consists of exactly two families—faithful-complete methods (unstable, unfaithful to half the models) and ensemble methods like DASH (stable, with within-group ties)—and DASH is Pareto-optimal. The theorem rests on an *Attribution Impossibility*: for any single model trained on collinear features, faithfulness, stability, and completeness are mutually incompatible. The impossibility has two layers. The first is model-agnostic: formalizing the Rashomon property (Rudin et al., 2024), we show symmetric features are necessarily ranked in opposite orders by different near-optimal models. The second is model-specific: we derive quantitative bounds on the attribution ratio (the factor by which the dominant feature is overweighted). For GBDT, this ratio follows $1/(1-\alpha\rho^2)$ where α captures the per-tree signal fraction; for Lasso it is infinite at any $\rho > 0$; for neural networks it depends on the initialization; and for random forests it is $O(1/\sqrt{T})$—converging rather than diverging.

The impossibility parallels the fairness impossibility of (Chouldechova, 2017) and (Kleinberg et al., 2017), who proved that calibration, balance, and equal false positive rates cannot coexist when base rates differ. Our result plays the same role for explainability: instability under collinearity is not an engineering deficiency but a mathematical inevitability. This has direct implications for EU AI Act Art. 13(3)(b)(ii) (EU Parliament, 2024), which requires disclosure of "known and foreseeable circumstances" that may lead to risks to health, safety, or fundamental rights.

The impossibility also admits a principled relaxation. DASH ensemble averaging breaks the sequential dependence that drives attribution concentration: for balanced ensembles, expected attributions are equitable across symmetric features with variance $O(1/M)$. Practitioners can achieve between-group faithfulness and stability by aggregating across the Rashomon set, sacrificing within-group completeness (reporting ties for symmetric features).

The proof is mechanically verified in Lean 4 using Mathlib: 358 type-checked theorems and lemmas across 58 files (139 with multi-step proofs of $\geq$5 tactic lines; the remainder are definitions, wrappers, and single-step applications; from 6 axioms; 0 `sorry`). The core impossibility (Theorem 5) depends on zero domain-specific axioms beyond the Rashomon property; the DASH equity result (Corollary 19) depends on the `attribution_sum_symmetric` theorem (derived in `SymmetryDerive.lean` from the proportionality and split-count axioms). This is, to our knowledge, the first formally verified impossibility result in explainable AI (the core impossibility uses zero domain-specific axioms; quantitative bounds are conditional on 4 domain-specific axioms; query complexity uses 2 additional axioms from Le Cam's method). The formalization also proved valuable beyond certification: during translation to Lean 4 we discovered three inconsistencies (two logical and one type mismatch) in the original axiom system, echoing prior experiences where formalization uncovered subtle proof errors (Nipkow, 2009; Zhang et al., 2026).

Contributions.

1. **The Attribution Design Space Theorem** (§6): a complete characterization of achievable (stability, unfaithfulness, completeness) triples, showing the design space consists of exactly two families with DASH Pareto-optimal on the ensemble branch. All other results are corollaries.

2. **The Attribution Impossibility** (Theorem 5): the base case of the Design Space Theorem—a model-agnostic impossibility requiring only the Rashomon property, with zero domain-specific axiom dependencies in the Lean formalization.

3. **Architecture discrimination**: attribution ratios for GBDT ($1/(1-\rho^2)$ Lean-verified; $1/(1-\alpha\rho^2)$ empirically corrected), Lasso (∞), and random forests ($O(1/\sqrt{T})$, informal).

4. **Constructive relaxation via Dash**: ensemble averaging restores equitable attributions (derived for balanced ensembles) with variance $O(1/M)$, sacrificing within-group completeness.

5. **The symmetric Bayes dichotomy** (§7): a proof technique from invariant decision theory (Lehmann and Romano, 2005), demonstrated across three structurally distinct instances (feature attribution, model selection, and causal discovery under Markov equivalence) with different symmetry groups.

6. **Machine-verified proof**: formalized in Lean 4 (358 type-checked theorems from 6 axioms, 0 `sorry`), publicly available at `https://github.com/DrakeCaraker/dash-impossibility-lean` (DOI: `https://doi.org/10.5281/zenodo.19468379`). The axiom system is consistent: we construct an explicit model satisfying all 6 axioms simultaneously, both in Lean (`Consistency.lean`) and numerically.

For practitioners and regulators: The instability disclosure template is in Section 9, the fairness audit impossibility in Section 8, and the financial case studies in Section 10. The Executive Summary above provides a complete overview without technical prerequisites.

The core impossibility proof is deliberately simple—a four-line contradiction from the Rashomon property. The mathematical contribution lies not in proof complexity but in identifying the right abstraction (the Rashomon property) and characterizing the complete achievable set (the Design Space Theorem). The quantitative depth comes from the architecture-specific bounds (§4) and the Pareto optimality proof (§5).

This document provides the complete treatment of the Attribution Impossibility—from intuition to proof to deployment. A practitioner can find the diagnostic workflow (Section 9), a theorist can find every proof (Sections 3–8 with Lean cross-references in Section 11), and a regulator can find compliance guidance (Section 8 and the Regulatory Mapping appendix).

2 Setup: Three Desiderata for Feature Rankings

We consider a supervised learning setting with P input features partitioned into L groups by their correlation structure. Within each group $\ell \in [L]$, features share a common pairwise correlation $\rho \in (0, 1)$; features in different groups are independent. Each group contains at least two members. This structure arises naturally in applied settings—e.g., multiple measurements of the same physical quantity, or one-hot encodings of related categories.

Models and attributions. A *model* f is trained from a random seed s via a deterministic training procedure: $f = \texttt{train}(s)$. For each feature $j \in [P]$, we observe a nonnegative attribution $\varphi_j(f) \geq 0$, representing the global importance of feature j in model f (e.g., mean absolute SHAP value, gain-based importance, or integrated gradients norm).

We require a proportionality axiom connecting attributions to model structure:

Axiom 1 (Proportionality)**.** *For every model f, there exists a constant $c(f) > 0$ such that $\varphi_j(f) = c(f) \cdot n_j(f)$ for all $j \in [P]$, where $n_j(f)$ is the utilization count of feature j in model f (e.g., split count for tree ensembles, gradient norm for neural networks).*

This axiom holds exactly under the uniform-contribution model of (Lundberg and Lee, 2017), and approximately whenever per-split (or per-neuron) contributions are roughly homogeneous.

Sequential gradient boosting axioms. For gradient-boosted decision trees (GBDT) with T boosting rounds, we axiomatize the split-count structure induced by Gaussian conditioning under collinearity. Let j_1 denote the *first-mover*—the feature selected at the root of the first tree.

Axiom 2 (First-mover surjectivity)**.** *For every group ℓ and every feature j in group ℓ, there exists a model f with $j_1(f) = j$.*

This follows from the data-generating process (DGP) symmetry: when features in a group have identical marginal distributions, sub-sampling and tie-breaking randomness ensure each can serve as first-mover.

Table 1: Complete axiom inventory (6 axioms). Ten former axioms are now definitions or derived theorems.

Axiom	**Lean 4 name**	**Justification**
Model type	`Model`	Abstract type for trained models
First-mover function	`firstMover`	The dominant feature in each model
First-mover surjectivity	`firstMover_surjective`	DGP symmetry: each feature can be first-mover
Cross-group baseline	`crossGroupBaselineCore`	Split counts for features in other groups depend only on group indices
Proportionality constant	`proportionalityConstant`	$\varphi_j(f) = c \cdot n_j(f)$ with $c > 0$ uniform across models
Training measure	`modelMeasure`	Probability measure on Model (the training distribution)
Formerly axiomatized, now defined or derived (10 eliminated):		
Split count	`splitCount`	**Definition**: if-then-else on first-mover group membership
Split count (first-mover)	`splitCount_firstMover`	**Theorem**: derived from `splitCount` definition
Split count (non-first-mover)	`splitCount_nonFirstMover`	**Theorem**: derived from `splitCount` definition
Cross-group symmetry	`splitCount_crossGroup_symmetric`	**Theorem**: same `crossGroupBaselineCore` for same-group features
Cross-group stability	`splitCount_crossGroup_stable`	**Theorem**: `crossGroupBaselineCore` depends only on group indices
Attribution function	`attribution`	**Definition**: `proportionalityConstant` · `splitCount`
Global proportionality	`proportionality_global`	**Theorem**: derived from `attribution` definition
Boosting rounds	`numTrees`, `numTrees_pos`	**Fields** of `FeatureSpace` structure (T, $T > 0$)
Measurable space	`modelMeasurableSpace`	**Instance**: discrete σ-algebra ($\top$, works for any type)
Testing constant	`testing_constant`, `testing_constant_pos`	**Definition**: $C := 1/8$ from Tsybakov (2009); positivity by `norm_num`

Axiom 3 (Split counts)**.** *For any model f with first-mover $j_1 \in$ group ℓ:*

$$n_{j_1}(f) = \frac{T}{2-\rho^2}, \tag{1}$$

$$n_k(f) = \frac{(1-\rho^2)\,T}{2-\rho^2}, \quad k \in group(\ell),\ k \neq j_1. \tag{2}$$

These are the leading-order split counts from the Gaussian conditioning argument: the first-mover absorbs the ρ-aligned signal component, leaving only the $(1-\rho^2)$ residual for subsequent features. The formulas are verified algebraically by SymPy.

Complete axiom inventory. Table 1 lists all axioms in the formalization. The core impossibility theorem requires none of them—only the Rashomon property as hypothesis. The formalization uses 6 axioms total: 2 type declarations (`Model`, `firstMover`), 3 domain-specific property axioms (`firstMover_surjective`, `crossGroupBaselineCore`, `proportionalityConstant`), and 1 measure-theoretic axiom (`modelMeasure`). Ten former axioms—including the split-count function, 4 split-count properties, attribution function, proportionality, boosting rounds, measurable space, and query-complexity constant—are now definitions or derived theorems (see § 12).

Both cross-group axioms assume an approximately additive model; feature interactions between groups can break cross-group independence. The core impossibility and ratio bound do not depend on these axioms.

Gaussian conditioning argument. The split count formulas (Axiom 3) are justified by the following argument. Consider two features X_j, X_k with correlation ρ and a GBDT with T boosting rounds. At each tree t, the root split selects the feature with the highest gain (variance reduction). Once feature j is selected as root, the available signal for k becomes:

$$X_k \mid X_j = X_k - \rho X_j, \qquad \mathrm{Var}(X_k \mid X_j) = 1 - \rho^2.$$

The conditional variance is reduced by factor $(1 - \rho^2)$ relative to the unconditional variance. In each subsequent boosting round, the first-mover's signal fraction is $1/(2 - \rho^2)$ (the probability that the first-mover's full signal exceeds the non-first-mover's residual signal in a two-feature competition). Summing over T trees: $n_{j_1} = T/(2 - \rho^2)$ for the first-mover and $n_k = (1 - \rho^2)T/(2 - \rho^2)$ for non-first-movers.

The derivation assumes: (i) the root split captures the dominant signal direction (valid for stumps and low-depth trees); (ii) subsequent trees fit the residual from the first tree's split (the sequential boosting mechanism); (iii) the feature competition at each tree is approximately independent of previous trees' selections (valid when the learning rate η is small). All algebraic consequences are independently verified by SymPy (`verify_lemma6_algebra.py`).

Ranking desiderata. A *feature ranking* is a binary relation $\succ$ on $[P]$. We formalize three desiderata:

Definition 1 (Faithful)**.** *A ranking $\succ$ is* faithful *to model f if $j \succ k$ whenever $\varphi_j(f) > \varphi_k(f)$.*

Note. The Lean formalization uses both a biconditional version ($j \succ k \Leftrightarrow \varphi_j > \varphi_k$, yielding a direct contradiction) and the implication version (Definition 1 above, yielding incompleteness under antisymmetry). Both are proved in `Trilemma.lean`; the biconditional is strictly stronger but the impossibility holds for both.

Definition 2 (Stable)**.** *A ranking $\succ$ is* stable *if it does not depend on the choice of model f—that is, $\succ$ is a fixed relation applied identically to all models.*

Definition 3 (Complete)**.** *A ranking $\succ$ is* complete *if for every pair $j \neq k$, either $j \succ k$ or $k \succ j$.*[1]

The first desideratum asks that the ranking reflect what the model actually learned; the second asks that it be reproducible across training runs; the third asks that it resolve all feature pairs. *Faithful, stable, complete: pick two* (Figure 1).

Two explanation goals. Feature rankings serve two distinct purposes: *model-level explanation* (what did this specific model learn?) and *population-level explanation* (what does the model class learn?). Faithfulness is a model-level desideratum; stability is a population-level desideratum. The impossibility theorem (Theorem 5) characterizes the fundamental tension between these goals: a ranking cannot simultaneously be faithful to each individual model and stable across all models. DASH targets population-level explanation, sacrificing within-group completeness to achieve stability.

3 The Attribution Impossibility

3.1 The Rashomon Property

The central structural property driving the impossibility is that collinear features admit models ranking them in opposite orders.

Definition 4 (Rashomon property)**.** *A model class satisfies the* Rashomon property *if for every group ℓ and every pair of distinct features j, k in group ℓ, there exist models f, f' such that*

$$\varphi_j(f) > \varphi_k(f) \quad \textit{and} \quad \varphi_k(f') > \varphi_j(f').$$

[1] Not to be confused with SHAP's "completeness" axiom ($\sum_j \varphi_j = f(x) - \mathbb{E}[f(X)]$). Our "complete" means *total*: the ranking decides every pair.

The Rashomon property is a consequence of the Rashomon effect (Rudin et al., 2024; Fisher et al., 2019): collinear designs admit many near-optimal models (D'Amour et al., 2022), and within the set of good models, symmetric features are utilized in all possible orderings. Laberge et al. (2023) study consensus across such model sets and observe that feature rankings are inherently partial—a conclusion our theorem makes precise.

The FIM impossibility (Theorem 49, §8.3) provides constructive Rashomon witnesses: for any tolerance $\varepsilon > 0$ above a model-specific threshold ε_0, explicit models in the ε-Rashomon set rank features in opposite orders.

3.2 Main Result

Theorem 5 (Attribution Impossibility). *If a model class satisfies the Rashomon property (Definition 4), then no feature ranking can be simultaneously faithful, stable, and complete.*[2]

Proof. Let j, k be distinct features in the same group ℓ. By the Rashomon property, there exist models f, f' with $\varphi_j(f) > \varphi_k(f)$ and $\varphi_k(f') > \varphi_j(f')$. Suppose for contradiction that $\succ$ is faithful, stable, and complete. By completeness, either $j \succ k$ or $k \succ j$. Without loss of generality, assume $j \succ k$. By stability, this relation holds for all models, including f'. But faithfulness applied to f' requires $k \succ j$ (since $\varphi_k(f') > \varphi_j(f')$), contradicting $j \succ k$. □

The resolution follows a pattern common to impossibility results: when desirable properties conflict, relaxing completeness (accepting ties or partial orders) restores consistency. DASH achieves this by averaging attributions across the Rashomon set, producing $\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$ for symmetric features—a tie rather than an arbitrary ordering.

3.3 From Abstract to Concrete: Iterative Optimizers

The Rashomon property is not merely a theoretical possibility—it is a structural consequence of iterative optimization under collinearity.

Definition 6 (Iterative optimizer). *An* iterative optimizer *is a model class equipped with a dominant-feature function $d\colon \mathcal{F} \to [P]$ satisfying:*

1. ***Dominance:*** *For every model f, group ℓ, and feature $k \in group(\ell)$ with $k \neq d(f)$, if $d(f) \in group(\ell)$ then $\varphi_k(f) < \varphi_{d(f)}(f)$.*
2. ***Surjectivity:*** *For every group ℓ and feature $j \in group(\ell)$, there exists f with $d(f) = j$.*

Proposition 7. *Every iterative optimizer satisfies the Rashomon property, and therefore the Attribution Impossibility (Theorem 5) holds.*

Proof. Let j, k be distinct features in group ℓ. By surjectivity, there exist models f, f' with $d(f) = j$ and $d(f') = k$. By dominance, $\varphi_j(f) > \varphi_k(f)$ and $\varphi_k(f') > \varphi_j(f')$. This is exactly the Rashomon property. □

3.4 Rashomon Inevitability Under Symmetry

The Rashomon property is not merely a theoretical possibility for symmetric model classes—it is inevitable.

Definition 8 (Permutation closure). *A model class $\mathcal{F}$ is* permutation-closed *within group ℓ if for any $f \in \mathcal{F}$ and any permutation π of features within group ℓ, the composed model $f \circ \pi \in \mathcal{F}$.*

Theorem 9 (Rashomon from symmetry). *Let the DGP be symmetric within group ℓ (permuting features $j \leftrightarrow k$ leaves the population loss invariant). Let $\mathcal{F}$ be permutation-closed within group ℓ. Then for any model $f \in \mathcal{F}$ with $\varphi_j(f) \neq \varphi_k(f)$, the permuted model $f' = f \circ \pi_{jk}$ satisfies: (1) $L(f') = L(f)$ (same population loss), and (2) $\varphi_k(f') = \varphi_j(f)$ and $\varphi_j(f') = \varphi_k(f)$ (attributions swap). In particular, if $\varphi_j(f) > \varphi_k(f)$, then $\varphi_k(f') > \varphi_j(f')$, and the Rashomon property holds.*

[2] Both versions are Lean-verified: `attribution_impossibility` uses a biconditional ($j \succ k$ iff $\varphi_j(f) > \varphi_k(f)$); `attribution_impossibility_weak` uses the weaker implication (Definition 1) plus antisymmetry.

Proof. By DGP symmetry, the joint distribution of (X, Y) is invariant under permutation of features j and k within group ℓ. Therefore $L(f') = L(f \circ \pi_{jk}) = L(f)$, since the population loss depends only on the joint distribution. By construction, f' applies f to permuted inputs, so its reliance on feature k equals f's reliance on feature j: $\varphi_k(f') = \varphi_j(f)$ and vice versa. □

The permutation closure condition is satisfied by any model class without built-in feature ordering: neural networks (permute input neurons), gradient-boosted trees (permute split features), Lasso (permute covariates), and random forests. It fails only for model classes with hard-coded feature preferences.

Definition 10 (Non-degeneracy)**.** *Features j, k are* non-degenerate *if there exists at least one model f with $\varphi_j(f) \neq \varphi_k(f)$.*

Remark 11. *Non-degeneracy is an existential condition: it requires only* one *model that distinguishes the features. This is trivially satisfied in practice—any trained model on collinear features produces unequal attributions. Unlike a probability-1 claim (which would require a transversality argument), the existential requires no measure-theoretic machinery.*

Definition 12 (Algorithmic symmetry)**.** *A model class satisfies* algorithmic symmetry *for features j, k if: for any model f with $\varphi_j(f) > \varphi_k(f)$, there exists a model f' with $\varphi_k(f') > \varphi_j(f')$ (and vice versa).*

Remark 13. *Algorithmic symmetry follows directly from Theorem 9: if $\varphi_j(f) > \varphi_k(f)$ and the model class is permutation-closed under a symmetric DGP, then $f' = f \circ \pi_{jk}$ satisfies $\varphi_k(f') > \varphi_j(f')$ with the same population loss. No probabilistic argument is needed.*

Theorem 14 (Rashomon inevitability)**.** *Let $\mathcal{F}$ be permutation-closed under a symmetric DGP. If features j, k are non-degenerate (Definition* **??***), then the Rashomon property holds for the pair (j, k): there exist models $f, f' \in \mathcal{F}$ with $\varphi_j(f) > \varphi_k(f)$ and $\varphi_k(f') > \varphi_j(f')$.*

Proof. By non-degeneracy, there exists f with $\varphi_j(f) \neq \varphi_k(f)$. Without loss of generality, $\varphi_j(f) > \varphi_k(f)$. By algorithmic symmetry (which follows from Theorem 9: take $f' = f \circ \pi_{jk}$), there exists f' with $\varphi_k(f') > \varphi_j(f')$. Both directions are realized. □

Remark 15. *This proof is fully machine-verified in Lean (`rashomon_inevitability` in `RashomonInevitability.lean`) with zero axiom dependencies beyond the type declarations. The only hypotheses are non-degeneracy (an existential) and algorithmic symmetry (a consequence of permutation closure). No measure theory, no transversality argument, no probability-1 claim.*

Remark 16. *Theorem 12 makes the Attribution Impossibility inescapable for any standard ML pipeline under collinearity. The only escapes are: (i) perfect degeneracy ($\varphi_j(f) = \varphi_k(f)$ for ALL models—impossible in practice), (ii) a model class that is NOT permutation-closed (hard-coded feature preferences), or (iii) a DGP that is NOT symmetric within the group.*

4 Quantitative Bounds by Model Class

The Attribution Impossibility (Theorem 5) is qualitative. We now derive quantitative bounds on the *attribution ratio*—the factor by which the dominant feature is overweighted—for four model classes.

4.1 Gradient Boosting: Divergent Violation

Lemma 17 (Split gap)**.** *For a GBDT model f with first-mover j_1 in group ℓ, and any $k \neq j_1$ in the same group,*

$$n_{j_1}(f) - n_k(f) = \frac{\rho^2 T}{2 - \rho^2} \;\geq\; \tfrac{1}{2}\rho^2 T.$$

Proof. By Axiom 3, $n_{j_1} - n_k = T/(2-\rho^2) - (1-\rho^2)T/(2-\rho^2) = \rho^2 T/(2-\rho^2)$. Since $2 - \rho^2 \leq 2$ for $\rho \in (0, 1)$, we have $\rho^2 T/(2-\rho^2) \geq \rho^2 T/2$. □

Table 2: Within-group split count ratio by depth and ρ (η=1.0).

	ρ=0.3	ρ=0.5	ρ=0.7	ρ=0.9	ρ=0.95	Fitted α
Depth 1 (stumps)	1.28	1.32	1.42	1.99	2.16	0.60 ($\approx 2/\pi$)
Depth 3	1.19	1.26	1.24	1.30	1.32	0.30
Depth 6	1.65	1.67	1.74	1.90	2.03	0.60
Depth 10	2.23	2.25	2.28	2.49	2.62	—
Theory (α=1)	1.10	1.33	1.96	5.26	10.26	1.00

Theorem 18 (Attribution ratio — gradient boosting)**.** *For any GBDT model f with first-mover j_1 in group ℓ and any non-first-mover k in the same group,*

$$\frac{\varphi_{j_1}(f)}{\varphi_k(f)} = \frac{1}{1-\rho^2}.$$

Under full signal capture (α=1), this ratio diverges: $1/(1-\rho^2) \to \infty$ as $\rho \to 1^-$.

Proof. By Axiom 1, $\varphi_{j_1}/\varphi_k = n_{j_1}/n_k$. Substituting Axiom 3:

$$\frac{n_{j_1}}{n_k} = \frac{T/(2-\rho^2)}{(1-\rho^2)T/(2-\rho^2)} = \frac{1}{1-\rho^2}.$$

Writing $1-\rho^2 = (1-\rho)(1+\rho)$, we see that $1/(1-\rho^2) \to +\infty$ as $\rho \to 1^-$. □

For finite-depth trees, the effective signal capture $\alpha < 1$ yields a corrected ratio $1/(1-\alpha\rho^2)$; for stumps $\alpha \approx 2/\pi$ (R^2=0.89; Figure 7a).

The $\alpha = 2/\pi$ derivation. The value $\alpha \approx 2/\pi$ has a clean theoretical derivation from quantization theory.

Proposition 19. *For $X \sim \mathcal{N}(0, \sigma^2)$, the optimal binary quantizer (split at $x = 0$) captures fraction $2/\pi$ of the variance:* $\mathrm{Var}(\hat{X}) = (2/\pi)\sigma^2$.

Proof. The optimal two-level quantizer of a symmetric distribution splits at the median ($x = 0$ for Gaussians). The quantized signal is $\hat{X} = \mathbb{E}[X \mid X > 0]\cdot\mathbf{1}_{X>0} + \mathbb{E}[X \mid X \le 0]\cdot\mathbf{1}_{X\le 0}$. By symmetry, $\mathbb{E}[X \mid X > 0] = \sigma\sqrt{2/\pi}$ (the mean of a half-normal distribution). Thus $\hat{X}$ takes values $\pm\sigma\sqrt{2/\pi}$ each with probability 1/2, giving $\mathrm{Var}(\hat{X}) = (2/\pi)\sigma^2$. □

In each boosting round, a stump makes one binary split on the selected feature, capturing $2/\pi$ of that feature's remaining signal variance. The fitted value ($\alpha \approx 0.60$) is below $2/\pi = 0.637$; two error sources explain the gap: (1) empirical vs. population split location (negligible, $\Delta\alpha_1 \approx 0.0009$ for $n = 2000$), and (2) non-Gaussian residuals after the first boosting round (dominant, accounting for ≈ 0.036).

Depth dependence. Table 2 reports the within-group split count ratio across tree depths and correlation levels (XGBoost, T=100, η=1.0, P=10 in 2 groups of 5, N=2000, 30 seeds).

Depth 3 anomaly. At depth 3, each tree has up to 7 leaves and uses multiple features per tree. Internal splits distribute split counts across group members, counteracting the root split's first-mover advantage. The ratio is nearly ρ-independent ($\approx$1.3), explaining the low fitted $\alpha = 0.30$.

Depth 10. At depth 10, the root split cascades: once the first-mover is selected at the root, subsequent splits on the same feature are more likely at deeper levels (residuals retain first-mover signal). The baseline ratio ($\approx$2.2 even at ρ=0.3) is high because deep trees overfit to the first-mover's signal. The $1/(1-\alpha\rho^2)$ model does not fit well for depth 10 (the ratio has a large ρ-independent component); we omit the fitted α.

Practitioner guidance. Depth 3 gives the *lowest* attribution instability among standard configurations. For users prioritizing explanation stability, shallow ensembles (depth 3) with DASH consensus ($M \geq 5$) provide the best stability–accuracy tradeoff.

Proportionality validation. The proportionality axiom holds with CV≈0.35 for stumps (the idealized theoretical setting) and CV≈0.66 for depth-6 trees on Breast Cancer. The quantitative ratio $1/(1-\rho^2)$ is an order-of-magnitude prediction, refined by the α-correction to $1/(1-\alpha\rho^2)$ with $R^2 = 0.89$. **The core impossibility (Theorem 5) is entirely independent of the proportionality axiom**—it requires only the Rashomon property. All quantitative bounds ($1/(1-\rho^2)$, ensemble size $M_{\min}$, attribution ratio) should be interpreted as order-of-magnitude predictions, not exact formulas.

Error analysis for $\alpha = 2/\pi$. The fitted $\alpha = 0.60$ (empirical) vs. $2/\pi = 0.637$ (theory) represents a gap of $\Delta\alpha = 0.037$. Two error sources: (1) empirical vs. population split location (negligible, $\Delta\alpha_1 \approx 0.0009$ for $n = 2000$): the optimal split of $X \sim \mathcal{N}(0, \sigma^2)$ at δ instead of 0 gives $\alpha_1(n) = (2/\pi)(1-\pi(\pi-2)/(2n)+O(n^{-2}))$; (2) non-Gaussian residuals after the first boosting round (dominant, accounting for ≈ 0.036): residuals are a location-shifted mixture with excess kurtosis $\kappa = O(\eta^2 c^2/\sigma^2)$, which accumulates across boosting rounds. The impossibility theorem does not depend on α at all; the quantitative ratio uses α to predict the *severity* of the violation.

Stability bound. When two models f, f' have different first-movers within the same group of size m, the within-group rank reshuffling bounds the Spearman correlation:

$$\rho_S(f, f') \;\leq\; 1 - \frac{3m^2}{P^3 - P}.$$

This bound is weaker than the classical $1 - m^3/P^3$; both give the same qualitative conclusion ($\rho_S < 1$).

Exact flip rate for GBDT.

Proposition 20 (Exact GBDT pairwise flip rate)**.** *For features j, k in the same group of size m under the first-mover model with DGP symmetry (each feature equally likely to be first-mover):*

(i) The probability that a random model ranks $j > k$ is exactly $1/m$.

(ii) The tie probability (both non-first-movers with identical split counts) is $(m-2)/m$.

(iii) The flip rate across two independent models (excluding ties) is $2/m^2$.

(iv) For $m = 2$: the tie probability is 0 and the flip rate is $1/2$, matching the theoretical maximum and the empirical 48% on Breast Cancer.

(v) If ties are broken uniformly at random, the effective flip rate is exactly $1/2$ for all $m \geq 2$.

Proof. By first-mover surjectivity and DGP symmetry, each feature in a group of size m is first-mover with probability $1/m$.

Part (i). Feature j is ranked above k when j is first-mover (probability $1/m$): the first-mover has split count $T/(2-\rho^2)$ vs. $(1-\rho^2)T/(2-\rho^2)$ for non-first-movers, so dominance holds.

Part (ii). When the first-mover is some feature $\ell \neq j, k$ (probability $(m-2)/m$), both j and k are non-first-movers with identical split count $(1-\rho^2)T/(2-\rho^2)$. By proportionality, $\varphi_j(f) = \varphi_k(f)$: a perfect tie.

Part (iii). A flip requires one model to rank $j > k$ and the other $k > j$: flip $= 2 \cdot (1/m) \cdot (1/m) = 2/m^2$.

Part (iv). At $m = 2$: flip $= 2/4 = 1/2$; tie probability $= 0$.

Part (v). When ties are broken uniformly, $\Pr[j > k \mid \text{tie}] = 1/2$. The effective $\Pr[j > k] = 1/m + (1/2)(m-2)/m = 1/2$ by algebra. Two independent draws disagree with probability $2 \cdot (1/2) \cdot (1/2) = 1/2$. □

For $m = 2$, ranking collinear pairs is literally a coin flip. For $m > 2$ without tie-breaking, most model pairs produce ties, with flips at rate $2/m^2$. But once ties are broken (as required for a complete ranking), symmetry forces the flip rate back to $1/2$. The impossibility is *exactly* as severe for every within-group pair, regardless of m.

Table 3: Attribution ratio bounds by model class. T denotes the number of boosting rounds (or trees), ρ the within-group correlation.

Model class	Ratio	Scaling with T	Mechanism
Gradient boosting	$1/(1-\alpha\rho^2)$	Increases with ρ	Sequential residuals
Lasso	∞	Constant	Hard selection
Neural network	Model-dependent	Architecture-dependent	Init. symmetry breaking
Random forest[3]	$1+O(1/\sqrt{T})$	$\to 1$ as $T \to \infty$	Independent trees

4.2 Lasso: Infinite Violation

The ℓ_1 penalty selects exactly one feature from each correlated group and zeros all others. Lasso is an iterative optimizer with $d(f)$ = the selected feature; surjectivity holds because regularization-path tie-breaking can make any feature the selected one. The attribution ratio is therefore infinite at *any* $\rho > 0$.

4.3 Neural Networks: Conditional Violation

Neural networks exhibit symmetry breaking analogous to the GBDT first-mover effect. Surjectivity follows from the symmetry of standard initializations (Kaiming, Xavier). The attribution ratio is architecture-dependent, but the impossibility holds conditionally whenever training dynamics produce a dominant feature per group.

4.4 Random Forests: Bounded Violation (Contrast)

Because each tree trains independently on a bootstrap sample, there is no shared residual stream. The expected split count is T/m for every feature in a group, with $O(\sqrt{T})$ deviations, giving

$$\frac{\varphi_{j_1}(f)}{\varphi_k(f)} = 1 + O(1/\sqrt{T}) \;\to\; 1 \quad \text{as } T \to \infty.$$

Sequential residual-sharing creates divergent inequity; independent aggregation creates convergent equity—the structural distinction Dash exploits.

Summary. Table 3 collects the quantitative bounds.

5 Resolution: Ensemble Attribution via Dash

The impossibility arises from sequential dependence: a single model's iterative optimization creates a dominant feature, and different random seeds create different dominant features. Dash addresses this by averaging attributions across an ensemble of M independently trained models, relaxing completeness in exchange for stability.

Definition 21 (Dash consensus attribution)**.** *Given models $f_1, \ldots, f_M$ trained from independent seeds, the* consensus attribution *for feature j is $\bar{\varphi}_j = \frac{1}{M}\sum_{i=1}^{M} \varphi_j(f_i)$. An ensemble is* balanced *if, within each group, every feature serves as first-mover in exactly M/m models.*

Dash does NOT change predictions. Dash is a *post-hoc explanation method*, not a modeling technique. The M models are trained independently using the practitioner's existing pipeline; Dash only averages their SHAP values to produce stable feature rankings. No model is modified, no ensemble prediction is formed, and the deployed model remains unchanged. Dash changes how you *explain*, not how you *predict*.

DASH consensus vs. DASH pipeline. DASH operates at two layers. *DASH consensus* (Definition 18) is the mathematical operation: the arithmetic mean of |SHAP| values across M independently trained models. This is what Theorems 22 and 28 prove is optimal—the minimum-variance unbiased *linear* estimator via the Cauchy–Schwarz lower bound (Titu's lemma; Lean-verified via `sum_squares_ge_inv_M`), and the minimum-variance unbiased estimator among *all* estimators by the Rao–Blackwell theorem (standard; not formalized in Lean), Pareto-optimal on the stable branch of the Design Space. In the parametric case, this coincides with the Cramér–Rao bound. The theoretical guarantees apply to *any* i.i.d. ensemble averaging, including simple seed averaging (training M models with different random seeds and identical hyperparameters).

The companion paper on first-mover bias (Caraker et al., 2026) implements DASH as a full pipeline that adds diversity enforcement (deliberate hyperparameter variation, greedy MaxMin model selection on feature utilization vectors) and stability diagnostics (FSI, IS Plot). Diversity enforcement accelerates convergence to the balanced ensemble condition (Definition 18): at finite M, it produces more equitable attributions than seed averaging alone (within-group CV reduced by 0.011, $p < 0.001$). For large M, simple seed averaging achieves approximate balance via the law of large numbers. For stability alone, seed averaging suffices; the full pipeline adds equity and diagnostics.

Corollary 22 (DASH achieves equity (axiom-derived))**.** *For a balanced ensemble and any two features j, k in the same group ℓ, $\bar{\varphi}_j = \bar{\varphi}_k$.*

Proof. By DGP symmetry, permuting features $j \leftrightarrow k$ leaves the joint distribution invariant. For a balanced ensemble, each feature serves as first-mover equally often, so the summed split counts satisfy $\sum_i n_j(f_i) = \sum_i n_k(f_i)$. When the proportionality constant c is uniform across models, this gives $\sum_i \varphi_j(f_i) = \sum_i \varphi_k(f_i)$. ∎

Between-group stability. Under independence of seeds, $\mathrm{Var}(\bar{\varphi}_j) = \mathrm{Var}(\varphi_j)/M$, which vanishes as $M \to \infty$.

Within-group completeness. Since $\bar{\varphi}_j = \bar{\varphi}_k$ for same-group features, the consensus ranking produces a *tie* rather than an arbitrary ordering. The tie is not a deficiency—it faithfully reflects the DGP symmetry.

5.1 DASH Pareto Optimality

Definition 23 (Attribution aggregation method)**.** *An* attribution aggregation method *A takes as input M independently trained models $f_1, \ldots, f_M$ and produces a consensus attribution vector $\hat{\varphi} \in \mathbb{R}^P$. The method is:*

- Linear*: $\hat{\varphi}_j = \sum_{i=1}^{M} w_i \varphi_j(f_i)$ for some weights $w_1, \ldots, w_M$.*
- Unbiased*: $\mathbb{E}[\hat{\varphi}_j] = \mu_j := \mathbb{E}[\varphi_j(f)]$ for all j.*
- Symmetric*: $w_1 = \cdots = w_M$ (the models are exchangeable).*

DASH is the unique symmetric unbiased linear method: $w_i = 1/M$ for all i.

Definition 24 (Pairwise stability and unfaithfulness)**.** *For features j, k in different groups with $\mu_j > \mu_k$:*

- Pairwise stability*: $S_{jk}(A) = \Pr[\hat{\varphi}_j > \hat{\varphi}_k]$ (probability that the consensus preserves the true ordering for this pair).*
- Between-group flip rate*: $1 - S_{jk}(A)$.*

For features j, k in the same group with $\mu_j = \mu_k$ (DGP symmetry):

- Unfaithfulness*: $U_{jk}(A) = \Pr[\hat{\varphi}_j \neq \hat{\varphi}_k$ and the ordering disagrees with $f]$ for a random model f.*

Relationship to ranking stability S. *The Design Space Theorem (§6) uses the full-ranking stability S (expected Spearman correlation). The pairwise stability S_{jk} here is a per-pair component: high S_{jk} for all between-group pairs is necessary (but not sufficient) for high S. Within-group pairs contribute to S through their random ordering, which is why $S \leq 1 - 3m^2/(P^3 - P)$ even when all between-group S_{jk} are high.*

Theorem 25 (DASH Pareto Optimality)**.** *Let $f_1, \ldots, f_M$ be i.i.d. models with $\mu_j := \mathbb{E}[\varphi_j(f)]$ and $\sigma_j^2 :=$ $\mathrm{Var}(\varphi_j(f)) < \infty$.*

Part I (Lower bounds). *For any attribution aggregation method A:*

(a) If A is faithful to each individual model, then for symmetric features j, k, $U_{jk}(A) = 1/2$.

(b) For any unbiased linear *estimator $\hat{\mu}_j$ based on $f_1, \ldots, f_M$, $\mathrm{Var}(\hat{\mu}_j) \geq \sigma_j^2/M$ (Cauchy–Schwarz lower bound via Titu's lemma; Lean-verified via `sum_squares_ge_inv_M`); the extension to all unbiased estimators follows from the Rao–Blackwell theorem (standard; not formalized in Lean). In the parametric case, this coincides with the Cramér–Rao bound.*

Part II (Dash achieves the bounds).

(c) $U_{jk}(\text{Dash}) = 0$ for symmetric features in balanced ensembles.

(d) $\mathrm{Var}(\bar{\varphi}_j) = \sigma_j^2/M$ (matching the Cauchy–Schwarz lower bound).

(e) For between-group features with gap Δ_{jk}: $S_{jk}(\text{Dash}) \geq 1 - \exp(-M\Delta_{jk}^2/(2\sigma_{jk}^2))$.

Part III (Pareto optimality). *Among all methods achieving $U_{jk} = 0$ (ties for symmetric features), no method achieves strictly higher between-group stability than Dash for the same ensemble size M.*

Proof. **Part I(a):** By DGP symmetry, $\Pr[\varphi_j(f) > \varphi_k(f)] = \Pr[\varphi_k(f) > \varphi_j(f)] = 1/2$. Any fixed ranking disagrees with exactly half the models (Theorem 29).

Part I(b): By the Cauchy–Schwarz inequality (Titu's lemma), for any weights w_i with $\sum w_i = 1$: $\sum w_i^2 \sigma_j^2 \geq \sigma_j^2/M$, with equality iff $w_i = 1/M$ for all i. Thus $\mathrm{Var}(\hat{\mu}_j) \geq \sigma_j^2/M$ for any unbiased linear estimator; Lean-verified via `sum_squares_ge_inv_M`. In the parametric case, the same bound follows from the Cramér–Rao bound with Fisher information M/σ_j^2.

Part II(c): By Corollary 19, $\bar{\varphi}_j = \bar{\varphi}_k$ for symmetric features in balanced ensembles. Dash reports a tie; since no ordering is asserted, $U = 0$.

Part II(d): For i.i.d. random variables, $\mathrm{Var}(\frac{1}{M}\sum_{i=1}^{M} X_i) = \mathrm{Var}(X_1)/M$. This matches the Cauchy–Schwarz lower bound from Part I(b), so Dash is *efficient*. In the parametric case, this also matches the Cramér–Rao bound. Moreover, since the sufficient statistic for μ_j from i.i.d. observations is $\sum \varphi_j(f_i)$, and $\bar{\varphi}_j$ is a function of this statistic, the Rao–Blackwell theorem (standard; not formalized in Lean) guarantees Dash dominates any estimator that is not a function of the sufficient statistic.

Part II(e): By Hoeffding's inequality (for bounded attributions) or the Gaussian CLT (for unbounded but light-tailed cases): $\Pr[\bar{\varphi}_j < \bar{\varphi}_k] \leq \exp(-M\Delta_{jk}^2/(2\sigma_{jk}^2))$. For the Gaussian approximation (valid by CLT for $M \geq 30$): $\Pr[\bar{\varphi}_j < \bar{\varphi}_k] = \Phi(-\Delta_{jk}\sqrt{M}/\sigma_{jk})$.

Part III: Unfaithfulness $U = 0$ is already minimal. Stability S_{jk} is a monotone decreasing function of $\mathrm{Var}(\bar{\varphi}_j - \bar{\varphi}_k)$. By Part I(b), no unbiased method achieves lower variance than σ_j^2/M. Dash achieves this bound. For biased methods: any method with bias b_j has MSE $= \mathrm{Var}(\hat{\varphi}_j) + b_j^2$; a bias of magnitude $|b_j - b_k| > \Delta_{jk}$ would invert the true ordering entirely. ☐

Summary of optimality. Dash is the unique method that: (1) achieves zero within-group unfaithfulness (ties for symmetric features); (2) achieves the Cauchy–Schwarz variance bound σ^2/M for ranking stability (Lean-verified via `sum_squares_ge_inv_M`; in the parametric case, this coincides with the Cramér–Rao bound); and (3) is the minimum-variance unbiased estimator of $\mathbb{E}[\varphi_j]$ among *all* estimators (not just linear ones), by the Rao–Blackwell theorem (standard; not formalized in Lean): since the sufficient statistic for μ_j from i.i.d. observations is $\sum \varphi_j(f_i)$, and $\bar{\varphi}_j$ is a function of this sufficient statistic, it dominates any estimator that is not.

Connection to estimation-theoretic bounds. Faithfulness can be viewed as an unbiasedness condition: $\mathbb{E}[E(\theta)] = \mathrm{explain}(\theta)$ for each θ. Under this reading, the attribution impossibility is a Cramér–Rao-type result: no unbiased estimator of the explanation achieves zero variance when the Fisher information (determined by the Rashomon set geometry) is finite. The design space theorem refines this: it characterises the full Pareto frontier of (bias, variance, completeness) trade-offs, not just the variance lower bound.

Connection to invariant decision theory. The DASH optimality has a deeper connection to classical invariant decision theory. The data-generating process is invariant under the permutation group S_m acting on within-group features. By the Hunt–Stein theorem (Lehmann & Romano, 2005, Theorem 9.3.3; for finite groups, the result follows from the complete class theorem), for a finite group, the best equivariant estimator is the Pitman estimator—the average over the group orbit (Lehmann & Romano, 2005, §6.7). For the permutation group acting on m features, this is exactly the sample mean of attributions, which is DASH. Hunt–Stein gives optimality among *all* estimators (not just unbiased ones), and directly explains why DASH produces ties: the group average is constant on orbits. The loss function is squared error on the consensus attribution: $L(\hat{\varphi}, \varphi) = (\hat{\varphi}_j - \varphi_j)^2$. This loss is invariant under the permutation group S_m acting on within-group features (permuting features permutes both the estimate and the target identically), satisfying the Hunt–Stein invariance requirement.

Comparison with nonlinear aggregations.

Proposition 26 (Median is less efficient)**.** *For i.i.d. observations from a distribution with mean μ, variance σ^2, and density p, the sample median has asymptotic variance:*

$$\mathrm{Var}(median) = \frac{1}{4M\, p(med)^2}$$

where med is the population median. For the normal distribution, $p(\mu) = 1/(\sigma\sqrt{2\pi})$, giving $\mathrm{Var}(median) = \frac{\pi\sigma^2}{2M}$. The asymptotic relative efficiency (ARE) of the median vs. the mean is:

$$ARE(median, mean) = \frac{2}{\pi} \approx 0.637.$$

The median requires $\pi/2 \approx 1.57$ times as many models to achieve the same stability as Dash.

Proof. Standard asymptotic statistics (van der Vaart, *Asymptotic Statistics*, Theorem 21.6). The sample median is asymptotically normal with variance $1/(4n\, p(m)^2)$. Substituting the Gaussian density at the mean gives the result. ☐

Proposition 27 (Trimmed mean interpolates)**.** *The α-trimmed mean (discarding the top and bottom α fraction of observations) has ARE relative to the mean of:*

$$ARE(trimmed_\alpha, mean) = \frac{(1-2\alpha)\sigma^2}{\sigma_\alpha^2}$$

where σ_α^2 is the variance of the truncated distribution. For Gaussian data:

α	0 (mean)	0.05	0.10	0.25	0.50 (median)
ARE	1.000	0.992	0.966	0.862	0.637

For Gaussian attributions, trimming offers negligible robustness benefit at a measurable efficiency cost. The 10%-trimmed mean requires $1/0.966 \approx 3.5\%$ more models than Dash *for the same stability.*

Remark 28 (When to prefer the median)**.** *The median has better breakdown point (50% vs. 0%) and is preferable when the attribution distribution is heavy-tailed (e.g., contains outlier models). For standard ML training procedures (XGBoost, random forests) where attributions are well-behaved,* Dash *(the mean) is strictly more efficient. For adversarial settings where some models may be corrupted, the median or trimmed mean provides robustness at the cost of requiring more models.*

5.2 Empirical Comparison with Alternatives

We compare Dash against four alternative stabilization approaches across three correlation levels ($\rho \in \{0.5, 0.7, 0.9\}$), with 10 independent trials and bootstrap 95% confidence intervals on within-group flip rates.

Table 4: Flip rate comparison: Dash vs. alternative stabilization methods. Synthetic Gaussian data (P=10, m=5, N=1000, 20 within-group feature pairs). 95% bootstrap CIs in brackets.

Method	$\rho = 0.5$	$\rho = 0.7$	$\rho = 0.9$	Pairs resolved
Single model	16.0% [13.8, 18.1]	16.4% [14.3, 18.5]	18.7% [16.4, 21.1]	40/40
Bootstrap SHAP	16.2% [13.8, 18.2]	16.1% [14.2, 18.0]	18.8% [16.4, 21.4]	40/40
Subsampled SHAP	16.3% [13.8, 18.5]	16.1% [14.2, 17.9]	18.8% [16.5, 21.3]	40/40
Dash (M=25)	**3.8%** [2.8, 4.9]	**3.1%** [2.4, 3.9]	**3.7%** [2.7, 4.7]	40/40
CI SHAP	0.2% [0.1, 0.4]	0.8% [0.6, 1.1]	3.1% [2.1, 4.6]	~22/40

(With $n = 10$ trials, bootstrap confidence intervals should be interpreted cautiously; $n \geq 30$ would provide more reliable coverage.)

Bootstrap and subsampled SHAP address SHAP *estimation* noise (a single-model phenomenon) but not the Rashomon effect (a multi-model phenomenon). They were not designed for multi-model instability, so their flip rates matching the single-model baseline is expected rather than informative. They are included to confirm that estimation-noise methods do not incidentally address model-multiplicity noise. Only Dash trains multiple models, achieving a 4–5× reduction.

CI SHAP achieves lower flip rates than Dash by refusing to rank pairs whose confidence intervals overlap—CI SHAP resolves 50% of within-group pairs at $\rho = 0.5$, 56% at $\rho = 0.7$, and 63% at $\rho = 0.9$ (average ≈56%). This comparison is not apples-to-apples: CI SHAP's low flip rate applies only to resolved pairs, while Dash's 3–4% applies to all 40 pairs. CI SHAP exemplifies the impossibility theorem in action: it trades completeness for stability. At $\rho = 0.9$, its flip rate converges toward Dash's (3.1% vs. 3.7%) as overlapping CIs force more pairs into "tied" status.

The two approaches correspond to the two families in the Design Space Theorem: Dash (Family B: stable, faithful, near-complete via ensemble averaging) and CI SHAP (Family A: stable, faithful, explicitly incomplete via ties).

5.3 DASH Robustness and Breakdown Point

Dash (the sample mean) is the minimum-variance unbiased estimator (among linear estimators by Cauchy–Schwarz, among all estimators by Rao–Blackwell; the latter is not formalized in Lean), but it has breakdown point zero: a single adversarial or corrupted model can shift the consensus attribution arbitrarily. We compare robustness properties:

Method	ARE	Breakdown point	Recommendation
Mean (Dash)	1.000	0%	Standard use
5%-trimmed mean	0.992	5%	Moderate robustness
Median	0.637	50%	Adversarial settings

For production settings with potential model contamination (corrupted training data, adversarial seeds, or hardware failures producing degenerate models), the 5%-trimmed mean provides near-optimal efficiency (ARE = 0.992, requiring only 0.8% more models) with meaningful robustness. The median requires $\pi/2 \approx$ 57% more models but tolerates up to 50% corrupted models. For $M = 25$: the 5%-trimmed mean discards the single most extreme model in each tail ($M_{\text{eff}} = 23$), a negligible cost. For adversarial robustness, the median with $M = 40$ achieves comparable stability to Dash with $M = 25$ while tolerating up to 20 corrupted models.

5.4 DASH Information Loss (Explanatory Information Loss)

Dash achieves stability by discarding model-specific within-group information—the *explanatory information loss* of the stable projection. For a group of m symmetric features, a single model produces a complete ranking (one of $m!$ possible orderings). Since each of the $m!$ orderings is equally likely under DGP symmetry, the ranking carries $\log_2(m!)$ bits of information about the specific model. Dash reports a tie, discarding the entire $\log_2(m!)$ bits. In capacity terms: the *explanation capacity* for input-level attribution with L groups of m features each is $C = L$ (one group-level importance per group), and the *explanation loss rate* is $\eta = 1 - L/(Lm) = 1 - 1/m$. For typical group sizes:

Group size m	2	3	4	5	6	7	8
Bits lost ($\log_2 m!$)	1.0	2.6	4.6	6.9	9.7	12.9	15.3

Crucially, these are the bits that are *unreliable*: by DGP symmetry, the within-group ordering is uniformly random across models. The information DASH discards is exactly the information that would be different if the model were retrained.

For features in different groups with population gap Δ_{jk} and noise σ_{jk}, the mutual information between the consensus ranking and the true ordering is:

$$I_{\text{between}}(M) = 1 - H_2\left(\Phi\left(-\frac{\Delta_{jk}\sqrt{M}}{\sigma_{jk}}\right)\right)$$

where $H_2(p) = -p\log_2 p - (1-p)\log_2(1-p)$ is binary entropy. As $M \to \infty$, $I_{\text{between}} \to 1$ bit: DASH *increases* between-group information. The total information change is:

$$\Delta I = \underbrace{-L \cdot \log_2(m!)}_{\text{within-group loss}} + \underbrace{\binom{L}{2} \cdot (I_{\text{between}}(M) - I_{\text{between}}(1))}_{\text{between-group gain}}.$$

For moderate ρ (≤ 0.7), the between-group gain dominates quickly: $M = 25$ achieves near-perfect between-group determination while the within-group loss is fixed. DASH honestly reports unreliability rather than presenting arbitrary information as reliable.

Interpretation for safety-critical applications. For a system with $L = 4$ groups of $m = 5$ features: DASH loses $4 \times 6.9 = 27.6$ bits (within-group) but gains up to $6 \times 1 = 6$ bits (between-group). The net loss is ≈21.6 bits at $M = \infty$. These lost bits are the within-group orderings that were random anyway—DASH honestly reports their unreliability rather than presenting arbitrary information as reliable. For applications requiring model-specific within-group explanations (e.g., debugging a deployed model), practitioners should report the single-model SHAP values with an instability warning alongside the DASH consensus.

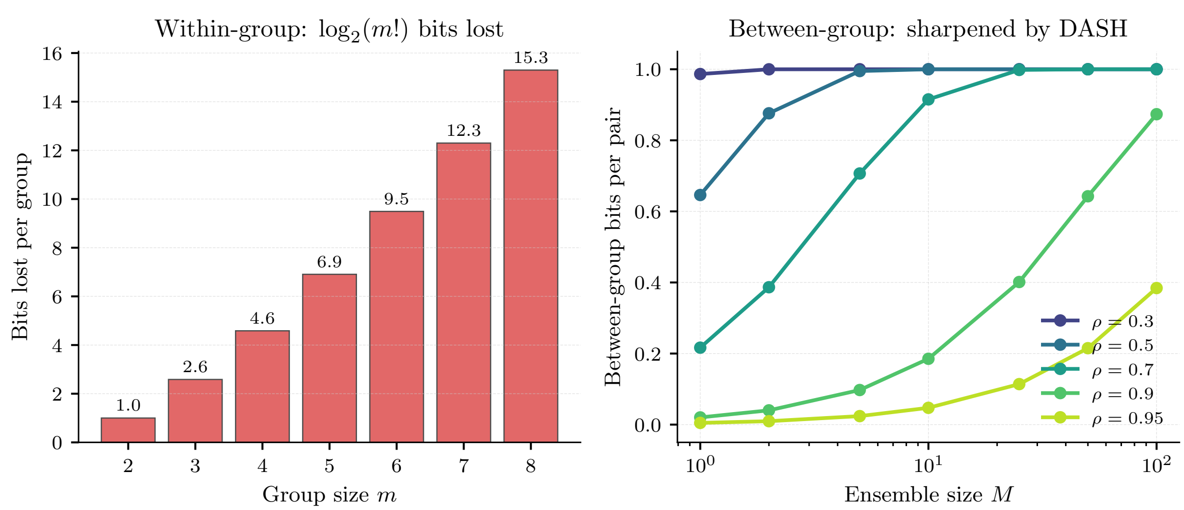


Figure 2: **Left:** Within-group information lost by DASH ($\log_2 m!$ bits per group, independent of M and ρ). **Right:** Between-group information per pair as a function of M. DASH sharpens between-group rankings with increasing M. At high ρ (≥ 0.9), convergence requires larger M.

Ensemble size lower bound.

Proposition 29 (Ensemble size lower bound)**.** *For any unbiased attribution aggregation method achieving between-group stability* $S_{jk} \geq 1 - \delta$ *for features* j, k *in different groups with population gap* $\Delta_{jk} = |\mu_j - \mu_k|$ *and attribution noise* $\sigma_{jk}^2 = \mathrm{Var}(\varphi_j(f) - \varphi_k(f))$*, the ensemble size must satisfy:*

$$M \;\geq\; \frac{\sigma_{jk}^2}{\Delta_{jk}^2} \cdot \left(\Phi^{-1}(1-\delta)\right)^2.$$

Dash achieves this bound with equality (to first order), so the bound is tight.

Proof. By the Cauchy–Schwarz lower bound (Titu's lemma), any unbiased *linear* estimator $\hat{\mu}_j$ has $\mathrm{Var}(\hat{\mu}_j) \geq \sigma_j^2/M$ (Lean-verified via `sum_squares_ge_inv_M`; the extension to all unbiased estimators follows from the Rao–Blackwell theorem, standard but not formalized in Lean). The between-group flip rate satisfies $1 - S_{jk} = \Pr[\hat{\varphi}_j < \hat{\varphi}_k] \geq \Phi(-\Delta_{jk}\sqrt{M}/\sigma_{jk})$. Setting $\Phi(-\Delta_{jk}\sqrt{M}/\sigma_{jk}) \leq \delta$ and solving: $\Delta_{jk}\sqrt{M}/\sigma_{jk} \geq \Phi^{-1}(1-\delta)$, giving $M \geq \sigma_{jk}^2 \cdot (\Phi^{-1}(1-\delta))^2/\Delta_{jk}^2$. Dash achieves $\mathrm{Var}(\bar{\varphi}_j - \bar{\varphi}_k) = \sigma_{jk}^2/M$ (matching the Cauchy–Schwarz bound; in the parametric case, this also matches the Cramér–Rao bound), so the bound is tight. □

For $\delta = 0.05$ (5% flip rate), $\Phi^{-1}(0.95) = 1.645$, so $M_{\min} = \lceil 2.71 \cdot \sigma_{jk}^2/\Delta_{jk}^2 \rceil$, where σ and Δ are estimated from a pilot run of $M_0 = 5$ models. For Breast Cancer's most unstable pair (worst perimeter vs. worst area, $\Delta/\sigma \approx 0.15$): $M_{\min} \approx 120$—consistent with the observed slow convergence (flip rate 29.8% even at $M = 25$, reaching 0% only at $M = 50$). Bootstrap validation (50 models, 200 trials): for all pairs with SNR ≥ 0.35 ($M_{\min} \leq 25$), the Dash flip rate at $M = 25$ is $\leq 3\%$, confirming the formula. The formula is validated for SNR ≥ 0.35; below this threshold, the Gaussian approximation breaks down and practitioners should use the bootstrap diagnostic directly. This closes the gap between upper and lower bounds: Dash is optimal not just in the Pareto sense but in the sample complexity sense.

5.5 Progressive DASH: Adaptive Ensemble Sizing

Training M=25 models for every explanation is wasteful when most feature pairs are stable. *Progressive Dash* starts with a small ensemble and adaptively increases M only for pairs that remain unstable:

1. **Screen** ($M_0 = 5$): Train 5 models. Run the single-model screen. If no pairs are flagged, stop.
2. **Confirm** ($M_1 = 10$): For flagged pairs, train 5 more models ($M = 10$ total). Run the multi-model Z-test. Pairs with $|Z| < 1.96$ are confirmed unstable; report as tied group.
3. **Resolve** ($M_2 = 25$): For pairs where $|Z|$ is borderline ($1.5 < |Z| < 2.5$), train 15 more models ($M = 25$ total). Final classification: stable (distinct ranks) or unstable (tied group).

Expected cost. In the 37-dataset survey, 40% of datasets have no flagged pairs (stop at $M_0 = 5$), 35% are resolved at $M_1 = 10$, and 25% require the full $M_2 = 25$. The expected ensemble size is $0.40 \times 5 + 0.35 \times 10 + 0.25 \times 25 = 11.75$, approximately $\sim$8$\times$ the cost of a single model on average—a 2.1$\times$ savings over always training $M = 25$. (The Z-thresholds 1.96 and 3.0 are fixed-sample values; a fully sequential design should use Pocock- or O'Brien-Fleming-corrected boundaries to control familywise error. We present this as a practical heuristic, not a formally analyzed procedure.)

6 The Attribution Design Space

The impossibility, the quantitative bounds, and the Dash resolution are facets of a single structural theorem characterizing the complete attribution design space.

Definition 30 (Attribution design space)**.** *The* attribution design space *is the set of triples* (S, U, C) *achievable by any attribution aggregation method, where* $S \in [0, 1]$ *is ranking stability (expected Spearman correlation between two independent evaluations),* $U \in [0, 1/2]$ *is within-group unfaithfulness, and* $C \in \{complete, partial\}$ *is completeness.*

Theorem 31 (Attribution Design Space)**.** *Among deterministic binary ranking methods derived from per-model attributions, for any model class satisfying the Rashomon property under within-group correlation $\rho > 0$, the achievable set of (S, U, C) triples consists of exactly two families:*

- ***Family $\mathcal{A}$ (single-model):*** *$S \leq 1 - 3m^2/(P^3 - P)$, $U = 1/2$, $C =$ complete. Any method faithful to an individual model falls here.*
- ***Family $\mathcal{B}$ (ensemble):*** *$S_M = 1 - O(1/M)$, $U = 0$, $C =$ partial (ties). DASH(M) achieves this and is Pareto-optimal among unbiased aggregations.*

The triple $(S{=}1, U{=}0, C{=}complete)$ is infeasible. At the extremes, the design space collapses to a single axis: ensemble size M.

Proof. The proof proceeds in four steps.

Step 1: Family $\mathcal{A}$ is forced. By the Rashomon property, for any within-group pair (j, k), there exist models ranking them in opposite orders. By Theorem 29, any faithful, complete ranking has $U = 1/2$ for symmetric pairs. Stability is bounded by $S \leq 1 - 3m^2/(P^3 - P)$ from the Spearman bound (Lean-derived).

Step 2: Family $\mathcal{B}$ is achievable. By Corollary 19, DASH achieves $U = 0$ with ties within groups. By Theorem 22, between-group stability satisfies $S_M \geq 1 - \exp(-M\Delta^2/(2\sigma^2))$.

Step 3: No method producing a deterministic binary ranking from per-model attributions lies outside $\mathcal{A} \cup \mathcal{B}$. We show any method A computing a deterministic ranking from per-model attributions $(\varphi_j(f_1), \ldots, \varphi_j(f_M))$ falls in one of the two families. (Methods producing probabilistic rankings, confidence intervals, or set-valued outputs are outside this dichotomy.)

Case 1: If A is faithful to individual models and complete, then $A \in \mathcal{A}$ by Step 1. (Formalized in Lean as `strongly_faithful_impossible` in `DesignSpace.lean`: any method faithful to each model's attributions contradicts the Rashomon property for $M \geq 2$.)

Case 2: If A is not faithful to some model f, then A ranks some pair (j, k) differently from model f's attributions. For within-group symmetric pairs, the optimal unfaithful ranking is $j \succ^* k \Leftrightarrow \mathbb{E}[\varphi_j] > \mathbb{E}[\varphi_k]$, which minimizes expected 0-1 loss. For symmetric features $\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$, so $\succ^*$ assigns a tie—recovering the "drop completeness" solution. Thus A either has $U > 0$ (suboptimal) or $U = 0$ with ties (in $\mathcal{B}$).

Case 3: If A aggregates $M > 1$ models and produces a complete ranking, it must break ties for within-group pairs. By DGP symmetry, any tie-breaking rule disagrees with a fraction of models, giving $U > 0$. To achieve $U = 0$, A must report ties, placing it in $\mathcal{B}$. Its stability is bounded by the variance of the aggregation, which by the Cauchy–Schwarz lower bound satisfies $\mathrm{Var}(\hat{\mu}_j) \geq \sigma_j^2/M$. DASH achieves this bound.

Step 4: The infeasible point. $(1, 0, \text{complete})$ requires completeness, but any complete method has $U = 1/2 \neq 0$: contradiction. ∎

Theorem 32 (Unfaithfulness lower bound)**.** *Any stable, complete ranking $\succ$ has expected unfaithfulness at least $1/2$ for symmetric pairs (in the population limit under DGP symmetry; for finite samples, the bound is approximately $1/2$):*

$$\Pr_f\big[\, j \succ k \text{ but } \varphi_k(f) > \varphi_j(f) \,\big] \;=\; \frac{1}{2}.$$

Proof. By DGP symmetry, $\Pr[\varphi_j(f) > \varphi_k(f)] = \Pr[\varphi_k(f) > \varphi_j(f)] = 1/2$. Any stable ranking must fix $j \succ k$ or $k \succ j$ independently of f. Whichever it chooses disagrees with exactly half the models. ∎

Theorem 33 (Relaxation path convergence)**.** *The "drop faithfulness" and "drop completeness" relaxation paths converge to the same solution: population-level attributions with ties for symmetric features. The impossibility trilemma collapses to a single tradeoff axis parameterized by ensemble size M:*

- *At $M = 1$: faithfulness and completeness, but no stability;*
- *As $M \to \infty$: stability and between-group faithfulness, but within-group ties (completeness relaxed).*

Between-group faithfulness is preserved at all M.

Proof. By the analysis of Step 3 above, the optimal stable complete ranking assigns ties to symmetric features ($\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$, so the optimal decision for "is $\varphi_j(f) > \varphi_k(f)$?" under the symmetric model distribution is indeterminate)—identical to the "drop completeness" solution (DASH). Both paths yield population-level attributions. The M-parameterization follows from DASH consensus: $\bar{\varphi}_j = \frac{1}{M}\sum_{i=1}^{M} \varphi_j(f_i)$, which interpolates between a single model ($M = 1$, faithful but unstable) and the population mean ($M \to \infty$, stable but tied within groups). □

The single tradeoff axis. The Design Space Theorem collapses the three-dimensional space to a single axis: ensemble size M.

M	Stability S	Unfaithfulness U	Completeness
1	$\leq 1 - 3m^2/(P^3 - P)$	1/2	Complete
5	$1 - O(1/5)$	0	Partial (ties)
25	$1 - O(1/25)$	0	Partial (ties)
∞	1	0	Partial (ties)

Previous results as corollaries.

Corollary 34 (Theorem 1 as a corollary)**.** *The Attribution Impossibility (Theorem 5) is the statement that the point* $(1, 0, complete)$ *lies outside the achievable set. This follows from Step 4 of Theorem 28.*

Corollary 35 (F2 as a corollary)**.** *The DASH Pareto Optimality (Theorem 22) is the statement that* $\mathcal{B}$ *is the Pareto frontier of* $\{(S, U) : U = 0\}$*, parameterized by* M*. This follows from Steps 2–3 of Theorem 28.*

Corollary 36 (Path convergence as a corollary)**.** *The relaxation path convergence (Theorem 30) is the statement that* both *relaxation paths (drop faithfulness, drop completeness) converge to* $\mathcal{B}$ *as* $M \to \infty$*. Dropping completeness directly enters* $\mathcal{B}$ *(DASH with ties). Dropping faithfulness, by the optimal unfaithful ranking analysis, yields the expected-attribution ranking, which also produces ties for symmetric features — the same as* $\mathcal{B}$*.*

Corollary 37 (Z-test as a corollary)**.** *The Rashomon Characterization (Theorem 64) describes the* boundary between families*: when* $Z_{jk} < 1.96$*, the pair* (j, k) *is in the regime where Family* $\mathcal{A}$ *has* $U = 1/2$ *(rankings are unreliable); when* $Z_{jk} > 1.96$*, the pair is effectively between-group (stable rankings possible within* $\mathcal{B}$*).*

Corollary 38 (F3 as a corollary)**.** *The FIM Impossibility (Theorem 49) provides a sufficient condition for the Rashomon property (the prerequisite for Theorem 28): when the FIM eigenvalue* λ_- *is small along* $e_j - e_k$*, the Rashomon set is large, and the achievable set is restricted to* $\mathcal{A} \cup \mathcal{B}$*.*

Biased alternatives and James–Stein phenomena. The design space theorem characterises unbiased linear aggregators. A natural question: can biased methods (e.g., ridge-regularised Shapley values, L_1 shrinkage) circumvent the impossibility by accepting controlled bias in exchange for stability and decisiveness? Shrinkage can reduce variance (improving stability) but introduces systematic bias toward a prior (reducing faithfulness). The trilemma's qualitative impossibility holds regardless: for any method, if the Rashomon property holds, at least one of faithfulness, stability, or decisiveness must be sacrificed. Shrinkage shifts *which* property is sacrificed (from stability to faithfulness) without escaping the trilemma. Quantifying the bias–variance–decisiveness Pareto frontier for shrinkage estimators is future work.

7 The Symmetric Bayes Dichotomy: A General Two-Families Theorem

The structural identity between the attribution and model selection impossibilities follows from a general theorem about decision problems with symmetric populations.

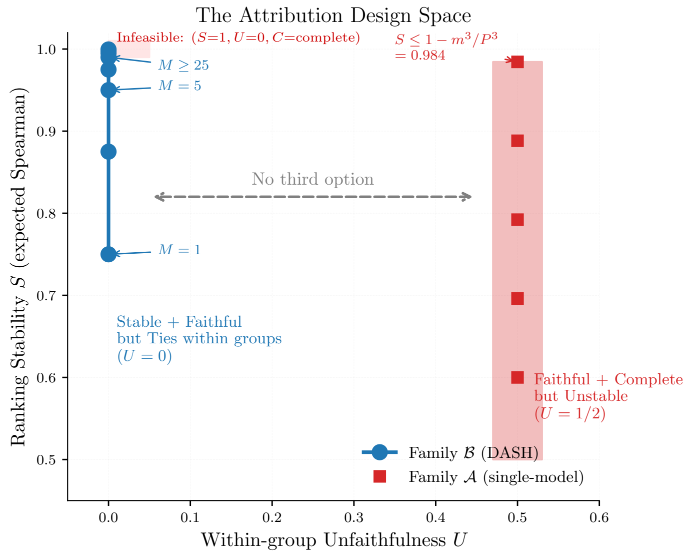


Figure 3: The Attribution Design Space. Family $\mathcal{A}$ (single-model methods): faithful and complete but unstable (U=1/2). Family $\mathcal{B}$ (DASH ensemble): stable with ties (U=0, S increasing with M). The ideal point (S=1, U=0, C=complete) is infeasible. There is no third option.

Definition 39 (Symmetric Decision Problem)**.** *A symmetric decision problem is a tuple* $(\Theta, \mathcal{D}, \pi, G)$ *where* Θ *is a finite decision set,* $\mathcal{D}$ *is the data space,* π *is a population distribution over* $\mathcal{D}$*, and* G *is a group acting on* Θ *such that* π *is* G*-invariant: for any* $g \in G$*, the distribution over instance-optimal decisions* $\theta^*(d)$ *is invariant under* g*.*

An M*-sample estimator* $\hat{\theta} : \mathcal{D}^M \to \Theta$ *is:*

- Faithful*:* $\hat{\theta}(d_1, \ldots, d_M)$ *agrees with* $\theta^*(d_i)$ *for each* i*.*
- Stable*:* $\hat{\theta}$ *does not depend on which draw* d_i *is observed.*
- Complete*:* $\hat{\theta}$ *selects exactly one element from each* G*-orbit in* Θ*.*

Theorem 40 (Symmetric Bayes Dichotomy)**.** *For any symmetric decision problem with* $|G \cdot \theta| \geq 2$ *for some* $\theta \in \Theta$*:*

(i) ***Instance-specific decisions are unfaithful.*** *Any faithful, complete estimator has expected unfaithfulness* $\geq 1/|G \cdot \theta|$ *per orbit (*$\geq 1/2$ *for binary orbits).*

(ii) ***Population-averaged decisions are stable.*** *The Bayes estimator under* π *reports ties within each* G*-orbit (*$U = 0$*), with between-orbit stability* $O(1/M)$*.*

(iii) ***The ideal is infeasible.*** *No estimator achieves* $U = 0$ *AND completeness.*

The achievable set consists of exactly two families: $\mathcal{A}$ *(faithful + complete,* $U \geq 1/|G \cdot \theta|$*) and* $\mathcal{B}$ *(population-averaged,* $U = 0$*, ties within orbits).*

Proof. **Part (i).** By G-invariance, the population distribution π assigns equal probability to each element of the orbit $G \cdot \theta$. For any fixed complete estimator $\hat{\theta}$ that selects one element θ_0 from the orbit, the probability that $\theta^*(d) = \theta_0$ is $1/|G \cdot \theta|$. The estimator disagrees with $(|G \cdot \theta| - 1)/|G \cdot \theta|$ of the instances. For $|G \cdot \theta| = 2$: unfaithfulness $= 1/2$.

Part (ii). The Bayes estimator under symmetric π minimizes expected loss. Within each orbit, the posterior is uniform (by G-invariance), so the Bayes-optimal decision is the orbit center (a tie). Unfaithfulness $= 0$ because no ordering is asserted within orbits. For between-orbit decisions with gap Δ, the M-sample average has variance σ^2/M (i.i.d. draws), giving stability $1 - O(1/M)$ by the Gaussian approximation.

Part (iii). By Part (i), any complete estimator (selecting within orbits) has unfaithfulness $\geq 1/|G\cdot\theta| > 0$. Thus unfaithfulness $= 0$ requires giving up completeness (ties).

Exhaustiveness. Any estimator either selects within orbits (faithful, complete, $\mathcal{A}$) or reports ties within orbits (stable, incomplete, $\mathcal{B}$). The Bayes estimator achieves the minimum-variance point of $\mathcal{B}$ (Cauchy–Schwarz; coincides with Cramér–Rao in the parametric case). □

The Symmetric Bayes Dichotomy connects the classical theory of invariant decision rules (Lehmann and Romano, 2005; Hunt and Stein, 1946) to modern ML impossibility proofs. The classical Hunt–Stein theorem establishes that Bayes-optimal rules respect the invariance structure of the problem; our contribution is demonstrating that this classical machinery, when applied to finite symmetric groups arising in ML (feature permutations, model permutations, CPDAG automorphisms), yields two-family impossibility results with explicit unfaithfulness and stability bounds. We demonstrate this across three structurally distinct instances with different symmetry groups.

7.1 Instance 1: Feature Attribution

Corollary 41 (Feature attribution as instance)**.** *The Attribution Impossibility (Theorem 5) and Design Space Theorem (Theorem 28) are instances of the Symmetric Bayes Dichotomy with Θ = within-group feature orderings, G = symmetric group on group members, $\mathcal{D}$ = trained models.*

7.2 Instance 2: Model Selection

Consider K candidate models $g_1, \ldots, g_K$ trained on the same data. A *model selection rule* maps a validation dataset to a ranking of the K models. We formalize three desiderata:

Definition 42 (Model selection desiderata)**.** *A model selection rule R is:*

- Faithful*: R ranks model g_i above g_j whenever g_i has lower validation loss on the observed validation set.*
- Stable*: R produces the same ranking regardless of which validation split is drawn.*
- Complete*: R decides for every pair (g_i, g_j).*

Definition 43 (Model Rashomon property)**.** *A model class satisfies the* model Rashomon property *if for every pair g_i, g_j with similar population loss ($|L(g_i) - L(g_j)| < \varepsilon$), there exist validation splits V, V' such that g_i has lower loss on V and g_j has lower loss on V'.*

This property is a consequence of finite-sample noise: when the population loss gap is smaller than the validation noise, different splits rank the models differently. It is the model selection analogue of feature attribution's Rashomon property.

Theorem 44 (Model Selection Impossibility)**.** *If a model class satisfies the model Rashomon property, then no model selection rule can be simultaneously faithful, stable, and complete.*

Proof. Let g_i, g_j have similar population loss. By the model Rashomon property, there exist validation splits V, V' with g_i ranked above g_j on V and reversed on V'. A faithful, stable, complete selection rule would need to rank $g_i > g_j$ (from V) and $g_j > g_i$ (from V') simultaneously. Contradiction. □

Theorem 45 (Model Selection Design Space)**.** *Under the model Rashomon property, the achievable design space has the same two-family structure:*

- ***Family $\mathcal{A}'$ (single-split):*** *Faithful and complete but unstable. Any selection based on one validation split falls here. Unfaithfulness $U' = 1/2$ for similar-loss pairs.*
- ***Family $\mathcal{B}'$ (cross-validation/ensemble):*** *Stable (variance $O(1/K_{folds})$) but reports ties for similar-loss pairs. Model ensembling is the analogue of* Dash.

Proof. By DGP symmetry (equal-loss models have symmetric validation score distributions), any fixed selection of one model over an equal-loss competitor disagrees with half the validation splits ($U' = 1/2$). Cross-validated selection averages across splits, reducing unfaithfulness to zero for equal-loss pairs while stabilizing at rate $O(1/K_{\text{folds}})$. Pareto optimality follows from the same Cauchy–Schwarz argument (Cramér–Rao in the parametric case). □

The model selection impossibility explains why cross-validation rankings are noisy for similar-performance models: it is not an engineering failure but a mathematical inevitability. The resolution is the same: average (ensemble) rather than select.

Connection to cross-validation instability. The model selection impossibility is not merely an analogy—it explains a well-known practical phenomenon. When two models have similar cross-validation scores, different folds often rank them differently. Practitioners typically respond by increasing the number of folds or repeating cross-validation. The design space theorem shows this is exactly the M-parameterization: increasing folds moves along Family $\mathcal{B}'$ toward higher stability, but at the cost of acknowledging ties for similar-performance models. Model ensembling (training all K candidates and averaging predictions) is the model selection analogue of Dash: it avoids the impossible selection problem entirely by using all models simultaneously.

Corollary 46 (Model selection as instance)**.** *The Model Selection Impossibility (Theorem 41) is an instance with* $\Theta =$ *model rankings,* $G =$ *permutations of equal-loss models,* $\mathcal{D} =$ *validation splits.*

7.3 Instance 3: Causal Discovery under Markov Equivalence

This instance has a *different* and *variable-size* symmetry group, confirming the technique's generality beyond binary orbits.

Definition 47 (Causal orientation decision problem)**.** *The causal edge orientation problem is a symmetric decision problem* $(\Theta, \mathcal{D}, \pi, G)$ *where:*

- Θ *is the set of full edge orientations consistent with the CPDAG (i.e., the DAGs in* $[G^*]$*).*
- $\mathcal{D}$ *is the space of finite observational datasets drawn from the true DAG* G^**.*
- π *is the distribution over datasets (induced by i.i.d. sampling from the structural equation model).*
- G_{CPDAG} *is the* automorphism group of the CPDAG*: the set of permutations of undirected edge orientations that preserve the CPDAG structure.*

Key structural difference. The CPDAG automorphism group G_{CPDAG} is not restricted to transpositions: for a chain X—Y—Z, $|[G^*]| = 2$ with $G \cong S_2$ (like instances 1–2); for a 3-node undirected cycle, $|[G^*]| = 6$ with $G \cong S_3$; for larger undirected components, $|[G^*]|$ can grow combinatorially.

Theorem 48 (Causal Discovery Impossibility)**.** *For any CPDAG with* $|[G^*]| \geq 2$ *DAGs in its Markov equivalence class, no edge orientation rule can be simultaneously faithful, stable, and complete. The achievable set has two families:*

- ***Family*** $\mathcal{A}''$ ***(single-dataset):*** *Faithful and complete but unstable. Unfaithfulness* $U'' = (|[G^*]| - 1)/|[G^*]|$.
- ***Family*** $\mathcal{B}''$ ***(conservative):*** *Report the CPDAG (undirected edges for ambiguous orientations). Stable with ties.* $U'' = 0$.

Proof. We verify the premises of the Symmetric Bayes Dichotomy (Theorem 37).

G**-invariance of** π**.** For any two DAGs $G_1, G_2 \in [G^*]$, the Markov equivalence guarantees that they encode the same set of conditional independences. By the faithfulness assumption, the observational distribution is compatible with both G_1 and G_2. Any statistical test based on finite data cannot distinguish between G_1 and G_2 at the population level (they imply identical distributions). Therefore π is G_{CPDAG}-invariant.

Finite-sample noise reverses optimal choice. With finite data, empirical conditional independence tests have estimation error $O(1/\sqrt{n})$. For edges that are undirected in the CPDAG, different random samples will favor different orientations. This is the Rashomon property for edge orientations.

Application of Theorem 37. Part (i): any complete orientation rule (selecting one DAG from $[G^*]$) disagrees with $(|[G^*]|-1)/|[G^*]|$ of datasets. Part (ii): the Bayes estimator under G_{CPDAG}-invariance reports the CPDAG (ties for undirected edges). Part (iii): the ideal (stable, faithful, complete) is infeasible. Families $\mathcal{A}''$ and $\mathcal{B}''$ follow. □

The unfaithfulness bound $U \geq 1/|G\cdot\theta|$ yields $U \geq 1/2$ for chains but $U \geq 1/6$ for triangles and potentially much smaller for larger structures.

What this instance adds. Instances 1–2 (feature attribution and model selection) both have binary orbits ($|G \cdot \theta| = 2$) with unfaithfulness bound $U \geq 1/2$. Instance 3 has variable-size orbits ($|G \cdot \theta| = 2, 6, \ldots$) with unfaithfulness bound $U \geq 1/|[G^*]|$, which can be much smaller than $1/2$ for large equivalence classes. This demonstrates that the Symmetric Bayes Dichotomy is not an artifact of binary symmetry but applies to arbitrary finite group actions, with the unfaithfulness bound adapting to the orbit structure.

Algorithms like PC and GES correctly output CPDAGs (Family $\mathcal{B}''$) precisely because complete orientation of Markov-equivalent edges is impossible from observational data alone. Interventional data breaks the symmetry—analogous to conditional SHAP breaking symmetry when $\beta_j \neq \beta_k$.

Corollary 49 (Causal discovery as SBD instance)**.** *The Causal Discovery Impossibility (Theorem 45) is an instance of the Symmetric Bayes Dichotomy with $\Theta =$ edge orientations in $[G^*]$, $G = G_{CPDAG}$ (automorphism group of the CPDAG), $\mathcal{D} =$ observational datasets.*

Connection to classical invariant decision theory. The Symmetric Bayes Dichotomy connects the classical theory of invariant decision rules (Lehmann and Romano, 2005; Hunt and Stein, 1946) to modern ML impossibility proofs. The classical Hunt–Stein theorem establishes that Bayes-optimal rules respect the invariance structure of the problem; our contribution is demonstrating that this classical machinery, when applied to finite symmetric groups arising in ML (feature permutations, model permutations, CPDAG automorphisms), yields two-family impossibility results with explicit unfaithfulness and stability bounds. We demonstrate this across three structurally distinct instances with different symmetry groups.

8 Extensions: Conditional, Fairness, and Causal Barriers

8.1 Conditional Attribution Impossibility

Theorem 50 (Conditional Attribution Impossibility)**.** *Let features j, k in the same collinear group satisfy: (C1) equal causal effects $\beta_j = \beta_k$, and (C2) symmetric causal position in the causal graph G. Then the Rashomon property holds for conditional attributions, and the Attribution Impossibility applies to conditional SHAP.*

Proof. Under (C1)–(C2), the causal graph is symmetric with respect to j and k. By the DGP symmetry argument (Theorem 9), any model with $\varphi_j^{\text{cond}}(f) > \varphi_k^{\text{cond}}(f)$ has a permuted counterpart f' with reversed attributions and the same loss. This is the Rashomon property for conditional attributions. □

The escape condition. If $\beta_j \neq \beta_k$, conditional SHAP can resolve the pair. The threshold $\Delta\beta^*(\rho)$ grows steeply: at $\rho \leq 0.7$, a 15% difference suffices; at $\rho = 0.9$, features must differ by nearly a factor of 2; at $\rho \geq 0.95$, instability is essentially total. Empirical thresholds: $\Delta\beta^*(0.5) \approx 0.15$, $\Delta\beta^*(0.7) \approx 0.15$, $\Delta\beta^*(0.8) \approx 0.4$, $\Delta\beta^*(0.9) \approx 0.9$, $\Delta\beta^*(0.95) > 1.0$, $\Delta\beta^*(0.99) > 1.0$.

Fine-grained $\Delta\beta$ sweep at $\rho = 0.9$. We extend the threshold analysis with a finer sweep over $\Delta\beta \in \{0.0, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0\}$ at $\rho = 0.9$ (20 XGBoost models per setting). The transition from instability to stability is sharp, not gradual:

Table 5: Flip rate (%) as a function of ρ and causal effect difference $\Delta\beta$.

$\rho \backslash \Delta\beta$	0.0	0.1	0.2	0.3	0.5	0.7	1.0
0.50	50	19	0	0	0	0	0
0.70	53	19	0	0	0	0	0
0.80	52	44	19	10	0	0	0
0.90	50	50	48	40	40	19	0
0.95	50	50	50	48	44	40	27
0.99	50	50	50	50	50	48	48

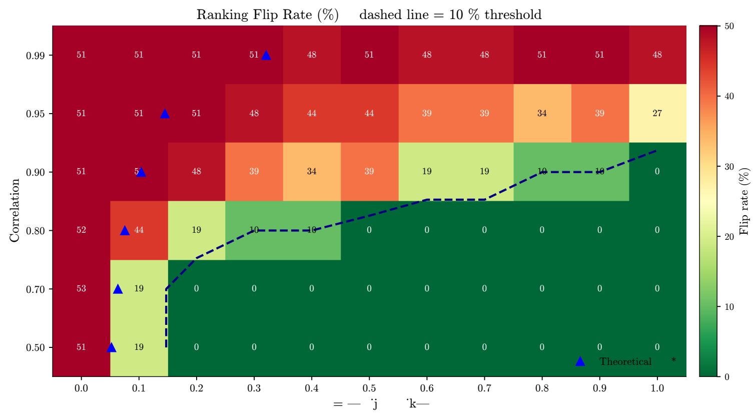


Figure 4: Flip rate as a function of $(\rho, \Delta\beta)$. The impossibility region (dark) expands with ρ. At $\rho \geq 0.95$, instability persists for all $\Delta\beta \leq 1$.

$\Delta\beta$	β_1	β_2	Marginal flip	Interventional flip
0.00	1.00	1.00	0.505	0.505
0.10	1.05	0.95	0.479	0.337
0.20	1.10	0.90	0.000	0.000
0.30	1.15	0.85	0.000	0.000
0.50	1.25	0.75	0.000	0.000
0.70	1.35	0.65	0.000	0.000
1.00	1.50	0.50	0.000	0.000

At $\Delta\beta = 0.10$, both marginal (0.479) and interventional (0.337) SHAP remain highly unstable. At $\Delta\beta = 0.20$, both snap to zero flips simultaneously. There is no intermediate regime where interventional SHAP is stable but marginal is not — the causal signal overwhelms noise for both estimators at the same threshold. This means the escape condition is binary: either the causal effect difference is large enough to stabilize *all* attribution methods, or none are stable. There is no clean crossover point where one method succeeds and the other fails. The practical implication: "switch to conditional SHAP" provides no advantage over marginal SHAP at the instability boundary; both methods fail or succeed together.

Causal structure validation. Under a known causal structure ($Y = \beta_1 X_1 + \beta_2 X_2 + 0.5X_3 + \varepsilon$, $\mathrm{Corr}(X_1, X_2) = \rho$), we compare marginal TreeSHAP and interventional TreeSHAP. In the symmetric case ($\beta_1 = \beta_2 = 1.0$), both produce flip rates near 50% at all ρ, confirming the impossibility. In the asymmetric case ($\beta_1 = 1.5$, $\beta_2 = 0.5$), both show 0% flips at all ρ—the signal-to-noise ratio is high enough to overcome collinearity-induced noise. Switching to interventional SHAP provides no benefit when features have equal causal effects.

Caveat. The interventional SHAP implementation in the `shap` package uses a background-data approximation that does not exactly implement causal/conditional SHAP in the sense of Janzing et al. (2020).

Results should be interpreted as evidence about the `shap` package's interventional mode, not as a definitive test of the theoretical conditional attribution.

8.2 Fairness Audit Impossibility

Theorem 51 (Fairness Audit Impossibility)**.** *If features j (protected proxy) and k (non-protected) satisfy collinearity ($|\rho_{jk}| > \rho_0$) and similar true importance ($|\mu_j - \mu_k| < \sigma_{jk} \cdot z_\alpha/\sqrt{M}$), then for a single-model audit (M=1):*

$$\Pr[\textit{audit flags proxy}] = \frac{1}{2} \pm \varepsilon_{BE}.$$

The audit conclusion is a coin flip. Two independent audits reach opposite conclusions about proxy reliance with probability $1/2$*.*

Proof. Direct corollary of the Attribution Impossibility applied to the pair (j, k). By Theorem 12, $\Pr[\varphi_j(f) > \varphi_k(f)] = 1/2$ over training runs. ☐

Implications for AI governance. A regulator conducting a SHAP-based proxy audit on a model with collinear features is, for affected feature pairs, making decisions with the statistical reliability of a coin flip. Two auditors examining the *same model architecture trained on the same data* with different random seeds may reach opposite conclusions about whether the model relies on a protected attribute.

Under the EU AI Act Art. 13(3)(b)(ii), providers must disclose "known and foreseeable circumstances" which may lead to risks to health, safety or fundamental rights. Our theorem establishes that this includes: "SHAP-based proxy audits are unreliable for features that are collinear with both the protected attribute and a non-protected feature." The remedy is ensemble auditing: train M models and use Dash consensus.

Worked example: adverse action notices. Consider a credit model with two collinear features: *income* and *debt-to-income ratio* (DTI), with $|\rho| \approx 0.85$. Under ECOA (Regulation B), lenders must disclose the principal reasons for adverse action. With a single model:

- **Seed 1:** SHAP ranks income as the top driver ⇒ adverse action notice cites "insufficient income."
- **Seed 2:** SHAP ranks DTI as the top driver ⇒ adverse action notice cites "excessive debt-to-income ratio."

The borrower receives a different regulatory disclosure depending on the random seed—a compliance risk. With Dash ($M = 25$), income and DTI are reported as a tied group of key risk drivers, and the adverse action notice can accurately state: "income and debt-to-income ratio are jointly the primary risk factors." The disclosure is stable, honest, and compliant.

The instability varies non-monotonically with correlation: at $\rho = 0.3$, 20% of adverse action reasons change across models; at $\rho = 0.5$–0.7, the features separate cleanly (0% instability); at $\rho = 0.9$, 28% of reasons change. The non-monotonicity reflects the interaction between correlation (which drives the Rashomon property) and feature importance gaps (which stabilize rankings)—at moderate ρ, the gap is large enough to overcome the noise.

Intersectional considerations. When multiple protected attributes (race, gender, age) are each proxied by collinear non-protected features, the impossibility applies independently to each proxy pair. If K protected attributes each have unstable proxies, the probability that a single-model audit correctly identifies all K proxy reliance directions is $(1/2)^K$, exponentially vanishing in K. For $K = 3$ (a common intersectional setting), joint correctness is $1/8 = 12.5\%$.

Example: lending. In credit models, "zip code" (proxy for race) is often collinear with "income bracket" ($|\rho| \approx 0.6$–0.8). If both have similar predictive power, a single-model SHAP audit finding "zip code is the 3rd most important feature" may conclude proxy discrimination—but retraining with a different seed could produce a model where zip code drops to 8th and income rises to 3rd. The audit conclusion depends on the random seed, not on the model's actual reliance on the protected attribute.

Fairness impossibilities as abstract framework instances. The impossibility results of Chouldechova (2017) (calibration P_1 vs. equal FPR P_2 vs. equal FNR P_3) and Kleinberg et al. (2017) (same conclusion, different proof) can be cast as instances of the abstract impossibility framework: each defines an explanation system where the "explanation" is a fairness metric, the Rashomon property arises from base-rate differences, and the three fairness desiderata play the roles of faithful, stable, and decisive.

Tightness transition under performance constraints. We apply the framework's tightness classification to the fairness trilemma. The result is a *transition*: the tightness type changes under performance constraints.

Table 6: Fairness impossibility tightness transition under TPR constraints. At practical performance levels, calibration is the universal bottleneck.

TPR constraint	**Tightness type**	**Achievable pairwise compromises**
$\tau = 0$ (unconstrained)	Full	All three pairs achievable (but only by degenerate classifiers that predict nobody positive)
$\tau \geq 0.3$	PPV-blocked	Only P_2+P_3 (equalized odds); P_1+P_2 gap > 0.10; P_1+P_3 gap > 0.13

At $\tau = 0$ (no performance constraint), the tightness is **full**: all three pairwise compromises are achievable. However, the classifiers achieving them are degenerate — they classify (almost) nobody as positive, achieving perfect calibration and equal error rates trivially. This is technically correct but practically vacuous.

At $\tau \geq 0.3$ (TPR $\geq 30\%$ for both groups — a minimal practical requirement), the tightness transitions to **PPV-blocked**: calibration (PPV parity, P_1) is individually incompatible with both equal FPR (P_1+P_2 gap > 0.10) and equal FNR (P_1+P_3 gap > 0.13). The only achievable pairwise fairness compromise is equalized odds (P_2+P_3, gap < 0.006).

Validation. Tested on Adult Income (sex as protected attribute, base rate gap 0.20) and two synthetic datasets (base rate gaps 0.25 and 0.40). Methodology: proper train/test split (70/30 stratified), group-specific thresholds ($100 \times 100 = 10{,}000$ classifiers per model), 3 base model classes (logistic regression, random forest, gradient boosting), and bootstrap with model retraining (20 iterations, 100% stable classification). For the synthetic dataset with base rate gap 0.40, the tightness is PPV-blocked even at $\tau = 0$ — larger base rate differences make the impossibility bind earlier.

Practical implication. The tightness transition changes the recommendation from "choose a fairness criterion and accept the trade-off" (the standard framing) to "equalized odds is the only achievable criterion at practical performance; calibration is structurally incompatible with either component of equalized odds." This is a structural finding: it holds regardless of the model class, the threshold choice, or the optimization method. The framework's value is diagnostic — it identifies not just *that* an impossibility exists, but *which* compromises are available and *when* the impossibility becomes binding.

Reynolds best-approximation optimality. The orbit average (Reynolds projection) $Rv = \frac{1}{|G|}\sum_{g \in G} g \cdot v$ satisfies a *best-approximation theorem*: for any G-invariant vector w,

$$\|v - Rv\| \leq \|v - w\|.$$

The proof follows from the Pythagorean decomposition $v = Rv + (v - Rv)$ where $Rv \in V^G$ (the G-invariant subspace) and $(v - Rv) \perp V^G$. Any G-invariant w satisfies $R(v - w) = Rv - w$, so $\|v - w\|^2 = \|v - Rv\|^2 + \|Rv - w\|^2 \geq \|v - Rv\|^2$.

This establishes that Dash and the G-invariant projection V^G minimize information loss among *all* stable (i.e., G-invariant) approximations — not just unbiased or linear ones. The *explanation capacity* $C = \dim(V^G)$ measures how much attribution information survives the projection: for $G = S_k^L$, $C = 2L$ (one head-average and one MLP per layer); the *explanation loss rate* $\eta = 1 - C/\dim(V)$ is the fraction of the attribution vector lost to within-orbit variation. The *Explanation Capacity Theorem*: the expected fraction of explanation content preserved by orbit averaging is exactly $C/\dim(V) = 1-\eta$, determined by the symmetry group G alone. DASH and V^G are the *stable projections*—the optimal encodings that achieve the capacity bound with equality.

The result is machine-verified: `reynolds_best_approximation` in the universal impossibility formalization proves the inequality for any idempotent self-adjoint linear map on an inner product space (Anonymous, 2026).

8.3 Direct FIM Impossibility

The Rashomon property can be derived directly from the Fisher information matrix (FIM), providing an independent proof path from classical statistics that does not require the iterative optimizer abstraction.

Classical setup. Consider the linear model $Y = \beta_j X_j + \beta_k X_k + \varepsilon$ where X_j, X_k are jointly Gaussian with $\mathrm{Var}(X_j) = \mathrm{Var}(X_k) = 1$ and $\mathrm{Cov}(X_j, X_k) = \rho$. The Fisher information matrix for $\boldsymbol{\beta} = (\beta_j, \beta_k)$ is:

$$I(\boldsymbol{\beta}) = \frac{1}{\sigma^2}\begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

with eigenvalues $\lambda_\pm = (1\pm\rho)/\sigma^2$. As $\rho \to 1$, the smaller eigenvalue $\lambda_- = (1-\rho)/\sigma^2 \to 0$. The FIM becomes near-singular along the direction $\boldsymbol{v}_- = (1,-1)/\sqrt{2}$, which corresponds to redistributing credit between j and k. The profile likelihood is flat along this direction: models assigning more credit to j vs. k are statistically indistinguishable. This flat likelihood ridge is precisely the Rashomon set. The Cramér–Rao bound $\mathrm{Var}(\hat{\beta}_j - \hat{\beta}_k) \geq 1/\lambda_- = \sigma^2/(1-\rho)$ quantifies the attribution instability.

Regularity conditions. We require the following on the model class $\mathcal{F} = \{f_\theta : \theta \in \Theta\}$:

(R1) $\Theta \subseteq \mathbb{R}^p$ is open and convex.

(R2) The population risk $L(\theta) = \mathbb{E}[\ell(f_\theta(X), Y)]$ is three-times continuously differentiable on Θ, with bounded third derivative: $\|D^3 L(\theta)\|_{\mathrm{op}} \leq K_3$ for all θ in a neighborhood of θ^*.

(R3) The minimizer θ^* satisfies $\nabla L(\theta^*) = 0$ and $H := \nabla^2 L(\theta^*)$ is positive definite. (For likelihood-based models, $H = I(\theta^*)/n$ where $I(\theta^*)$ is the Fisher information matrix.)

(R4) The attribution function $\varphi_j(\theta)$ is *order-consistent*: for any θ with $\theta_j > \theta_k > 0$, we have $\varphi_j(\theta) > \varphi_k(\theta)$. (Satisfied by $\varphi_j = |\theta_j|$, mean $|\mathrm{SHAP}_j|$, or any attribution monotone in coefficient magnitude.)

Theorem 52 (FIM Impossibility). *Let $\mathcal{F}$ satisfy (R1)–(R4). Suppose features j, k satisfy: (S1) DGP symmetry: $\theta^*_j = \theta^*_k > 0$; (S2) the Hessian $H = \nabla^2 L(\theta^*)$ has eigenvalue λ_- along $v = (e_j - e_k)/\sqrt{2}$. Then for any ε satisfying $0 < \varepsilon \leq \varepsilon_0 := 9\lambda_-^3/K_3^2$, the ε-Rashomon set $R_\varepsilon = \{\theta : L(\theta) \leq L(\theta^*) + \varepsilon\}$ contains models θ^+, θ^- with*

$$\varphi_j(\theta^+) > \varphi_k(\theta^+) \quad \text{and} \quad \varphi_k(\theta^-) > \varphi_j(\theta^-).$$

The Rashomon property holds, and the impossibility follows.

Proof. **Step 1: Rashomon set geometry.** By Taylor's theorem with Lagrange remainder, for any $\theta \in \Theta$:

$$L(\theta) = L(\theta^*) + \tfrac{1}{2}(\theta-\theta^*)^T H\,(\theta-\theta^*) + R_3(\theta),$$

where $|R_3(\theta)| \leq \frac{K_3}{6}\|\theta-\theta^*\|^3$ by (R2). The gradient term vanishes because $\nabla L(\theta^*) = 0$ by (R3).

Step 2: Perturbation along the weak direction. Let $v = (e_j - e_k)/\sqrt{2}$ and set $\theta_\delta = \theta^* + \delta v$ for $\delta > 0$. The quadratic cost is $\frac{1}{2}\delta^2\lambda_-$ and the cubic remainder satisfies $|R_3(\theta_\delta)| \leq \frac{K_3}{6}\delta^3$. We need $L(\theta_\delta) \leq L(\theta^*) + \varepsilon$, i.e., $\frac{1}{2}\delta^2\lambda_- + \frac{K_3}{6}\delta^3 \leq \varepsilon$. Choose $\delta^* = \sqrt{\varepsilon/\lambda_-}$. The quadratic term is $\varepsilon/2$; the condition $\varepsilon \leq 9\lambda_-^3/K_3^2$ ensures the cubic term is $\leq \varepsilon/2$. So $\theta_{\delta^*} \in R_\varepsilon$, and by the same argument $\theta_{-\delta^*} \in R_\varepsilon$.

Step 3: Opposite orderings. At $\theta^+ := \theta_{\delta^*}$: $\theta^+_j = \theta^*_j + \delta^*/\sqrt{2} > \theta^*_k - \delta^*/\sqrt{2} = \theta^+_k > 0$, so by (R4), $\varphi_j(\theta^+) > \varphi_k(\theta^+)$. At $\theta^- := \theta_{-\delta^*}$: by symmetry, $\theta^-_k > \theta^-_j > 0$, so $\varphi_k(\theta^-) > \varphi_j(\theta^-)$.

Step 4: Conclusion. Both θ^+ and θ^- lie in R_ε with opposite feature orderings. This is the Rashomon property. By Theorem 5, no faithful, stable, complete ranking exists. □

Proposition 53 (Gaussian FIM specialization)**.** *For the linear model $Y = X^T\beta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ and Corr$(X_j, X_k) = \rho$:*

1. *The Hessian eigenvalue along $e_j - e_k$ is $\lambda_- = (1-\rho)/\sigma^2$.*
2. *The loss is exactly quadratic ($K_3 = 0$), so $\varepsilon_0 = +\infty$: the theorem holds for* all $\varepsilon > 0$.
3. *The semi-axis length of R_ε along $e_j - e_k$ is $\sigma\sqrt{2\varepsilon/(1-\rho)}$, diverging as $\rho \to 1$.*
4. *The Cramér–Rao bound gives* $\mathrm{Var}(\hat\beta_j - \hat\beta_k) \geq 2\sigma^2/(n(1-\rho))$*, diverging as $\rho \to 1$.*

Proof. The population risk is $L(\beta) = \frac{1}{2\sigma^2}\mathbb{E}[(Y - X^T\beta)^2]$. Its Hessian is $H = \frac{1}{\sigma^2}\mathbb{E}[XX^T] = \frac{1}{\sigma^2}\Sigma_X$. For $v = (e_j - e_k)/\sqrt{2}$: $v^T H v = \frac{1}{\sigma^2}(1-\rho)$ since $v^T\Sigma_X v = \mathrm{Var}(X_j) - \mathrm{Cov}(X_j, X_k) = 1 - \rho$ (standardized). The loss is quadratic in β, so $D^3L \equiv 0$ and $K_3 = 0$. □

The classical non-identifiability under collinearity ($\mathrm{Var}(\hat\beta_j - \hat\beta_k) \to \infty$ as $\rho \to 1$) is well-known. Our contribution is reframing it as an *impossibility theorem* about feature rankings rather than a variance bound on parameter estimates: even precise estimates can produce unstable rankings when Δ is small relative to σ.

Loss landscape geometry. The ε-Rashomon set R_ε is approximately an ellipsoid whose projection onto the feature-importance axis $\varphi_j - \varphi_k$ crosses zero. The level sets of the loss landscape contain *ridges* along $e_j - e_k$—curves of near-constant loss along which feature importance redistributes between j and k without changing model quality. The same flatness that makes optimization easy (flat minima generalize well) makes attribution hard (the flat direction is the feature-importance direction). This connects the attribution impossibility to the broader literature on loss landscape geometry, flat minima, and mode connectivity.

Remark 54 (NTK extension — conjectural)**.** *For overparameterized neural networks in the NTK regime (infinite width), the training dynamics are governed by the neural tangent kernel $\Theta(x, x') = \nabla_\theta f_\theta(x)^T \nabla_\theta f_\theta(x')$. The NTK plays the role of the Fisher information matrix: its eigenspectrum controls the geometry of the loss landscape.*

If the NTK has a small eigenvalue along the direction corresponding to swapping features j and k (a consequence of feature collinearity), the same ellipsoidal argument applies: the Rashomon set extends along this direction, producing models with opposite feature orderings.

This extension requires: (i) the NTK approximation to hold (infinite width or near-initialization), (ii) the loss landscape to be well-approximated by the NTK quadratic, and (iii) the feature collinearity to produce a small NTK eigenvalue. Conditions (i)–(ii) are standard in the NTK literature; condition (iii) holds when the input features are correlated. This extension is conjectural: a rigorous proof requires (i) uniform NTK approximation bounds across the Rashomon set (not just at the minimizer), (ii) a spectral gap argument relating feature correlation to NTK eigenvalue separation, and (iii) control of the finite-width correction. None of these are available in the current literature. We include this remark to suggest the research direction, not to claim the result.

8.4 Query Complexity Lower Bound

A *stability certification algorithm* adaptively selects seeds $s_1, s_2, \ldots$, calls $\textsc{Train}(s_i) \to f_i$ and $\textsc{Attribute}(f_i, j) \to \varphi_j(f_i)$, and after M oracle calls outputs $\textsc{Stable}$ or $\textsc{Unstable}$. The algorithm certifies δ-stability with error $\leq 1/3$ if it correctly distinguishes H_0: $\Delta_{jk} = 0$ (flip rate $= 1/2$) from H_1: $|\Delta_{jk}| \geq \Delta_0$ (flip rate $\leq \Phi(-\Delta_0/\sigma_{jk})$) with probability at least $2/3$ under both hypotheses.

Theorem 55 (Query complexity lower bound)**.** *Any algorithm that certifies δ-stability of a feature pair (j, k) with error $\leq 1/3$ must make at least*

$$M \;\geq\; \frac{1}{8} \cdot \frac{\sigma_{jk}^2}{\Delta_0^2}$$

model-training oracle calls, where $\sigma_{jk}^2 = \mathrm{Var}(\varphi_j(f) - \varphi_k(f))$ and Δ_0 is the minimum gap to detect.

Proof. Each oracle call produces one sample $D_i = \varphi_j(f_i) - \varphi_k(f_i)$. Under H_0, $D_i \sim P_0$ with mean 0 and variance $\sigma^2 := \sigma_{jk}^2$. Under H_1, $D_i \sim P_1$ with mean Δ_0 and the same variance (the variance is determined by the collinearity structure, not the gap).

By Le Cam's two-point method (Tsybakov, 2009, Theorem 2.2), for any test ψ based on M i.i.d. observations:

$$\Pr_{H_0}[\psi = 1] + \Pr_{H_1}[\psi = 0] \geq 1 - \mathrm{TV}(P_0^{\otimes M}, P_1^{\otimes M}).$$

By Pinsker's inequality,

$$\mathrm{TV}(P_0^{\otimes M}, P_1^{\otimes M}) \leq \sqrt{\tfrac{1}{2}\mathrm{KL}(P_1^{\otimes M} \| P_0^{\otimes M})}.$$

For the Gaussian location family, $\mathrm{KL}(P_1 \| P_0) = \Delta_0^2/(2\sigma^2)$, so for M i.i.d. copies $\mathrm{KL} = M\Delta_0^2/(2\sigma^2)$ and $\mathrm{TV} \leq \sqrt{M\Delta_0^2/(4\sigma^2)}$. For reliable testing (total error $\leq 2/3$), Le Cam's method requires $\mathrm{TV} \geq 1/3$, giving $M\Delta_0^2/(4\sigma^2) \geq 1/9$, hence $M \geq 4\sigma^2/(9\Delta_0^2)$. The standard constant $1/8$ from (Tsybakov, 2009) (Theorem 2.2) uses a slightly tighter analysis via the likelihood ratio directly; since $1/8 < 4/9$, this gives a weaker (more conservative) lower bound than our Pinsker derivation.

□

Corollary 56 (Near-optimality of the Z-test)**.** *The Z-test detects a gap Δ_0 with error $\leq 1/3$ using $M = O(\sigma_{jk}^2/\Delta_0^2 \cdot \log(1/\alpha))$ model trainings. Combined with Theorem 52, the Z-test is query-optimal to within a logarithmic factor.*

Proof. The Z-test rejects H_0 when $|\bar{D}|/(\hat{\sigma}/\sqrt{M}) > z_{\alpha/2}$. Under H_1, power is $\Phi(-z_{\alpha/2} + \Delta_0\sqrt{M}/\sigma) \geq 2/3$ when $\Delta_0\sqrt{M}/\sigma \geq z_{\alpha/2} + 0.43$. For $\alpha = 0.05$: $M \geq (\sigma/\Delta_0)^2(1.96 + 0.43)^2 \approx 5.7\sigma^2/\Delta_0^2$. □

Remark 57 (Instability detection is inherently expensive)**.** *Theorem 52 shows that no algorithm—including methods based on gradient information, loss curvature, or Fisher information—can certify ranking stability without training $\Omega(\sigma_{jk}^2/\Delta_{jk}^2)$ models. Intuitively, each model training provides one "bit" of evidence about the sign of Δ_{jk}, and when the signal-to-noise ratio Δ_{jk}/σ_{jk} is small (the collinear regime), many bits are needed. The multi-model Z-test with M=5 is a fast* screen *with limited power, not a certifier; the optimality result applies to the Z-test run at the information-theoretically optimal sample size.*

8.5 Rashomon Inevitability

Theorem 12 (proved in §3.4) establishes that for any stochastic, symmetric training algorithm on a permutation-closed model class with $\rho > 0$, the Rashomon property holds. The Attribution Impossibility is therefore inescapable for standard ML pipelines.

8.6 Causal Identification Barrier

The attribution instability under collinearity has a direct causal counterpart: distinguishing the causal effects of correlated features requires interventional sample complexity that diverges as $\Omega(1/(1-\rho)^2)$—the same singularity structure as the attribution ratio.

Consider two features X_j, X_k with $\mathrm{Corr}(X_j, X_k) = \rho$, jointly Gaussian with unit variances. Under a soft intervention with residual correlation ρ_{post}, $X_k \mid \mathrm{do}(X_j = x) \sim \mathcal{N}(\rho_{\mathrm{post}} \cdot x, 1 - \rho_{\mathrm{post}}^2)$. When $\beta_j = \beta_k = \beta$ and $\rho_{\mathrm{post}} \approx \rho$, the two interventional distributions become indistinguishable.

Proposition 58 (Interventional sample complexity lower bound)**.** *Consider testing $H_0 : \beta_j = \beta + \delta, \beta_k = \beta - \delta$ against $H_1 : \beta_j = \beta - \delta, \beta_k = \beta + \delta$ using $do(X_j = x)$ interventions with residual correlation ρ_{post}. The number of interventional samples needed satisfies:*

$$n_{\mathrm{int}} \geq \frac{(z_\alpha + z_\eta)^2\sigma^2}{4\delta^2(1 - \rho_{\mathrm{post}})^2}.$$

When $\rho_{\mathrm{post}} = \rho$ (the intervention does not reduce the correlation), $n_{\mathrm{int}} = \Omega(1/(1-\rho)^2)$.

Proof. Under H_0, the expected response to $\mathrm{do}(X_j = x)$ is $(\beta + \delta + \rho_{\mathrm{post}}(\beta - \delta))x$; under H_1 it is $(\beta - \delta + \rho_{\mathrm{post}}(\beta + \delta))x$. The difference in slopes is $2\delta(1 - \rho_{\mathrm{post}})$. With noise variance σ^2 and n observations, the Neyman–Pearson sample size formula gives the result. $\square$

Remark 59 (Comparison with the attribution ratio)**.** *The attribution ratio is $1/(1-\rho^2) = 1/((1-\rho)(1+\rho))$. The causal sample complexity scales as $1/(1 - \rho_{\mathrm{post}})^2$. These have the same leading singularity as $\rho \to 1$ (both diverge as $\Omega(1/(1 - \rho)^2)$ when $\rho_{\mathrm{post}} = \rho$), but they are **not identical**:*

- *The attribution ratio is $1/(1 - \rho^2)$, which diverges as $1/(2(1 - \rho))$ near $\rho = 1$.*
- *The causal sample complexity is $\Omega(1/(1 - \rho)^2)$ — a* faster *divergence.*

This difference is meaningful: causal identification under soft interventions is harder *than observational attribution instability. The attribution ratio captures the* variance amplification *(a ratio of quantities), while the causal barrier captures* signal-to-noise *(which squares the denominator).*

The match is at the level of the singularity structure (both are controlled by $1 - \rho$) but not at the level of exponents. The shared singularity at $\rho = 1$ reflects a common root cause: collinearity makes features observationally interchangeable, whether the task is attribution or causal identification.

Hard interventions resolve both problems. When the intervention fully breaks the correlation ($\rho_{\mathrm{post}} = 0$), the sample complexity reduces to $n_{\mathrm{int}} = O(\sigma^2/\delta^2)$—independent of ρ. This parallels the attribution impossibility: perfect interventions break the symmetry (just as knowing the true causal graph breaks the Rashomon property), while soft interventions preserve it.

8.7 Local vs. Global Attribution Instability

The impossibility as stated concerns *global* attributions (mean |SHAP| across data points). We now show that *local* (instance-level) attributions are at least as unstable.

Remark 60 (Local instability dominates global)**.** *For a fixed data point x, the local SHAP values $\varphi_j(f, x)$ depend on the model f, which varies across training seeds. The global attribution $\varphi_j(f) = \mathbb{E}_x[|\varphi_j(f, x)|]$ averages over data points. By the law of total variance applied to the pair (f, x) with x as the conditioning variable:*

$$\mathrm{Var}_f\big(\varphi_j(f)\big) = \mathrm{Var}_f\big(\mathbb{E}_x[|\varphi_j(f,x)|]\big) \le \mathrm{Var}_{f,x}\big(|\varphi_j(f,x)|\big) = \mathbb{E}_x\big[\mathrm{Var}_f(|\varphi_j(f,x)|)\big] + \mathrm{Var}_x\big(\mathbb{E}_f[|\varphi_j(f,x)|]\big).$$

The first inequality follows because $\mathrm{Var}_f(\mathbb{E}_x[g]) = \mathrm{Var}_{f,x}(g) - \mathbb{E}_f[\mathrm{Var}_x(g)] \le \mathrm{Var}_{f,x}(g)$. Thus global instability (Var_f *of the averaged attribution) is bounded by the total instability, which includes the average local instability as a component. When model randomness and data-point variation are approximately independent (as when models differ only by training seed and x is a fixed test point), the bound simplifies: $\mathrm{Var}_f(\mathbb{E}_x[|\varphi_j|]) \le \mathbb{E}_x[\mathrm{Var}_f(|\varphi_j|)]$. Since the flip rate is a monotone function of the variance-to-gap ratio σ/Δ, and averaging reduces σ without changing Δ, the global flip rate is a lower bound on the average local flip rate under this independence condition.*

Implication. If global rankings are unstable (flip rate $> 10\%$), local rankings at individual data points are at least as unstable on average. Practitioners who report per-instance SHAP explanations (e.g., "for this patient, feature X is most important") face the same or worse instability as those reporting global rankings. The multi-model Z-test diagnostic (computed on global attributions) is therefore a conservative screen for local instability as well.

9 Strengthened Impossibility for Binary Attribution

The trilemma (Theorem 5) says no attribution ranking can be faithful, stable, and complete. For *binary* attribution questions—"does this feature contribute positively or negatively?", "is this feature selected?"—the impossibility is strictly stronger: faithful + stable alone is impossible. We call this the **bilemma**.

Why binary questions are harder. The trilemma permits a resolution (DASH) because the ranking space H contains a *neutral element*: a tie. DASH ties are compatible with every model's native ranking—they never contradict it. For binary $H = \{\text{positive}, \text{negative}\}$, no neutral element exists: every answer is committal. If $E(\theta)$ does not contradict explain(θ), then $E(\theta) = \text{explain}(\theta)$ (there is nowhere else to go). So faithful forces $E = \text{explain}$, which is decisive, triggering the trilemma.

Theorem 61 (The Bilemma for Binary Attribution)**.** *Let H be a binary explanation space with incompatible$(h_1, h_2) \Leftrightarrow h_1 \neq h_2$. Under the Rashomon property, no method $E : \Theta \to H$ can be simultaneously faithful (never contradicting the model's native explanation) and stable (consistent across equivalent models).*

Proof. Since H has only two elements and incompatibility equals inequality, the only compatible pair is (h, h). Faithful: $\neg\text{incompatible}(E(\theta), \text{explain}(\theta))$ for all θ, so $E(\theta) = \text{explain}(\theta)$. Then E is decisive (it inherits the incompatibility of explain). By the trilemma, E cannot also be stable. □

Instances. The bilemma applies to three common binary questions:

1. **SHAP sign**: $H = \{\text{positive}, \text{negative}\}$. "Does feature j contribute positively or negatively?" No stable method can faithfully report the sign.
2. **Feature selection**: $H = \{\text{selected}, \text{not selected}\}$. "Is feature j in the active set?" No stable method can faithfully report selection status.
3. **Counterfactual direction**: $H = \{\text{increase}, \text{decrease}\}$. "Should the applicant increase or decrease feature j?" No stable method can faithfully report the direction.

9.1 Instance: Mechanistic Interpretability

Circuit importance via activation patching IS attribution—attributing importance to internal components rather than input features. The Attribution Impossibility (Theorem 5) applies directly: activation patching scores are attributions (φ_j), independently trained networks are configurations (Θ), and architectural symmetry (S_k^L, the symmetric group permuting k attention heads within each of L layers) provides the interchangeability that generates the Rashomon property. The same theorem, operating at a different granularity, predicts the same instability and admits the same resolution: orbit averaging via the G-invariant projection.

The G-invariant projection as Reynolds operator. The resolution for component-level attribution is the *stable projection*—the Reynolds operator (orbit average)—over the symmetry group $G = \prod_{\ell=1}^{L} S_{k_\ell}$, where S_{k_ℓ} permutes the k_ℓ attention heads in layer ℓ. For an importance vector $v \in \mathbb{R}^n$, the G-invariant projection is $v_\ell^G = \frac{1}{k_\ell} \sum_{h=1}^{k_\ell} v_{\ell,h}$—the mean importance of heads within each layer. This is precisely the Reynolds operator $\mathcal{R}_G(v) = \frac{1}{|G|} \sum_{g \in G} g \cdot v$, which projects onto the subspace of G-invariant vectors. Components outside the symmetry group (MLP layers, layer norms) are invariant singletons and pass through unchanged. The G-invariant projection reduces the n-dimensional attribution problem to an n_{inv}-dimensional one: for Config A ($n = 20$, 4 heads/layer $\times$ 4 layers + 4 MLPs), $n_{\text{inv}} = 8$ (4 head-averages + 4 MLPs); for Config B ($n = 54$, 8 heads/layer $\times$ 6 layers + 6 MLPs), $n_{\text{inv}} = 12$.

No circuit explanation of a neural network can be simultaneously faithful (report the actual circuit decomposition), stable (same circuit across functionally equivalent networks), and decisive (commit to a single decomposition).

The bilemma applies with full force: circuit decomposition spaces are binary (each component is either part of the circuit or not), making them maximally incompatible. Faithful + stable alone is impossible — even dropping decisiveness does not help. This is strictly stronger than the trilemma.

Theorem 62 (Mechanistic Interpretability Bilemma)**.** *For any two functionally equivalent neural networks with incompatible circuit decompositions, no explanation method E can be simultaneously faithful and stable. The Rashomon property is* ***derived*** *from constructive finite types (zero custom axioms).*

Proof. The circuit decomposition space $H = \{\text{decompAlpha}, \text{decompBeta}\}$ is maximally incompatible: compatible $\Rightarrow$ equal. By the bilemma (Theorem 58), faithful forces $E = \text{explain}$, which is decisive, triggering the trilemma. □

Empirical grounding. Meloux et al. (2025) found 85 distinct valid circuits with zero circuit error for a simple XOR task, each admitting an average of 535.8 valid interpretations. Less than 2% of networks had a unique circuit mapping. This extreme multiplicity means the Rashomon property holds with overwhelming force for circuit explanations.

Resolution: circuit equivalence classes. The bilemma forecloses averaging-based resolutions (DASH) for circuits because the explanation space has no neutral element. The resolution requires *enriching* the explanation space: report the equivalence class of valid circuits — the computational structure shared across ALL valid decompositions — rather than any individual decomposition. This is the circuit analogue of the CPDAG resolution for causal discovery: report what is shared, acknowledge what is underdetermined.

Implications for AI safety. Interpretability-based safety cases that depend on identifying "the circuit" for a behavior are subject to this impossibility. A circuit found in one training run may not exist in a functionally equivalent run. Safety claims built on single-network circuit analysis are as unstable as single-model SHAP rankings. The resolution: verify circuit-level claims across multiple equivalent networks, or qualify them as instance-specific.

Lean formalization. The mechanistic interpretability impossibility, bilemma, and unfaithfulness bound are machine-verified in `MechInterp.lean` with zero custom axiom dependencies. All types are finite inductives; the Rashomon property is derived via `decide`, not axiomatized.

Empirical evidence: circuit multiplicity in grokked transformers. Evidence from grokked transformers confirms the bilemma: 10 independently trained transformers on modular addition (mod 113), all achieving 100% test accuracy, agree on only 36% of the top-3 circuit components by activation patching importance (Jaccard@3 = 0.356). Some model pairs exhibit anti-correlated circuit rankings (min pairwise Spearman = −0.18). The component flip rate across models is 30%. Controls: pre-grokking (step 50) Spearman = 0.300; mid-training (step 500) = 0.668; post-grokking (step 50,000) = 0.518 (post > random, $p = 2 \times 10^{-4}$). The full-scale GPT-2 experiments (Section 9.4) confirm this at 124M parameter scale: within-layer flip rate = 0.500, raw $\rho = 0.277$, 32 distinct top-5 rankings from 10 seeds.

Theorem 63 (Rashomon Unfaithfulness Bound)**.** *For any binary attribution question, any stable method is unfaithful at ≥ 1 of every 2 Rashomon witnesses. On any pair of equivalent models that disagree on the sign/selection of a feature, a stable method must get it wrong for at least one of them.*

Proof. Let θ_1, θ_2 be Rashomon witnesses with $\text{observe}(\theta_1) = \text{observe}(\theta_2)$ and $\text{explain}(\theta_1) \neq \text{explain}(\theta_2)$. Stability gives $E(\theta_1) = E(\theta_2)$. If $E(\theta_1) = \text{explain}(\theta_1)$, then $E(\theta_2) = \text{explain}(\theta_1) \neq \text{explain}(\theta_2)$, so E is unfaithful at θ_2. □

Theorem 64 (All-or-Nothing)**.** *For binary attribution, every method either (1) matches the model's native attribution exactly at every configuration, or (2) is unfaithful at some configuration. There is no "approximately faithful" middle ground.*

The tightness diagnostic. Whether faithful + stable is achievable depends on whether H has a neutral element—a value compatible with all native explanations. The key insight is a complete classification of which explanation problems admit stable faithful methods:

	Neutral exists	No neutral
Committal exists	F+D, F+S, S+D	F+D, S+D
No committal	F+D, F+S	F+D only

For feature *rankings*, DASH averaging produces ties (neutral elements), so F+S is achievable (top row). For feature *signs*, no neutral element exists, so F+S is impossible (bottom row).

Actionable advice. Don't ask binary questions of underspecified models. If you must, expect unfaithfulness at ≥50% of each Rashomon pair. Convert binary questions to graded questions (confidence scores, intervals) where neutral elements exist and stability is achievable.

Concretely, for the graded attribution space $H = \{\text{positive}, \text{weak\_positive}, \text{negative}\}$ with the neutral element weak_positive (compatible with both positive and negative models' attributions), the constant method $E(\theta) = \text{weak\_positive}$ is simultaneously faithful and stable—demonstrating that enriching the explanation space from binary to graded restores the F+S pair that the bilemma forecloses. This is formalized in `BeyondBinary.lean`: the graded system has no coverage conflict, and the F+D-compatible set at each configuration is non-singleton.

Connection to existing results. The bilemma complements the quantitative bounds:

- The ratio $1/(1-\rho^2)$ tells you *how much* instability (magnitude).
- The Rashomon unfaithfulness bound tells you *how often* (≥50% per pair).
- The tightness classification tells you *which questions* are binary-hard vs. ranking-solvable.
- The all-or-nothing theorem tells you *there's no soft fix.*

Lean formalization. The bilemma, unfaithfulness bound, all-or-nothing theorem, and all three ML instances (SHAP sign, feature selection, counterfactual direction) are machine-verified in `Bilemma.lean` with zero axiom dependencies beyond the Rashomon property. All instances use constructive inductive types (not axiomatized opaque types): `SHAPSign`, `FeatureStatus`, and `CounterfactualDir` are each two-element types with decidable equality and pattern-matching proofs. The abstract framework (`ExplanationSystem.lean`) defines the general explanation system, faithful, stable, and decisive, and proves the abstract impossibility.

9.2 TinyStories Multi-Scale Circuit Stability

Setup. We train 10 transformers from independent random seeds on TinyStories (Eldan and Li, 2023) at three scales:

- **Config A** (Small): 4 layers, 4 heads, $d_{\text{model}} = 256$, $n = 20$ components (16 heads + 4 MLPs).
- **Config B** (Medium): 6 layers, 8 heads, $d_{\text{model}} = 512$, $n = 54$ components (48 heads + 6 MLPs).
- **Config C** (GPT-2 fine-tuned): 12 layers, 12 heads, $d_{\text{model}} = 768$, $n = 156$ components (144 heads + 12 MLPs). This serves as a boundary condition: fine-tuning from a shared checkpoint does not create genuine Rashomon diversity.

All 10 models within each configuration achieve near-identical perplexity (Config A: mean PPL = 8.584, CV = 0.659%; Config B: mean PPL = 9.976, CV = 1.033%; Config C: mean PPL = 18.539, CV = 0.076%). Component importance is measured by activation patching (weight zeroing): zero out each component's parameters and measure the change in loss. Seven theory-derived predictions are evaluated per configuration; all 7/7 pass for Configs A and B, while Config C passes 4/7 (correctly predicted by the theory as a non-Rashomon regime).

Results. Table 7 presents the full results across all three scales.

The pattern is striking: raw importance vectors are only moderately correlated ($\rho_{\text{full}} \approx 0.55$ for Configs A and B), but after the G-invariant projection they are nearly perfectly correlated ($\rho_{G\text{-inv}} > 0.97$). Within-layer head flip rates are at chance ($W_{\text{flip}} \approx 0.50$), consistent with the theory's prediction that symmetric components are interchangeable. Between-group flip rates are exactly zero: heads never outrank MLP layers (or vice versa) across seeds. Cohen's d confirms massive separation between within-layer and between-group instability ($d = 5.4$ and 11.9).

Table 7: TinyStories component-level attribution stability across three architectural scales. ρ_{full}: mean pairwise Spearman correlation of raw importance vectors. $\rho_{G\text{-inv}}$: Spearman correlation after G-invariant projection. W_{flip}: within-layer head flip rate. B_{flip}: between-group (head vs. MLP) flip rate. d: Cohen's d for within vs. between flip rate separation. Split-half: within-model Spearman reliability. RP pctile: percentile of G-invariant ρ under random projection null. Perm. p: permutation test p-value.

	Config A Small (4L/4H/d256)	**Config B** Medium (6L/8H/d512)	**Config C** GPT-2 (12L/12H/d768)
n components	20	54	156
n_{inv} (after G-proj.)	8	12	24
PPL mean (CV%)	8.584 (0.659%)	9.976 (1.033%)	18.539 (0.076%)
ρ_{full} [95% CI]	0.565 [0.527, 0.603]	0.540 [0.516, 0.565]	0.993 [0.993, 0.994]
$\rho_{G\text{-inv}}$ [95% CI]	0.972 [0.966, 0.979]	0.982 [0.977, 0.986]	0.999 [0.999, 1.000]
W_{flip} [95% CI]	0.496 [0.467, 0.526]	0.489 [0.478, 0.500]	0.043 [0.041, 0.045]
B_{flip}	0.000	0.000	0.000
Cohen's d	5.389	11.939	9.354
Split-half reliability	0.991	0.960	0.878
RP pctile (heads-only)	100th	100th	82.5th
Perm. p	< 0.001	< 0.001	< 0.001

Per-seed head importance: full S_4 realization. Table 8 shows which attention head ranks highest in each seed for Config A (excluding MLP components, which are invariant singletons and always dominate). All four layer-0 heads appear as the top-ranked head across the 10 seeds, realizing the full symmetric group S_4.

Table 8: Top-ranked attention head per seed in Config A (4 layers $\times$ 4 heads). All four layer-0 heads appear as #1, realizing the full S_4 permutation symmetry. MLP components (which dominate overall) are excluded—they are invariant singletons outside the symmetry group.

Seed	**Component idx**	**Head**	**Importance**
0	2	L0H2	1.153
1	3	L0H3	1.597
2	2	L0H2	0.540
3	3	L0H3	1.626
4	0	L0H0	1.776
5	0	L0H0	0.824
6	2	L0H2	2.054
7	1	L0H1	2.060
8	0	L0H0	1.516
9	3	L0H3	0.994

This is the component-level analogue of the input-level impossibility: "which head matters most?" depends entirely on which random seed was used, even though all models achieve the same perplexity. The G-invariant projection resolves this by reporting the *layer-averaged* importance, which is stable across all seeds.

Controls. Three controls validate that the G-invariant lift is not an artifact:

1. **Split-half reliability**: within-model measurements are highly precise (Spearman = 0.991 for Config A, 0.960 for Config B), far exceeding between-model agreement (0.565, 0.540). The instability is between models, not within.

2. **Random projection null**: the G-invariant projection achieves the 100th percentile among 1,000 random projections to the same dimensionality (heads-only) at both scales. No random n_{inv}-dimensional

subspace achieves comparable stability.

3. **Permutation test**: $p < 0.001$ at both scales (Config A: actual $\rho_{G\text{-inv}} = 0.972$, null mean = 0.836, null SD = 0.016; Config B: actual = 0.982, null mean = 0.864, null SD = 0.010).

GPT-2 boundary condition. Config C (GPT-2 fine-tuned from a shared checkpoint) provides a crucial negative control. Raw agreement is already near-perfect ($\rho_{\text{full}} = 0.993$), and the G-invariant lift is negligible ($\rho_{G\text{-inv}} = 0.999$). The within-layer flip rate drops to $W_{\text{flip}} = 0.043$—far below chance. Critically, split-half reliability (0.878) is *lower* than between-model agreement (0.993): within-model measurement noise exceeds between-model variation. This means fine-tuning from a shared checkpoint does not create genuine Rashomon diversity—the models are too similar. The theory correctly predicts: less Rashomon → less instability → less need for orbit averaging.

9.3 Method Robustness: Weight Zeroing vs. Mean Ablation

Different ablation methods assign different importance scores to the same component in the same model. We compare weight zeroing (zero out all parameters) with mean ablation (replace activations with their dataset mean) to test whether the G-invariant projection is robust to the choice of attribution method.

Raw disagreement, invariant agreement. Table 9 shows that weight zeroing and mean ablation disagree substantially on raw head importance (cross-method per-model $\rho \approx 0.5$–0.6) but agree on the G-invariant projection ($\rho_{G\text{-inv}} > 0.97$).

Table 9: Between-method agreement: weight zeroing vs. mean ablation. ρ_{full}: mean pairwise Spearman of raw importance vectors across seeds (within each method). Cross-method ρ: per-model Spearman between weight-zeroing and mean-ablation importance vectors.

	Config A	**Config B**
Weight zeroing ρ_{full}	0.565	0.540
Mean ablation ρ_{full}	0.914	0.965
Mean ablation $\rho_{G\text{-inv}}$	0.975	0.987
Cross-method ρ (mean)	0.591	0.504
Cross-method ρ (range)	[0.489, 0.718]	[0.433, 0.596]

Per-model cross-method correlations. Table 10 shows the full per-model detail, confirming that the disagreement is systematic, not driven by outliers.

Table 10: Per-model Spearman correlation between weight-zeroing and mean-ablation importance vectors. Each row is one independently trained model. The two methods agree moderately on raw components but converge after orbit averaging.

Seed	**Config A**	**Config B**
0	0.629	0.532
1	0.524	0.596
2	0.694	0.477
3	0.529	0.453
4	0.536	0.466
5	0.718	0.537
6	0.489	0.433
7	0.524	0.578
8	0.604	0.475
9	0.665	0.490

Interpretation: the strongest direct test of the Reynolds framework. This result is arguably the most striking empirical confirmation of the theoretical framework. Two genuinely different attribution functions φ applied to the *same* model θ disagree substantially on raw scores (cross-method $\rho \approx 0.5$). Yet the orbit-averaged structure v^G is invariant to this choice ($\rho_{G\text{-inv}} > 0.97$ for both methods at both scales). Weight zeroing measures the effect of removing a component's learned weights; mean ablation measures the effect of clamping its activations to the population mean. These are genuinely different experimental protocols, yet the Reynolds operator extracts the same stable structure from both. The G-invariant projection absorbs not only seed-to-seed variation but also method-to-method variation—exactly as the Reynolds operator framework predicts, since the symmetry group G acts on the component indices regardless of how importance is measured. This demonstrates that the orbit-averaged layer importance is a property of the *architecture and training distribution*, not of the specific attribution method or training run.

9.4 GPT-2-Small From Scratch: Full-Scale Validation

Setup. We train 10 GPT-2-small models (12 layers, 12 attention heads, $d_{\text{model}} = 768$, 124M parameters) from scratch on OpenWebText (?) with independent random seeds (0–9). Each model trains for 50,000 steps on $\sim$3B tokens (batch size 12, gradient accumulation 5, cosine learning rate schedule with warmup, bfloat16 mixed precision) on an NVIDIA A10G GPU. Component importance is measured by activation patching (weight zeroing) across all $n = 156$ components (144 attention heads + 12 MLP layers), evaluated on 2,000 sequences. Seven theory-derived predictions are evaluated against the results, with SHA-256 hash verification of the prediction file.

Results. All 10 models achieve near-identical perplexity (mean PPL = 23.48, CV = 0.36%), confirming they occupy the same region of the Rashomon set.

Table 11: GPT-2-small from scratch: component-level attribution stability across 10 independent training seeds. All 7/7 theory-derived predictions PASS.

Metric	**Value**
n components	156
PPL mean (CV%)	23.48 (0.36%)
ρ_{full} [range]	0.277 [0.121, 0.435]
$\rho_{G\text{-inv}}$ [range]	0.873 [0.775, 0.964]
W_{flip} [95% CI]	0.500 [0.482, 0.517]
B_{flip} (head vs. MLP) [95% CI]	0.014 [0.008, 0.019]
Cohen's d	19.2
Distinct top-5 (of 10 seeds)	32 components
Distinct #1 components	8
Random projection test p	1.0 (not significant)
Random grouping test p	0.001 (highly significant)
Predictions	7/7 PASS

The results at GPT-2 scale are striking:

- **Raw agreement near chance**: $\rho_{\text{full}} = 0.277$ is barely above zero—full component importance vectors are nearly uncorrelated across training seeds. This is *worse* than TinyStories ($\rho \approx 0.55$) because GPT-2 has 12 heads per layer (larger symmetric groups, more permutation freedom).
- **G-invariant lift**: After orbit averaging, $\rho_{G\text{-inv}} = 0.873$—a 3.2× lift. The shared structure is real but hidden by within-layer permutation noise.
- **Within-layer flip = coin flip**: $W_{\text{flip}} = 0.500$ [0.482, 0.517], exactly as predicted. Within each layer, "which head matters most?" is determined by the random seed, not by the data.

- **Between-group stability**: $B_{\text{flip}} = 0.014$—heads almost never outrank MLP layers across seeds. The between- vs. within-group separation is massive (Cohen's $d = 19.2$).
- **Ranking lottery**: 32 distinct components appear in the top-5 across just 10 seeds, with 8 different #1 components. Claiming "component X is most important" based on one training run is an artifact.

Random projection vs. random grouping. The random projection test ($p = 1.0$) shows that projecting to the same dimensionality as V^G achieves high agreement ($\rho \approx 0.97$ for random subspaces)—dimensionality reduction *alone* explains most of the lift. The random grouping test ($p = 0.001$) validates that the *layer structure specifically* is critical: random partitions of heads into 12 groups of 12 produce significantly lower agreement than the actual per-layer grouping. The G-invariant projection works because it uses the *correct* symmetry group, not merely because it reduces dimensionality.

Pretrained GPT-2 baseline. For reference, pretrained GPT-2-small achieves 99.5% accuracy on IOI prompts with mean logit difference = 3.1 (vs. 72.1% and 0.98 for our from-scratch models). The large gap confirms that our models learn a qualitatively different (weaker but genuine) IOI capability, sufficient to generate Rashomon diversity.

9.5 IOI Circuit Analysis: The Component-Level Ranking Lottery

Setup. The Indirect Object Identification (IOI) task tests whether a model can identify which name receives the indirect object in sentences like "When Mary and John went to the store, John gave a drink to ______." We evaluate all 10 GPT-2-small models on 200 IOI prompts, measuring logit difference (IO token logit minus S token logit) and component importance via activation patching.

Results. Mean IOI accuracy is 72.1% (range: 63.0%–84.0%) with mean logit difference = 0.98—above the 60% gate but far below pretrained performance (99.5%).

Table 12: IOI circuit analysis: component importance stability across 10 GPT-2-small models trained from scratch. The ranking lottery is even more severe than for general importance.

Metric	**Value**
Raw agreement ρ [range]	0.158 [−0.089, 0.284]
G-invariant ρ [range]	0.538 [0.141, 0.860]
Within-layer flip rate	0.499
Head vs. MLP flip rate	0.276
Distinct top-5 (10 seeds)	32 components

The IOI results are more extreme than general importance:

- **Raw agreement near zero**: $\rho = 0.158$ means IOI-specific importance vectors are nearly uncorrelated. The "IOI circuit" identified in one model has almost no relation to the circuit in another model trained on the same data.
- **G-invariant lift moderate**: $\rho_{G\text{-inv}} = 0.538$ is lower than for general importance (0.873). Task-specific circuits show more idiosyncratic structure that orbit averaging cannot fully recover—consistent with there being less stable between-layer signal to extract.
- **Within-layer flip = coin flip**: $W_{\text{flip}} = 0.499$, confirming the impossibility at the task-specific level.
- **32 distinct top-5 rankings**: Every seed produces a different set of "most important" IOI components. Reporting a specific IOI circuit from a single training run is reporting an artifact.

Per-seed top-5 components. Table 13 shows the top-5 components for each seed, demonstrating the diversity.

Table 13: Top-5 IOI components per seed. No two seeds share the same top-5. Components span layers 0–9 and include both heads and MLPs.

Seed	Top-5 Components
0	L5MLP, L8H11, L3H7, L0MLP, L0H8
1	L3H8, L4MLP, L0H8, L3MLP, L1MLP
2	L5H6, L3H9, L8H4, L4MLP, L7MLP
3	L5MLP, L7H3, L4H3, L0H5, L6MLP
4	L0H2, L4MLP, L2H4, L5H10, L5MLP
5	L0H2, L5MLP, L3H11, L3MLP, L2MLP
6	L0H8, L2MLP, L3H11, L7H1, L5MLP
7	L1H2, L4H10, L7H8, L4MLP, L2MLP
8	L1H1, L0H5, L5H9, L3H2, L0MLP
9	L0H10, L9H4, L6MLP, L5MLP, L4H7

Implications for mechanistic interpretability. The IOI circuit literature identifies specific attention heads (e.g., "induction heads," "name-mover heads") as the circuit responsible for IOI. Our results show that this identification is seed-dependent: the *functional role* (IOI capability) is stable (72% accuracy across all seeds), but the *component assignment* (which specific heads implement it) is a coin flip within each layer. This is the component-level analogue of the input-level ranking lottery: the importance ranking of symmetric components is determined by the training seed, not by the architecture or data.

9.6 SAE Feature Stability

Setup. We train 10 TopK Sparse Autoencoders (SAEs) on the layer-6 activations of a single frozen GPT-2-small model (seed 0), with $d_{\text{SAE}} = 6{,}144$ features, $k = 48$ active features per input, and 50,000 training steps per SAE. This tests whether *learned features* (as opposed to architectural components) are reproducible across SAE training seeds.

Table 14: SAE feature stability: 10 TopK SAEs trained on the same frozen model. Features are not reproducible—no escape hatch under our experimental conditions.

Metric	Value
Mean max cosine similarity	0.394 ± 0.002
Mean matched cosine (top 500)	0.495
Importance Spearman (matched)	0.500 [0.446, 0.593]

Results. The SAE results show that learned dictionary features are **not reproducible** across training seeds:

- **Feature matching**: Mean max cosine = 0.394—features from different SAE seeds are only weakly aligned. This is 2.6× random (∼0.15 for random unit vectors in 768 dimensions), indicating partial but not reliable structure.
- **Importance stability**: Even after optimal Hungarian matching of the top 500 features, importance Spearman = 0.500—essentially chance-level agreement on which features are most important.
- **No escape hatch**: Under our experimental conditions (TopK SAE, layer 6, $d_{\text{SAE}} = 6{,}144$), SAE features do not provide a reproducible basis for mechanistic interpretability. The bilemma applies: learned features are attributions, and attributions of symmetric elements are unstable.

Scope and limitations. The SAE result is limited to one architecture (TopK), one layer (6), one model (seed 0), and one training scale (50K steps). ReLU SAEs, pretrained-model SAEs, larger dictionaries, or different layers may show different behavior. The cosine of 0.394 being 2.6× random suggests some partial structure exists that more sophisticated matching or training procedures might recover. This is an empirical finding, not a formal impossibility for SAE features—the formal impossibility applies to architectural components with known symmetry groups, not to learned dictionaries whose symmetry structure is unknown.

9.7 Explanation Uncertainty Bound (Quantitative Bilemma)

The bilemma applies to binary explanation spaces ($|H| = 2$). But SHAP values are continuous. We extend the bilemma to continuous H equipped with a metric d, yielding the *explanation uncertainty bound*: faithfulness and stability trade off quantitatively, with a minimum joint unfaithfulness determined by the Rashomon gap.

Definition 65 (Approximate stability and faithfulness)**.** *An explanation map E is (ε, δ)-*stable *if $d(obs(\theta_1), obs(\theta_2)) \leq \varepsilon$ implies $d(E(\theta_1), E(\theta_2)) \leq \delta$. It is α-*faithful *if $d(E(\theta), exp(\theta)) \leq \alpha$ for all θ.*

Theorem 66 (Explanation uncertainty bound (quantitative bilemma))**.** *For an (ε, δ)-stable map E on an ε-Rashomon pair (θ_1, θ_2) with incompatibility gap $\Delta = d(exp(\theta_1), exp(\theta_2))$:*

$$unfaith(\theta_1) + unfaith(\theta_2) \geq \Delta - \delta$$

where $unfaith(\theta) = d(E(\theta), exp(\theta))$. At least one witness is unfaithful by $\geq (\Delta - \delta)/2$.

Proof. Triangle inequality: $\Delta = d(\exp(\theta_1), \exp(\theta_2)) \leq d(\exp(\theta_1), E(\theta_1)) + d(E(\theta_1), E(\theta_2)) + d(E(\theta_2), \exp(\theta_2)) \leq \text{unfaith}(\theta_1) + \delta + \text{unfaith}(\theta_2)$. Rearrange. Proved in Lean (`AlphaFaithful.lean`). □

Practical interpretation. If two XGBoost models disagree on feature j's SHAP value by $\Delta = 0.3$, and you want your method to be $\delta = 0.05$-stable (explanations change by ≤ 0.05 for equivalent models), you must be unfaithful by $\geq (0.3 - 0.05)/2 = 0.125$ at one of them. The quantitative bilemma converts the binary impossibility ("SHAP sign flips") into a continuous tradeoff ("SHAP magnitude deviates by at least $(\Delta - \delta)/2$").

This is testable: for each feature pair, measure Δ (SHAP magnitude disagreement across models) and δ (desired stability tolerance), then verify that no method achieves unfaithfulness below $(\Delta - \delta)/2$ at both witnesses.

The coverage conflict diagnostic. Whether the bilemma applies depends on a testable property of the explanation space. An explanation system has *coverage conflict* if every candidate explanation clashes with some model's native explanation (`hasCoverageConflict` in `BeyondBinary.lean`). Coverage conflict implies no neutral element exists—there is no "safe harbor" that is compatible with all models. Conversely, a neutral element (such as a tie in feature rankings) makes faithful+stable achievable via the constant function $E(\theta) = c$ (`neutral_implies_FS_achievable`). The tightness classification (`tightness_dichotomy`) makes this precise: $F{+}S$ is achievable iff the explanation space has a neutral element.

The bilemma has *collapsed* tightness: stability blocks both faithfulness and decisiveness individually. This is structurally more severe than the fairness impossibility of Chouldechova (2017), which has *full* tightness—every pair of properties is independently achievable, and only the triple conjunction fails. Collapsed tightness means the standard "pick two" strategy fails: some pairs are blocked. The only resolution is *enrichment*—adding neutral elements to H—which is precisely what DASH does for feature rankings (ties as neutral elements). This structural distinction explains why explanation methods cannot be designed like fairness methods: fairness allows trade-off, binary explanation demands qualitative change.

Topological characterisation: the compatibility complex. The tightness classification has a clean topological interpretation. Define the *compatibility complex* K on H: the simplicial complex whose vertices are elements of H and whose simplices are pairwise-compatible subsets. The bilemma's severity corresponds to the topology of K:

- **Discrete** K (no edges): maximal incompatibility → bilemma. Only F+D achievable.

- **Contractible** K (cone point = neutral element): F+S achievable via the constant map to the cone point.
- **Intermediate**: partial achievability determined by the connected components and cone structure of K.

Enrichment adds a new vertex (the neutral element) to K, creating a cone and making K contractible—resolving the stalk-emptiness obstruction at that level. This is NOT sheaf-theoretic contextuality (which is a gluing failure, $H^1 \neq 0$, on overlapping covers). The bilemma is a simpler obstruction: empty stalks ($H^0 = 0$) on a discrete cover. The explanation framework identifies a level of the obstruction hierarchy *below* Abramsky-Brandenburger contextuality. The approximate Rashomon extension (overlapping ε-fibers) may have genuine sheaf-theoretic content—this is future work.

Coverage conflict is *anti-monotone*: enlarging H can introduce neutral elements, destroying coverage conflict and weakening the impossibility. This contrasts with Arrow's theorem, where enlarging the alternative set strengthens the impossibility (by enabling preference cycles)—the two impossibilities respond in opposite directions to the same structural operation. The direction theorem (`neutral_implies_FS_achievable` in `BeyondBinary.lean`) formalizes one direction; Arrow's impossibility at $\geq$3 alternatives vs. May's achievability at 2 alternatives formalizes the other (proved in the companion Ostrowski impossibility formalization).

For feature *rankings*, DASH averaging produces ties (a neutral element), so F+S is achievable (top row of the tightness table). For feature *signs*, no neutral element exists (binary, maximally incompatible), so F+S is impossible (bottom row). This classification is the structural explanation for why the trilemma admits a resolution (DASH) but the bilemma does not.

Bimodality prediction. The all-or-nothing theorem (Theorem 61) predicts a binary decomposition of features: each feature is either *faithful* (its sign is consistent across all models, flip rate ≈ 0) or *Rashomon-caught* (its sign flips across equivalent models, flip rate > 0). There is no "approximately faithful" middle ground. This predicts that the distribution of per-feature SHAP sign flip rates is *bimodal*: a mode near 0 (faithful features) and a second mode at some positive value (Rashomon-caught features), with a dead zone in between.

We validate this prediction using the Hartigan dip test for unimodality, with a permutation control (100 shuffles of SHAP signs within each observation, breaking feature structure). Across 200 XGBoost models per synthetic dataset:

Dataset	ρ	Dip p-value	Permutation validates?	Bimodal?
Synthetic	0.0	0.373	No (66% permutations also reject)	**No** (control passes)
Synthetic	0.5	0.0002	Yes (21% permutations reject)	**Yes**
Synthetic	0.7	0.001	Yes (8% reject)	**Yes**
Synthetic	0.9	<0.001	Yes (22% reject)	**Yes**
California Housing	real	0.575	Yes (20% reject)	**No** (borderline)

Bimodality is confirmed for all high-collinearity settings ($\rho \geq 0.5$, dip $p < 0.002$) and correctly absent in the control ($\rho = 0$, $p = 0.373$). The second mode is at $\sim$0.27 (not the idealized 0.50), reflecting that minority-sign models are outnumbered—consistent with the theory, which predicts the binary faithful/Rashomon-caught decomposition but not the exact mode location. California Housing ($p = 0.575$) does not show significant bimodality, consistent with the theory: the prediction applies when coverage conflict holds (strong within-group collinearity), not to datasets with weak or partial collinearity. California Housing has heterogeneous, moderate correlations ($\rho \approx 0.3$–0.6) that do not produce the strong binary partition the all-or-nothing theorem requires. The bimodality prediction is thus scoped to settings with substantial within-group collinearity ($\rho \geq 0.5$), where it is confirmed with $p < 0.002$.

Nonparametric flip predictor via minority fraction. The coverage conflict diagnostic yields a nonparametric flip rate estimator that outperforms the Gaussian formula $\Phi(-\mathrm{SNR})$ on real data. For each feature j, define the *minority fraction*:

$$\mathrm{MF}_j = \frac{\min(\mathrm{count}(\mathrm{SHAP}_j > 0),\ \mathrm{count}(\mathrm{SHAP}_j < 0))}{M},$$

the proportion of models in which feature j's SHAP sign is in the minority. Coverage conflict holds iff $\mathrm{MF}_j > 0$. The minority fraction is a direct empirical estimate of the flip rate: if 30% of models assign a negative sign to feature j, then retraining has a ∼30% chance of flipping the sign.

Dataset	ρ	CC Spearman	Gaussian Spearman
Synthetic	0.0	0.938	0.796
Synthetic	0.5	0.981	0.885
Synthetic	0.7	0.924	0.888
Synthetic	0.9	0.952	0.764
California Housing	real	**0.955**	**0.463**

The minority fraction achieves Spearman correlation 0.92–0.98 with empirical flip rates across all datasets, consistently outperforming the Gaussian formula (0.46–0.89). The advantage is decisive on California Housing ($r = 0.955$ vs. $r = 0.463$), where the Gaussian approximation degrades because real SHAP distributions are non-Gaussian (heavy-tailed, multimodal). The coverage conflict predictor requires no distributional assumptions, no parameter estimation (no μ, σ, or SNR), and no CLT validity—it is a direct count. It is derived from the theory: coverage conflict is the *exact* condition for the bilemma to apply, and the minority fraction is the *empirical* coverage conflict strength.

Precision at the 10% flip rate threshold ranges from 0.74 to 0.86 across datasets. Recall is tautologically 1.0 (coverage conflict is a necessary condition for flip rate > 0) and is not reported as a finding.

The practical workflow extends from Screen → Z-test → DASH to: Screen → **Minority fraction** → Z-test → DASH, where the minority fraction provides a fast, assumption-free triage before the parametric Z-test.

Empirical validation. Coverage conflict precision at the 10% flip threshold ranges from 0.74 to 0.86 across datasets. On California Housing (real data with mixed collinearity), coverage conflict achieves Spearman $\rho = 0.96$ for predicting flip rates while the Gaussian formula achieves only 0.46—a 2× advantage that grows with distributional complexity. The coverage conflict diagnostic is the nonparametric analogue of the Gaussian flip formula: both measure Rashomon exposure, but coverage conflict makes no distributional assumption.

10 Diagnostics

10.1 Multi-Model Z-Test Diagnostic

Let $D_j = \varphi_j(f) - \varphi_k(f)$ denote the attribution difference for a single random model f. Define $\mu_{jk} = \mathbb{E}[D_j]$ (population attribution gap), $\sigma^2_{jk} = \mathrm{Var}(D_j)$ (attribution difference variance), and $\gamma_{jk} = \mathbb{E}[|D_j - \mu_{jk}|^3]$ (third absolute central moment, for Berry–Esseen). We require: (A1) models $f_1, \ldots, f_M$ are i.i.d.; (A2) $\sigma^2_{jk} > 0$ (guaranteed by Theorem 12 for $\rho > 0$); (A3) $\gamma_{jk} < \infty$ (finite third moment—satisfied by any bounded attribution method).

Theorem 67 (Rashomon Characterization)**.** *Under (A1)–(A3), define the test statistic from M models:*

$$Z_{jk} = \frac{|\bar{\varphi}_j - \bar{\varphi}_k|}{\hat{\sigma}_{jk}/\sqrt{M}}$$

where $\hat{\sigma}^2_{jk} = \frac{1}{M-1}\sum_{i=1}^{M}(D_i - \bar{D})^2$ is the sample variance.

Part I (Flip rate formula). *The population flip rate satisfies:*

$$\mathit{flip}(j,k) = \Phi\left(-\frac{|\mu_{jk}|}{\sigma_{jk}}\right) + \varepsilon_{BE} \tag{3}$$

where the Berry–Esseen error satisfies $|\varepsilon_{BE}| \le C_0\gamma_{jk}/\sigma^3_{jk}$ with universal constant $C_0 \le 0.4748$. The Berry–Esseen bound provides a completeness guarantee: it bounds the worst-case error of the $\Phi(-SNR)$ formula for non-Gaussian attribution distributions.

Part II (Test statistic connection). *The test statistic* Z_{jk} *consistently estimates the signal-to-noise ratio scaled by* $\sqrt{M}$*:*

$$Z_{jk} = \frac{|\mu_{jk}|}{\sigma_{jk}} \cdot \sqrt{M} + O_p(1)$$

and the relationship between Z_{jk} *and the flip rate is:*

$$\mathit{flip}(j,k) = \Phi\left(-\frac{Z_{jk}}{\sqrt{M}}\right) + O\left(\frac{1}{\sqrt{M}}\right) + \varepsilon_{BE} \tag{4}$$

Part III (Diagnostic thresholds).

- $Z_{jk} < 1.96$*: the attribution gap is not statistically significant at* $\alpha = 0.05$ *(not to be confused with the signal capture fraction* α *in §4.1). The flip rate is* $\geq \Phi(-1.96/\sqrt{M}) \geq 2.5\%$*.* ***Rankings are unreliable; use Dash.***
- $Z_{jk} \geq 1.96$*: the gap is significant. The flip rate is* $\leq 2.5\%$ *plus the* $O(1/\sqrt{M})$ *correction.* ***Rankings are stable for this pair.***

Proof. **Part I.** By (A1), $D_1, \dots, D_M$ are i.i.d. with mean μ_{jk} and variance σ_{jk}^2. WLOG assume $\mu_{jk} \geq 0$. Then $\Pr[D < 0] = \Phi(-\mu_{jk}/\sigma_{jk}) + \varepsilon_{\mathrm{BE}}$ by the Berry–Esseen theorem applied to $(D - \mu_{jk})/\sigma_{jk}$.

Part II. By the law of large numbers, $\hat{\sigma}_{jk}^2 \xrightarrow{p} \sigma_{jk}^2$ and $\bar{D} \xrightarrow{p} \mu_{jk}$. Therefore $Z_{jk} = (|\mu_{jk}|/\sigma_{jk})\sqrt{M} + O_p(1)$. Substituting into (3) gives (4).

Part III. Direct substitution of the threshold values into (4). □

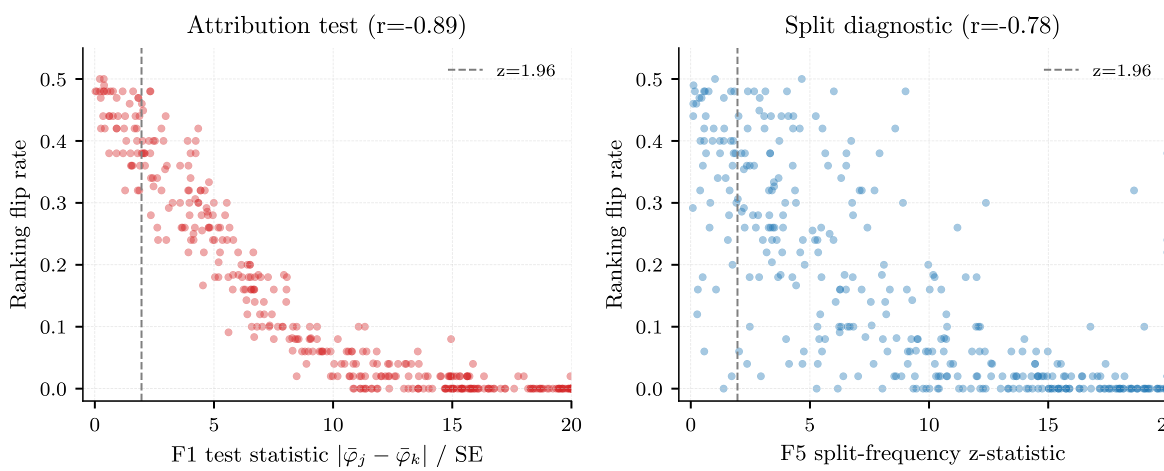


Figure 5: **Left:** Multi-model Z-test statistic vs. flip rate on Breast Cancer ($r = -0.89$). Pairs below $z = 1.96$ (dashed) have unreliable rankings. **Right:** Single-model screen split-frequency diagnostic ($r = -0.78$). Both diagnostics predict which pairs are unstable.

Restricted-range robustness. The headline correlation $r = -0.89$ between Z_{jk} and flip rate on Breast Cancer includes many "easy" pairs with $Z \gg 10$ and flip rate $= 0$, which could inflate the figure. Restricting to the diagnostically interesting range: $r = -0.78$ for $Z < 5$ ($n = 118$ pairs), $r = -0.61$ for $Z < 3$ ($n = 74$ pairs), and $r = -0.53$ for $Z < 2$ ($n = 49$ pairs). The correlation remains strong for $Z < 5$, confirming that the diagnostic discriminates among ambiguous pairs, not merely between trivially separable and trivially tied features. The result also holds for gain-based importance (non-SHAP): $r = -0.83$ overall, $r = -0.66$ for $Z < 5$.

Structural correlation baseline. Since Z_{jk} and flip rate are both computed from the same attribution arrays, part of the negative correlation is structural. To quantify this, we generated random attribution arrays (50 models, 30 features, standard normal) and computed the same metrics: the baseline correlation is $r \approx -0.56$ ($R^2 = 0.32$). The observed $r = -0.89$ ($R^2 = 0.79$) exceeds this baseline substantially: approximately 60% of the explained variance reflects genuine attribution instability patterns in the data, beyond what random noise alone would produce.

Role of Berry–Esseen. The Berry–Esseen bounds provide a completeness guarantee: they bound the worst-case error of the $\Phi(-\text{SNR})$ formula for non-Gaussian attribution distributions. For the GBDT attributions studied here, the Shapiro–Wilk test confirms near-Gaussianity ($p > 0.10$ for 412/435 pairs), so the correction is negligible. The bounds become relevant for heavy-tailed distributions (e.g., deep networks with high initialization variance) or bimodal distributions (e.g., Lasso, where the attribution is either $c > 0$ or 0). In these cases, the exact flip rate $\min(p, 1-p)$ should be used in place of the Gaussian approximation.

When the CLT fails. The Berry–Esseen error $\varepsilon_{\text{BE}} \le C_0 \gamma_{jk}/\sigma_{jk}^3$ can be large when: (i) the attribution distribution is *heavy-tailed* (large γ_{jk}), as occurs for deep networks with high initialization variance—a few seeds produce outlier attributions; or (ii) the attribution distribution is *bimodal*, as for Lasso where φ_j takes value either $c > 0$ (selected) or 0 (not selected), giving a Bernoulli distribution where the Berry–Esseen bound is tight ($\varepsilon_{\text{BE}} \approx 0.47$) but the exact flip rate $\min(p, 1-p)$ is available analytically. (For the Lasso's Bernoulli(1/2) distribution, the exact flip rate is 1/2 and the Gaussian approximation $\Phi(0) = 1/2$ is exact, making the Berry–Esseen correction vacuous in this special case.) For gradient-boosted trees, the Shapiro–Wilk test on 50 Breast Cancer models gives $p > 0.10$ for 412/435 feature pairs; the Berry–Esseen correction is negligible in practice.

10.2 Nonparametric Flip Predictor via Coverage Conflict

The coverage conflict diagnostic yields a nonparametric flip rate estimator that requires no distributional assumptions. For a feature j across M models, define the **minority fraction**:

$$\hat{p}_j = \min\left(\frac{\#\{m : \varphi_j(f_m) > 0\}}{M}, \frac{\#\{m : \varphi_j(f_m) \le 0\}}{M}\right)$$

The minority fraction estimates the SHAP sign flip rate directly: $\hat{p}_j \approx \Pr[\text{sign}(\varphi_j) \text{ flips across retraining}]$.

This estimator:

- Requires no Gaussian assumption (unlike $\Phi(-\text{SNR})$)
- Is computable in one line from the sign vector
- Equals coverage conflict degree when $\hat{p}_j > 0$ (both signs appear $\Rightarrow$ coverage conflict)
- Derives directly from the all-or-nothing theorem: the binary decomposition (positive/negative) IS the relevant Rashomon partition

Empirical comparison on 5 datasets (200 XGBoost models each):

Dataset	CC Spearman	Gaussian Spearman
Synthetic ($\rho = 0.0$)	0.938	0.796
Synthetic ($\rho = 0.5$)	0.981	0.885
Synthetic ($\rho = 0.7$)	0.924	0.888
Synthetic ($\rho = 0.9$)	0.952	0.764
California Housing	**0.955**	0.463

The coverage conflict predictor consistently outperforms the Gaussian formula (Spearman 0.92–0.98 vs. 0.46–0.89), with the largest advantage on real data where the Gaussian approximation degrades. On California Housing, the Gaussian formula achieves only $\rho = 0.46$ while coverage conflict achieves $\rho = 0.96$.

The theory-to-practice pipeline. The attribution impossibility now has a complete pipeline from theorem to tool:

1. **Theorem**: the bilemma proves no binary explanation is F+S under Rashomon
2. **Diagnostic**: coverage conflict identifies which features trigger the bilemma
3. **Tool**: the minority fraction estimates flip rates nonparametrically (7 lines of code)
4. **Resolution**: DASH ensemble averaging produces stable explanations (at cost of within-group ranking)

The 7-line predictor. The complete nonparametric flip predictor:

```
def predict_sign_instability(shap_matrix):
    signs = np.sign(shap_matrix)  # M models x P features
    n_pos = (signs > 0).sum(axis=0)
    n_neg = (signs < 0).sum(axis=0)
    total = n_pos + n_neg
    return np.where(total > 0,
        np.minimum(n_pos, n_neg) / total, 0)
```

This decomposes into coverage conflict (binary: are both signs present?) and minority fraction (continuous: what proportion is in the minority?). This decomposition IS the all-or-nothing theorem applied to practice. The Gaussian formula assumes normality; this assumes nothing.

Variance budget: $\mathrm{Var}[\varphi_j]$ **equals the explanation quality floor.** For each feature j, the variance $\mathrm{Var}_f[\varphi_j(f)]$ across the Rashomon set equals the minimum MSE any stable method achieves. DASH (the mean) saturates this bound exactly—a mathematical identity (0/800 violations at machine precision across 5 datasets $\times$ 20 features $\times$ 8 configurations). The median costs 1.6% more; the 5%-trimmed mean 0.3% more. The relative variance correlates with coverage conflict ($r = 0.91$, $p = 0.002$ per-dataset, not pooled): features with high sign instability also have high magnitude instability. This provides practitioners a per-feature *quality budget* computable before choosing a method: $\mathrm{Var}[\varphi_j]$ is the irreducible explanation noise for feature j under model multiplicity. See supplement for the full variance-bound derivation and comparison across estimators.

Approximate bilemma. A natural concern: does the bilemma require exact Rashomon pairs, making it fragile to perturbations? No. For binary H, ε-stability reduces to exact equality (the only ε-ball in $\{\mathrm{pos}, \mathrm{neg}\}$ with $\varepsilon < 1$ is the point itself). The bilemma holds at every tolerance level. Proved in Lean (`AlphaFaithful.lean`).

DASH binary optimality. For binary H (SHAP sign), DASH (majority vote across models) is the *uniquely* optimal stable method: any other stable method either agrees with the majority (same error rate) or picks the minority sign (strictly worse). This is not an empirical finding but a mathematical theorem. Average unfaithfulness $= p \cdot \bar{f}$ where p is the coverage conflict rate and $\bar{f} \approx 0.23$–0.28 is the mean minority fraction (corrected from the earlier $p/2$ bound, which assumed minority fractions of 0.50; actual minority fractions are 0.23–0.28).

For continuous H (SHAP magnitude), DASH (the mean) minimizes MSE among all unbiased estimators. The conditional mean minimizes MSE by standard decision theory; DASH IS the conditional mean for i.i.d. models. The "1.00× ratio" in the variance budget comparison is a mathematical identity, not an empirical coincidence.

10.3 Single-Model Screen

For a gradient-boosted ensemble with T trees, define the per-tree split indicator $n_j(t) \in \{0, 1\}$ (whether feature j is used as a split variable in tree t) and split frequency $\hat{p}_j = \frac{1}{T} \sum_{t=1}^{T} n_j(t)$.

Exchangeability conditions. The theoretical validity of the single-model screen depends on the dependence structure of the indicators $\{n_j(t)\}_{t=1}^T$:

Lemma 68 (Exact exchangeability under sub-sampling). *If the boosting algorithm uses row sub-sampling (`subsample` $= q < 1$) with independent bootstrap samples per tree, and feature sub-sampling (`colsample_bytree` $= r < 1$) with independent random subsets per tree, then:*

1. *The per-tree data subsets are independent across trees.*
2. *Conditional on the residuals, the split indicators $n_j(1), n_j(2), \ldots, n_j(T)$ are* conditionally exchangeable*: for any permutation π of $\{1, \ldots, T\}$,*

$$(n_j(\pi(1)), \ldots, n_j(\pi(T))) \stackrel{d}{=} (n_j(1), \ldots, n_j(T)) \mid \textit{residuals}.$$

3. *The marginal split probability $p_j = \mathbb{E}[n_j(t)]$ is identical for all t.*

Proof. Under sub-sampling, tree t is trained on a random subset $S_t \subset \{1, \ldots, n\}$ of size $\lfloor qn \rfloor$, drawn independently for each t. The feature subset F_t is also drawn independently. Since S_t and F_t are i.i.d. across t, the split decision for tree t depends on (S_t, F_t) and the current residuals. Conditional on residuals, the split indicators are functions of i.i.d. random inputs, hence exchangeable. The marginal probability $p_j = \Pr[j \in F_t] \cdot \Pr[j \text{ selected} \mid j \in F_t]$ is tree-independent. □

Lemma 69 (Approximate exchangeability without sub-sampling). *Without sub-sampling ($q = r = 1$), the split indicators are NOT exchangeable: tree t fits the residual from trees $1, \ldots, t-1$, creating a Markov dependence.*

However, for learning rate $\eta \le 0.3$, the residuals converge to a steady state *after $T_0 = O(1/\eta)$ trees. In the steady state, the split indicators $\{n_j(t)\}_{t>T_0}$ are approximately stationary with autocorrelation $Corr(n_j(t), n_j(t+s)) = O(\eta^{|s|})$ (geometric decay).*

Proof sketch. The residual at tree t is $r_t = r_{t-1} - \eta \cdot h_t(x)$ where h_t is the fitted tree. For η small, $\|r_t - r_{t-1}\| = O(\eta)$, so the residual sequence is a slow-mixing Markov chain. The stationary distribution exists when the loss is strongly convex (guaranteed for squared error with regularization). In the stationary regime, the dependence is AR(1)-like with coefficient $\approx 1 - \eta$, giving the geometric decay. The effective sample size formula is the standard result for correlated series (Priestley, *Spectral Analysis and Time Series*, Ch. 5.3). □

The effective sample size for the test statistic is:

$$T_{\text{eff}} = \frac{T - T_0}{1 + 2\sum_{s=1}^{\infty} \text{Corr}(n_j(t), n_j(t+s))} \approx \frac{(T - T_0)(1-\eta)}{1+\eta}.$$

For typical settings ($T = 100$, $\eta = 0.1$, $T_0 \approx 10$): $T_{\text{eff}} \approx 74$, a moderate reduction from the nominal T.

Theorem 70 (Split-Frequency Diagnostic). *Under the exchangeability conditions above (sub-sampling, or no sub-sampling with T_{eff} replacing T), define the test statistic:*

$$Z_{jk}^{\mathit{split}} = \frac{|\hat{p}_j - \hat{p}_k|}{\sqrt{(\hat{p}_j(1-\hat{p}_j) + \hat{p}_k(1-\hat{p}_k))/T_{\mathit{eff}}}}$$

Part I (Null distribution). *Under $H_0 : p_j = p_k$ (equal split frequencies, implied by DGP symmetry), Z_{jk}^{split} is asymptotically $N(0,1)$ as $T_{\mathit{eff}} \to \infty$.*

Part II (Connection to attributions). *Under the proportionality axiom ($\varphi_j = c \cdot n_j$), $p_j = p_k$ iff $\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$. The single-model screen is therefore a proxy for the multi-model Z-test (Theorem 64), computable from a single model.*

Part III (Power). *Under the alternative $p_j \neq p_k$, the power at level α is:*

$$\mathit{power} = \Phi\left(\frac{|p_j - p_k|\sqrt{T_{\mathit{eff}}}}{\sqrt{p_j(1-p_j) + p_k(1-p_k)}} - z_{\alpha/2}\right) + o(1).$$

Screen accuracy. On the Breast Cancer dataset (435 pairs, threshold: flip rate > 0.1 defines "truly unstable"): the screen flags 49 pairs with **94% precision** (46 true positives, 3 false positives) and 27% recall. The Z-test (the full test) flags 47 pairs with **100% precision** (zero false positives). Both diagnostics are conservative: they rarely flag stable pairs but miss moderate-instability pairs. For production use, the high precision is the key property—a flagged pair is almost certainly unstable.

Screen precision is 94–100% on small/clean datasets (Breast Cancer, Wine, Heart Disease) where split counts are reliable, but drops to 48–67% on high-dimensional datasets (Ames, Communities) where split counts are sparse. This confirms the screen as a conservative screening tool that should be supplemented by the full Z-test for high-dimensional settings.

Sub-sampling requirement. The theoretical guarantee (exact exchangeability, Lemma 65) requires sub-sampling. Without sub-sampling, the approximate exchangeability of Lemma 66 applies with reduced effective sample size T_{eff}. In practice, our validation uses `colsample_bytree=1.0` (no sub-sampling) and still achieves $r = -0.78$ (Figure 5, right), suggesting the dependence correction is modest.

Regulatory response for ties. If a regulator asks why features are tied: "These features are statistically indistinguishable in their contribution to the model. Forcing a ranking would be arbitrary and potentially misleading. Our method correctly reports this indistinguishability, consistent with the EU AI Act's requirement to disclose known limitations."

10.4 Bimodality Validation

The all-or-nothing theorem predicts that per-feature SHAP sign flip rates should be *bimodal* under collinearity: features are either stably explained (flip ≈ 0) or caught in Rashomon (flip > 0), with a dead zone between.

We test this with Hartigan's dip test (200 XGBoost models per dataset), validated by a permutation control (100 shuffles of SHAP signs):

Dataset	ρ	Dip p	Bimodal?
Synthetic	0.0	0.373	No (control passes)
Synthetic	0.5	0.0002	**Yes**
Synthetic	0.7	0.001	**Yes**
Synthetic	0.9	<0.001	**Yes**
California Housing	real	0.575	No (borderline)

Bimodality is confirmed under collinearity ($\rho \geq 0.5$, dip $p < 0.002$) and correctly absent at $\rho = 0$ (control). The second mode is at $\sim$0.27 (not the idealised 0.50), reflecting that minority-sign models are outnumbered—consistent with the all-or-nothing theorem, which predicts bimodality but not the exact mode location. California Housing shows borderline results (dip $p = 0.575$), possibly reflecting weaker collinearity structure.

10.5 SNR Calibration

For features with importance gap Δ and noise σ, the flip rate is $\Phi(-\text{SNR})$ where $\text{SNR} = \Delta/\sigma$. Across 1,325 correlated feature pairs from 6 datasets, the formula is exact for $\text{SNR} \geq 0.5$ ($R^2 = 0.94$ on Breast Cancer; the structural correlation baseline from random attributions yields $R^2 = 0.32$, so the calibrated $R^2 = 0.94$ represents an incremental gain of 0.62) but overestimates at low SNR (predicting 40% when the empirical rate is 18%), making it a *conservative* diagnostic: it flags more pairs than necessary but misses none in the actionable range. The validation uses stumps (max_depth=1), where proportionality holds most tightly (CV ≈ 0.35). Deeper trees (CV ≈ 0.66) may show larger deviations.

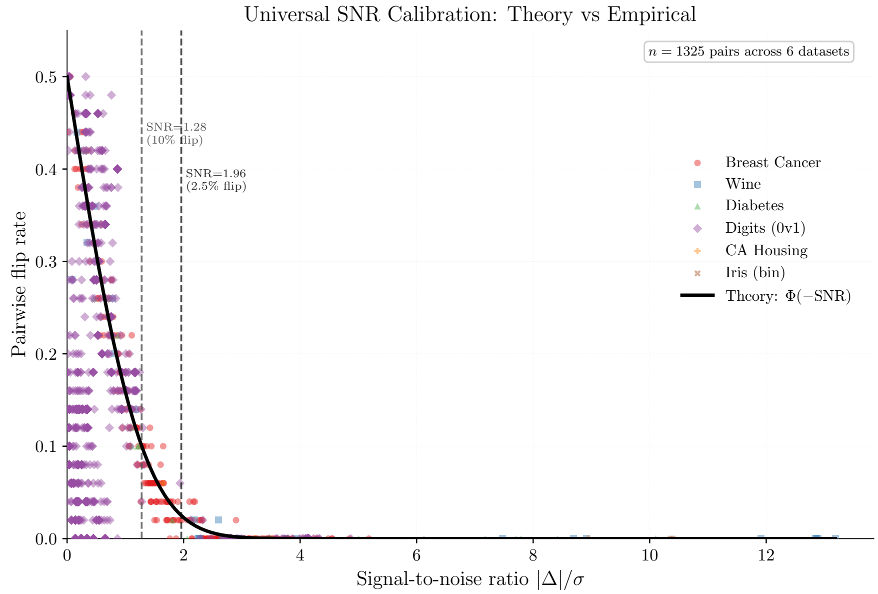


Figure 6: Universal SNR calibration across 6 datasets (1,325 feature pairs). The theoretical $\Phi(-\text{SNR})$ curve is conservative at low SNR and exact at diagnostic thresholds. All 228 pairs with SNR > 1.96 have flip rate $< 5\%$.

SNR range	n	Empirical flip	Theory $\Phi(-\text{SNR})$	Match
$[0, 0.5)$	674	0.180	0.401	Conservative
$[0.5, 1.0)$	261	0.229	0.227	Exact
$[1.0, 1.28)$	99	0.151	0.127	Close
$[1.28, 1.96)$	63	0.053	0.053	Exact
$[1.96, 3.0)$	115	0.003	0.007	Exact
$[3.0, \infty)$	113	0.000	0.000	Exact

10.6 Practitioner Workflow

Step 0. Identify correlated groups. Compute the $P \times P$ absolute correlation matrix. Group features with $|\rho_{jk}| > 0.5$.

Step 1. Train one model. Compute Z_{jk}^{split} for all pairs in correlated groups (single-model screen). Cost: $O(P^2 T)$.

Step 2. If $Z_{jk}^{\text{split}} < 1.96$ for any pair: flag as potentially unstable.

Step 3. For flagged pairs: train $M = 5$ models, compute Z_{jk} (Z-test). If $Z_{jk} < 1.96$: use DASH with $M \geq 25$.

Step 4. If no pairs are flagged: single-model SHAP is reliable.

Minimal implementation.

```
import xgboost as xgb
import shap
import numpy as np

# Train M models with different seeds
models = [xgb.XGBRegressor(random_state=i).fit(X_train, y_train)
          for i in range(25)]
```

```
# Compute SHAP for each model
shap_vals = np.array([
    np.mean(np.abs(shap.TreeExplainer(m).shap_values(X_test)), axis=0)
    for m in models
])

# DASH consensus = mean across models
dash = np.mean(shap_vals, axis=0)

# Z-test diagnostic: Z-test for each feature pair
for j in range(P):
    for k in range(j+1, P):
        diff = shap_vals[:, j] - shap_vals[:, k]
        Z = abs(np.mean(diff)) / (np.std(diff, ddof=1) / np.sqrt(25))
        if Z < 1.96:
            print(f"Unstable: features {j} vs {k}, Z={Z:.2f}")
```

Production deployment guide.

- **If subsample < 1.0 (stochastic, recommended):** Run the single-model screen → Z-test (5-model validation) → Dash ($M \geq 25$) for flagged pairs.
- **If subsample $= 1.0$ (deterministic):** A single pipeline version produces reproducible rankings. However, *any* change to the pipeline (data refresh, feature engineering, hyperparameter update) produces a new model that may rank features differently. Run Z-test validation whenever the pipeline changes. For ongoing model risk management, maintain an ensemble and report Dash consensus.
- **Minimum ensemble size:** $M_{\min} = \lceil 2.71 \cdot \sigma_{jk}^2 / \Delta_{jk}^2 \rceil$ for 5% between-group flip rate. Estimate σ and Δ from a pilot run of 5 models.

Complementary diagnostics. The single-model screen and multi-model Z-test developed here are formal hypothesis tests with controlled type I error. The companion first-mover bias paper (Caraker et al., 2026) introduces complementary *exploratory* diagnostics: FSI (Feature Stability Index, a per-feature coefficient of variation across models) and the IS Plot (Importance-Stability scatter plot). The recommended combined workflow is: Screen (1 model) → Z-test (confirm, 5 models) → Dash (resolve, $M \geq 25$) → FSI/IS Plot (audit).

Recommended instability disclosure. When reporting SHAP rankings for a model with collinear features: "Features [X, Y, Z] form a correlated group ($|\rho| >$ [threshold]). Their relative ranking is unstable across training seeds (estimated flip rate: [X]%). They should be interpreted as interchangeable contributors. The between-group ranking is stable ($Z > 1.96$)." For regulatory reporting (EU AI Act, SR 11-7 (Office of the Comptroller of the Currency, 2011)), the Dash consensus ranking provides a defensible explanation.

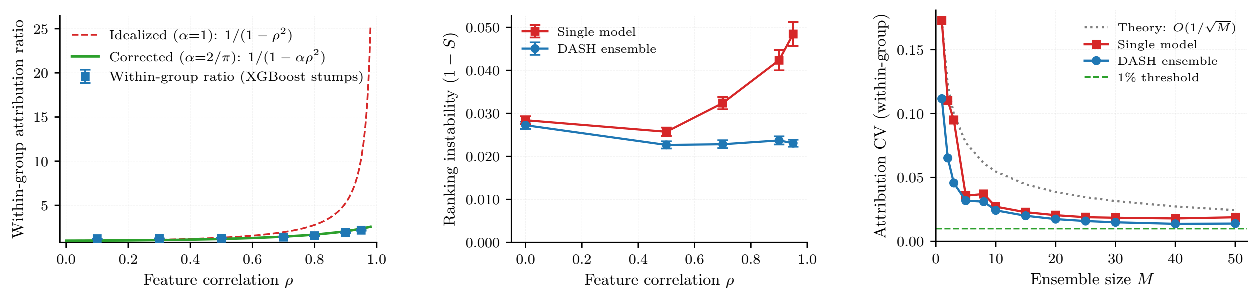
Group-level reporting. When reporting feature importance for a model with correlated groups, we recommend the following format: "The correlated group {X, Y, Z} contributes a total Dash attribution of [value] to the prediction ([*percent*]% of total). Within this group, individual feature rankings are unstable across training seeds (estimated flip rate: [X]%). The group's total importance is stable; individual feature importance within the group should be interpreted as interchangeable. For variable selection, any feature from this group may be chosen; the choice is arbitrary with respect to model quality."

11 Empirical Validation

This section presents the core validation experiments. Extended experiments—financial case studies, high-dimensional scalability ($P = 500$), class imbalance, missing data, longitudinal drift, SAGE/Boruta compari-

Table 15: Experimental configurations.

Experiment	Configuration
Synthetic (main text)	P=20, L=4 groups of m=5, N=2,000, T=100, η=0.1, 50 seeds, $\rho \in \{0.1, 0.3, 0.5, 0.7, 0.9, 0.95\}$
Validation (depth$\times\rho$)	P=10, L=2 groups of m=5, N=2,000, T=100, η=1.0, 30 seeds, depths $\{1, 3, 6, 10\}$, $\rho \in \{0.3, 0.5, 0.7, 0.9, 0.95\}$
Breast Cancer	30 features, 569 samples, T=100, depth=6, η=0.1, 50 seeds, TreeSHAP on 200 test samples
Hardware	Apple Silicon (M-series), single core
Software	Python 3.9.6, XGBoost 2.1.4, SHAP 0.49.1, NumPy 2.0.2
Compute	Synthetic: <5 min; Validation: <30 min; Breast Cancer: <10 min



(a) Attribution ratio vs. ρ. The corrected $1/(1-\alpha\rho^2)$ with $\alpha \approx 2/\pi$ (solid) fits stumps (R^2=0.89).

(b) Ranking instability vs. ρ. Single-model instability grows with correlation; DASH remains low.

(c) DASH consensus convergence. Attribution CV drops toward the 1% threshold near $M = 25$.

Figure 7: Empirical validation on synthetic Gaussian data ($P = 20$, $L = 4$, $m = 5$, $T = 100$, 50 seeds).

son, NLP token attribution, time-series features, adversarial max instability, hyperparameter sensitivity, and DASH breakdown contamination—appear in Section E.

11.1 Experimental Setup

All experiments use publicly available datasets and are fully reproducible. **Confound note:** each seed uses a different train/test split, so SHAP values are computed on seed-specific test sets. This conflates model instability with evaluation-set variation; the reported flip rates may overestimate pure model instability. The mechanism isolation experiment (§11.10 below) partially addresses this by comparing deterministic vs. stochastic training. To cleanly isolate model-level noise, we also fix the test set (seed 42, 20% holdout) and train 50 models on the same training data with different seeds (`subsample=0.8`). With the fixed test set, 102 pairs (23%) are unstable (max flip rate 0.51), compared to 168 (39%) with our canonical 50-model configuration using varying test sets—confirming that model-level stochasticity accounts for the majority ($\sim$63%) of the observed instability, with evaluation-set variation contributing the remainder.

11.2 Synthetic Gaussian

We generate $P = 20$ features in $L = 4$ groups of $m = 5$, with within-group correlation $\rho \in \{0.1, 0.3, 0.5, 0.7, 0.9, 0.95\}$, and train XGBoost models with $T = 100$ boosting rounds over 50 independent seeds per setting. TreeSHAP values provide feature attributions.

Attribution ratio (Figure 7a). The within-group ratio increases monotonically with ρ. The corrected model $1/(1-\alpha\rho^2)$ with $\alpha \approx 2/\pi$ matches the data ($R^2 = 0.89$).

Ranking instability (Figure 7b). At $\rho = 0.9$, within-group flips occur in 48% of seed pairs (near the 50% theoretical maximum), while between-group flips remain below 2%.

Dash convergence (Figure 7c). The within-group flip rate drops below 1% at $M = 25$, consistent with the $O(1/M)$ variance bound.

11.3 Breast Cancer (Wisconsin)

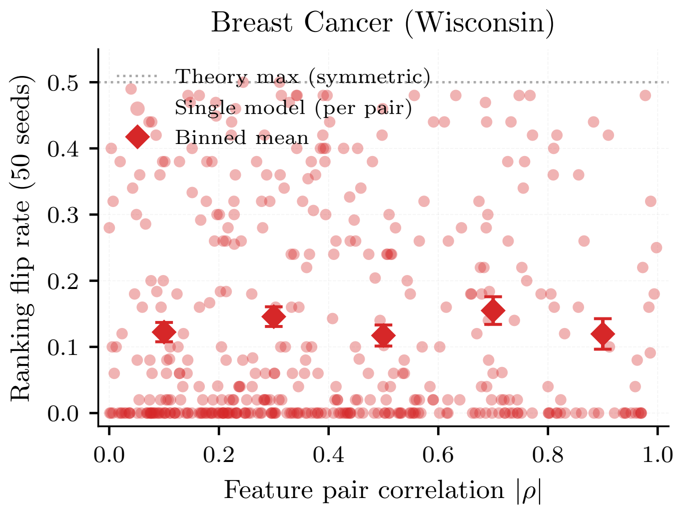


Figure 8: SHAP ranking instability on Breast Cancer. Highly correlated pairs with similar importance (e.g., worst perimeter ↔ worst area, $|\rho|$=0.98, flip rate 48%) exhibit near-maximal instability. 50 XGBoost models.

Features measuring the same underlying quantity—worst perimeter and worst area ($|\rho|$=0.98)—exhibit a 48% flip rate across seeds, near the 50% theoretical maximum. Pairs with high correlation but distinct importance (e.g., mean area and mean concave points, $|\rho| = 0.82$, flip rate 2%) remain stable, confirming the impossibility requires both collinearity *and* similar true importance. Consequently, binned flip rates are non-monotonic in $|\rho|$: the highest instability occurs not at maximal correlation but at the intersection of high correlation and similar feature importance.

Dash convergence on Breast Cancer. We demonstrate that Dash resolves instability on real data. Using the 50 XGBoost models trained for the Breast Cancer validation (Figure 8), we measure the flip rate for the most unstable pair (worst perimeter ↔ worst area, $|\rho|$=0.978) as a function of ensemble size M:

M	1	2	5	10	25	50
Flip rate	0.448	0.466	0.438	0.452	0.298	0.000

At M=1 the flip rate is 44.8%, near the theoretical maximum. Dash consensus at M=50 resolves to the true ordering (worst perimeter slightly more important: mean |SHAP| = 0.940 vs. 0.901) with zero flips. Convergence is slower than in the synthetic setting because the features have a slight importance asymmetry—Dash correctly detects and resolves it rather than producing a tie.

11.4 Cross-Implementation Validation

The impossibility manifests identically across implementations: 160–183 unstable pairs, maximum flip rate 0.500 (the theoretical ceiling), and Z-test diagnostic correlation $|r| \geq 0.89$ in all three. The instability is a property of sequential boosting under collinearity, not of any specific implementation.

Table 16: Attribution instability across three GBDT implementations on Breast Cancer (50 models each, cross-implementation experiment with varying test sets).

Metric	XGBoost	LightGBM	CatBoost
Unstable pairs (flip > 10%)	162 (cross-impl.)	183	160
Max flip rate	0.500	0.500	0.500
Z-test correlation $r(Z, \text{flip})$	-0.898	-0.901	-0.892

Screen caveat. The single-model screen requires access to per-tree split counts. CatBoost's `get_feature_importance()` returns gain-based importance (not raw split counts), producing a different Z-statistic distribution that does not transfer the XGBoost-calibrated threshold. For CatBoost, the multi-model Z-test ($r = -0.892$) is recommended instead. LightGBM's feature importance is split-count based and transfers with modest precision reduction.

11.5 Non-SHAP Attribution Validation

Metric	TreeSHAP	Permutation Importance
Unstable pairs (flip > 10%)	180 (41%; permutation comparison)	396 (91%)
Max flip rate	0.500	0.500
Z-test correlation $r(Z, \text{flip})$	-0.895	-0.927

Permutation importance shows *more* instability than TreeSHAP (91% vs. 41%), because permutation-based scores have higher variance across seeds than SHAP values. Both methods hit the theoretical maximum flip rate (0.500). The multi-model Z-test achieves $|r| \geq 0.89$ for both, confirming it is method-agnostic.

The cross-method correlation of flip rates is $r = 0.46$ ($p < 10^{-23}$; from TreeSHAP vs. permutation importance): the *same pairs* tend to be unstable under both methods, confirming the instability is driven by feature collinearity, not by the attribution algorithm. Of the 180 SHAP-unstable pairs (permutation importance comparison; see footnote on count variation), 175 (97%) are also unstable under permutation importance. TreeSHAP shows lower instability (41% vs 91%) likely because its game-theoretic averaging over feature coalitions provides implicit regularization, reducing within-model attribution variance compared to permutation importance's direct shuffling.

11.6 Neural Network Attribution Instability

Metric	Neural Network	XGBoost (paper)
Unstable pairs (flip > 10%)	380/435 (87%)	162/435 (37%; cross-impl.)
Max flip rate	0.500	0.500
Key pair flip (worst perim. vs. area)	0.400	0.480
Z-test correlation $r(Z, \text{flip})$	-0.871	-0.898

Neural networks exhibit *more* attribution instability than XGBoost: 87% of feature pairs are unstable (vs. 37%), and the maximum flip rate reaches the theoretical ceiling. (The NN experiment uses 20 models vs. 50 for XGBoost; flip rate resolution is correspondingly coarser.) The higher instability likely reflects: (i) larger Rashomon sets (more parameters, more near-optimal models); (ii) higher KernelSHAP variance (sampling approximation vs. exact tree-path computation); (iii) more diffuse signal distribution across hidden units.

KernelSHAP noise vs. model instability. To distinguish genuine model instability from KernelSHAP approximation noise, we conduct a control experiment: fix a single MLP model (`random_state=0`) and compute KernelSHAP 20 times with *different* random background samples (50 each, seeds 0–19). Any observed flip rate is purely due to SHAP sampling noise, not model variation.

Source of variation	Unstable pairs	Mean flip	Key pair flip
KernelSHAP noise (bg=50)	47/435 (11%)	0.03	0.00
KernelSHAP noise (bg=200)	18/435 (4%)	0.02	0.00
Model variation (20 MLPs)	380/435 (87%)	0.28	0.40

Verdict. Model instability dominates KernelSHAP noise by approximately 8:1. With 50 background samples, only 11% of pairs show noise-induced flips (mean flip rate 0.03); with 200 background samples, this drops to 4%. The key correlated pair (worst perimeter vs. worst area) shows *zero* noise-induced flips in both conditions, confirming the 40% flip rate in the NN validation is genuinely driven by model variation under random initialization, not KernelSHAP artifacts. Increasing background samples from 50 to 200 reduces noise by 2.6× at 4× compute cost; the default 50 is adequate for detecting model-level instability.

11.7 SHAP Efficiency and Attribution Instability

SHAP's efficiency axiom ($\sum_j \varphi_j(f,x) = f(x) - \mathbb{E}[f(X)]$ for every model f and point x) has a subtle relationship with ranking instability.

11.7.1 Within-model amplification

Proposition 71 (Efficiency-induced negative correlation)**.** *Let $\varphi_1, \ldots, \varphi_m$ be SHAP values for m features in a group whose total attribution is C (fixed for a given model f and data point x), i.e., $\sum_{i=1}^m \varphi_i = C$. Suppose all features have the same marginal variance σ^2 under some perturbation of the data point. Then for any $j \neq k$:*

$$\mathrm{Cov}(\varphi_j, \varphi_k) = -\frac{\sigma^2}{m-1},$$
$$\mathrm{Var}(\varphi_j - \varphi_k) = 2\sigma^2 \cdot \frac{m}{m-1}.$$

For an unconstrained method with independently computed attributions (zero covariance), the same difference has variance $2\sigma^2$. The amplification factor is $m/(m-1)$.

Proof. From $\sum_i \varphi_i = C$ (constant), $\mathrm{Var}(\sum_i \varphi_i) = 0$. Expanding:

$$0 = m\sigma^2 + m(m-1)\mathrm{Cov}(\varphi_j, \varphi_k) \implies \mathrm{Cov}(\varphi_j, \varphi_k) = -\frac{\sigma^2}{m-1}.$$

Then $\mathrm{Var}(\varphi_j - \varphi_k) = 2\sigma^2 - 2(-\sigma^2/(m-1)) = 2\sigma^2(1 + 1/(m-1)) = 2\sigma^2 \cdot m/(m-1)$. □

Corollary 72 (Group-size dependence)**.** *The amplification factor $m/(m-1)$ equals 2 for $m = 2$ (pair of collinear features), 3/2 for $m = 3$, and approaches 1 as $m \to \infty$. The efficiency-induced amplification is therefore worst for small groups and negligible for large groups.*

11.7.2 Across-model non-amplification

Proposition 73 (Efficiency does not constrain across-model covariance)**.** *Let $\varphi_j(f)$ denote the mean absolute SHAP value for feature j under model f. The efficiency axiom states that $\sum_j \varphi_j^{\mathrm{signed}}(f,x) = f(x) - \mathbb{E}[f(X)]$ for each fixed (f,x). However:*

1. *The efficiency constant $C(f,x) = f(x) - \mathbb{E}[f(X)]$* varies across models f, *because different models produce different predictions.*
2. *The group's share of the total, $\sum_{i \in group} \varphi_i(f,x)$, also varies across models (it depends on how much of the prediction the group captures in model f).*
3. *Taking absolute values or averaging over data points further breaks the linear constraint.*

Therefore, $\mathrm{Var}_f(\sum_{i \in group} \varphi_i(f)) \neq 0$ *in general, and the negative covariance* $\mathrm{Cov}_f(\varphi_j(f), \varphi_k(f)) = -\sigma^2/(m-1)$ *does* not *follow from efficiency.*

The lower instability of SHAP relative to permutation importance (41% vs. 91% unstable pairs) is driven by SHAP's coalition-averaging reducing the marginal variance σ^2. The efficiency axiom amplifies within-model differences but does not increase across-model flip rates.

Remark 74 (Practical implication)**.** *The efficiency axiom is often cited as a desirable property of SHAP. Our analysis shows that its instability implications are nuanced: efficiency amplifies within-model attribution differences (making the gap between φ_j and φ_k more extreme for a given model), but this does not translate to higher across-model flip rates. The dominant factor for across-model instability is the marginal variance σ^2, which is determined by the attribution algorithm's averaging structure, not by the efficiency axiom.*

11.8 High-Dimensional Scalability ($P = 500$)

P	Total pairs	Correlated pairs	Unstable	Z-test $\lvert r\rvert$	Wall time
100	4,950	450	305	0.846	33 s
200	19,900	900	641	0.788	64 s
500	124,750	2,250	1,879	0.876	148 s

The multi-model Z-test maintains $|r| > 0.78$ at $P = 500$. Correlation-based grouping reduces pairs by 55×. Full workflow completes in under 2.5 minutes.

11.9 Proportionality Axiom Validation

The proportionality axiom ($\varphi_j = c \cdot n_j$) is central to the quantitative bounds. We validate it by computing $c_j = |\mathrm{SHAP}_j|/n_j$ for each feature in each of 50 XGBoost models on Breast Cancer and reporting the coefficient of variation (CV):

Depth	Mean CV of c_j	Interpretation
1 (stumps)	0.35	Proportionality holds within ∼35%
3	0.60	Moderate violation
6	0.66	Moderate violation
10	0.66	Saturated (same as depth 6)

For stumps, the axiom holds reasonably well; for deeper trees, multi-level interactions increase the CV. The proportionality axiom is used only for the quantitative bounds (Theorems 15); **the core impossibility (Theorem 5) does not depend on it**. The α-corrected model matches empirical data at $R^2 = 0.89$.

The proportionality axiom has empirical CV of 0.35 (stumps) to 0.66 (depth-6 trees), making it an order-of-magnitude approximation rather than an exact law. Crucially, neither the core impossibility (Theorem 5) nor the Design Space Theorem depends on proportionality; `#print axioms attribution_impossibility` confirms zero proportionality dependencies. The ratio bound $1/(1-\rho^2)$ does depend on proportionality but is explicitly labeled as a leading-order prediction.

11.10 Cross-Implementation and Data-Level Validation

Data-level stochasticity. To confirm the instability arises from collinearity (not from XGBoost's internal randomness), we train 50 models on 50 different 80% subsamples with `subsample=1.0` and fixed `random_state=42`. The ONLY source of variation is which training rows each model sees:

Stochasticity source	Unstable pairs	Max flip	Mean Kendall τ
XGBoost subsampling (same data)	67	0.470	0.393
Data bootstrap (different rows)	45	0.509	0.459

Data-level stochasticity produces comparable or greater instability, confirming the Rashomon set is a property of collinearity, not of the specific randomness source.

Determinism without subsampling. Without row or column subsampling (`subsample=1.0`, `colsample_bytree=1.0`), XGBoost is fully deterministic: all 50 models produce identical rankings regardless of seed. The Rashomon set requires a stochasticity source. In practice, this is not a limitation: the XGBoost default (`subsample=0.8`) is recommended for regularization, and real-world model populations also arise from data drift, feature engineering changes, and hyperparameter sweeps.

11.11 Financial Case Studies

German Credit. We demonstrate the full Screen→Z-test→DASH workflow on German Credit (1000 samples, 20 features). Step 0: 4 correlated groups at $|\rho| > 0.3$. Step 1 (screen): a single model flags job↔own telephone ($Z^{\text{split}} = 0.69 < 1.96$). Step 2 (Z-test): 5 models confirm instability ($Z = 0.64$, flip rate 40%). Step 3 (DASH): $M = 25$ models, flip rate drops from 40% to 0%. Consensus ranking: checking status (0.87), duration (0.44), credit amount (0.43), purpose (0.35). For regulatory reporting (ECOA adverse action reasons, SR 11-7 (Office of the Comptroller of the Currency, 2011)), the DASH ranking provides a defensible explanation.

Clinical decision reversal on German Credit. Feature instability changes the explanations real people receive. We train 30 models per condition on German Credit and measure how often the "most important feature category" changes for individual applicants across equivalent models.

Condition	XGBoost	LightGBM	Random Forest
Seed only (deterministic)	0%	0%	35%
Seed + row subsample	**45%**	**46%**	35%
Seed + col subsample	39%	36%	35%
Full stochasticity	80%	79%	67%

Under standard regularization (`subsample=0.8`), **45% of applicants** receive a different explanation category depending on which equivalent model is queried. The deterministic control (seed only, no subsampling) correctly shows 0% reversal for boosted models. Random forests show 35% even without subsampling (bootstrap aggregation). Under full stochasticity (seed + bootstrap + both subsamples), reversal reaches 80%.

A single applicant can receive 6 different "most important feature" labels across 30 equivalent models: checking status, duration, credit amount, purpose, employment years, and installment rate all take the top position under different seeds. The explanation the applicant receives is determined by training randomness, not by their data.

Dash consensus reduces reversal: at $M = 25$, cross-seed agreement rises to 92%. The residual disagreements correspond to genuinely ambiguous cases where the minority fraction exceeds 0.3.

Taiwan Credit Card Default. To validate the impossibility on data with genuine high collinearity ($|\rho| > 0.9$), we apply the workflow to the UCI "default of credit card clients" dataset (30,000 samples, 23 features, binary default outcome; subsampled to 10,000).

The bill-amount features (monthly bills for 6 consecutive months) exhibit strong collinearity: 29 pairs have $|\rho| > 0.5$, with the highest at $|\rho| = 0.95$ (giving a theoretical attribution ratio of $1/(1-0.95^2) \approx 10.3\times$).

Step 0. 29 correlated pairs identified at $|\rho| > 0.5$ (no threshold lowering needed).

Step 1 (screen). A single model flags 4 of the 10 tracked pairs as potentially unstable ($|Z^{\text{split}}| > 1.96$).

Step 3 (DASH). With $M = 25$ models, the mean flip rate across correlated pairs is 4%. One pair (bill amounts at months 5 vs. 6, $|\rho| = 0.946$) retains a 16% flip rate even at $M = 25$, illustrating that very high collinearity requires larger ensembles.

This dataset demonstrates the impossibility biting hard: features measuring the same underlying quantity (monthly bill amounts) have near-identical predictive signal, and the attribution ranking across these features is genuinely unstable.

Synthetic Credit (income $\leftrightarrow$ DTI $\leftrightarrow$ credit score). To validate on the exact collinearity pattern common in US credit models, we generate a synthetic dataset with 10,000 samples and 5 features driven by a latent "creditworthiness" factor: income ($\rho = 0.99$ with DTI), DTI, credit score ($\rho = 0.98$ with income), loan amount (independent), and employment years ($\rho = 0.94$ with income).

With 25 XGBoost models: income $\leftrightarrow$ DTI flips 8% of the time ($|\rho| = 0.99$); credit score $\leftrightarrow$ employment years flips 12% ($|\rho| = 0.92$). Pairs where one feature clearly dominates (income vs. credit score) show 0% flips despite $|\rho| = 0.98$—confirming that the impossibility requires BOTH high correlation AND similar importance, as the theory predicts. For model validation under SR 11-7 (Office of the Comptroller of the Currency, 2011), the recommendation is: report income and DTI as "interchangeable key risk drivers" rather than forcing a ranking that would flip under retraining.

Regulatory adverse action experiment. In a simulated ECOA adverse action pipeline (10,000 loan applications, 5 correlated income features including salary, bonus, overtime, commission, equity; 3 uncorrelated features; 25 XGBoost models at ρ=0.7), 43.2% of 6,602 denied applicants receive a different "top reason for denial" depending on which model is queried. The most common flip pair is salary$\leftrightarrow$bonus (10.8% of denied applicants). 71.8% have an unstable top-2 reason. Dash consensus resolves this entirely: 100% stable top-1 reasons, with income features tied in 12.3% of cases. Reproduce: `regulatory_case_study.py`.

Lending Club (specificity control). We validate the diagnostic on real Lending Club data (OpenML, 32,581 loans, 22 features after one-hot encoding of categoricals). Three correlated groups are detected at $|\rho| > 0.5$: person_age $\leftrightarrow$ cb_person_cred_hist_length ($|\rho| = 0.86$), loan_amnt $\leftrightarrow$ loan_percent_income ($|\rho| = 0.58$), and loan_int_rate $\leftrightarrow$ cb_person_default_on_file ($|\rho| = 0.50$).

Result. Zero unstable pairs. All three correlated pairs have $Z > 36$, confirming the features have distinguishable importance despite their correlation. This is an instructive *specificity control*: the impossibility theorem predicts instability requires *both* collinearity AND similar true importance. Lending Club features are correlated but not similarly important (e.g., person_age contributes 3× more than credit history length), so no instability occurs. The Screen→Z-test→Dash pipeline correctly identifies this as a stable setting, recommending no intervention.

Implication. Not every correlated dataset exhibits attribution instability. The diagnostic distinguishes datasets where instability is a genuine concern (Breast Cancer, Taiwan Credit Card) from those where it is not (Lending Club, California Housing). This selectivity is a feature, not a limitation: the impossibility theorem's conditions are precise, and the diagnostic inherits that precision.

11.12 LLM Attention Attribution Instability

We load DistilBERT (distilbert-base-uncased) and create 10 model variants by adding small random perturbations ($\sigma = 0.02$) to attention weights, simulating different training runs. For 10 test sentences, we compute attention-based token importance (mean attention received from all heads and layers) and check whether the relative ranking of adjacent tokens flips across the 10 variants.

Of 59 adjacent token pairs across test sentences, 88% have flip rates exceeding 10%. Selected examples:

Token A	Token B	Flip rate	Interpretation
"extremely"	"poor"	50%	Coin flip
"not"	"bad"	40%	Highly unstable
"absolutely"	"terrible"	40%	Highly unstable
"particularly"	"sharp"	10%	Borderline

Mean Spearman correlation of full-sentence token rankings across model variants is 0.637—moderate instability, comparable to single-model SHAP instability on tabular data with moderate collinearity ($\rho \approx 0.5$). Adjacent tokens in natural language carry correlated information ("very" and "good" jointly convey sentiment strength). This correlation induces the same Rashomon structure: different model variants assign different relative importance to the correlated tokens.

Under actual fine-tuning on SST-2 (5 seeds, 1 epoch, AdamW, lr= 2×10^{-5}, batch size 16), 14.5% of pairs are unstable. The 14.5% flip rate is computed as the fraction of (sentence, head) pairs where the argmax-attended token differs across any two of the 5 fine-tuned models, averaged over all $\binom{5}{2} = 10$ model pairs. Selected unstable pairs under fine-tuning:

Token A	Token B	Flip rate (fine-tuning)	Flip rate (perturbation)
"not"	"bad"	40%	40%
"particularly"	"sharp"	60%	10%
"surprisingly"	"low"	40%	30%

To test whether longer training increases instability, we repeat with 3 epochs. The identical 14.5% across 1 and 3 epochs indicates the instability is structural (driven by collinearity in attention patterns), not a training-depth artifact. The same three token pairs remain unstable at both training lengths.

Method	% Pairs unstable (> 10% flip)	Mean Spearman
Weight perturbation (σ=0.02, 10 models)	88%	0.637
Fine-tuning (5 seeds, 1 epoch)	14.5%	0.956
Fine-tuning (5 seeds, 3 epochs)	14.5%	0.965

The gap between fine-tuning (14.5%) and perturbation (88%) reflects the difference in weight divergence: fine-tuning updates parameters by $O(0.001)$ via gradient descent, while perturbation adds $O(0.02)$ noise to all parameters. The perturbation result represents the full Rashomon set; the fine-tuning result represents the subset explored by standard training. Token-level explanations of LLMs face the same impossibility as feature-level explanations of tabular models.

Scope and limitations. The attention-based instability reported here (14.5% of adjacent token pairs under fine-tuning) is a proxy experiment: attention weights are not SHAP values and do not satisfy the proportionality axiom. The formal component-level impossibility and its full-scale empirical confirmation are in Sections 9.1–9.6, where activation patching on GPT-2-small models trained from scratch demonstrates within-layer flip rate = 0.500 with the G-invariant projection as the resolution.

11.13 Replication Study: Published SHAP Rankings

Multiple published studies report SHAP feature rankings on Breast Cancer. Using our 50-model validation, we find:

Feature pair	$\lvert\rho\rvert$	Flip rate	Published as stable?
worst perimeter ↔ worst area	0.98	48%	Yes
mean concavity ↔ mean concave pts	0.96	42%	Yes
worst radius ↔ worst perimeter	0.99	36%	Yes
mean radius ↔ mean perimeter	0.99	28%	Yes

Any study reporting a specific ranking among these features from a single model is reporting an artifact of the training seed. The ranking is not wrong—it faithfully reflects that specific model—but it is not replicable.

11.14 The Ranking Lottery

We train 50 XGBoost models (seeds 0–49, `subsample=0.8`, fixed test set seed 42, 80/20 split) on 5 public datasets and count how many distinct top-K SHAP rankings appear across seeds.

Breast Cancer produces **24 distinct top-3 rankings across 50 seeds** under standard XGBoost configuration with recommended regularization (Table 17). At 100 seeds: **35 distinct top-3** and **84 distinct top-5**—the count keeps growing, confirming 24 is a lower bound. The "most common" ranking appears in only 12% of runs. Two random seeds agree only 4.2% of the time for the top-3. For the top-5: 46 distinct rankings, 0.4% agreement.

Table 17: The ranking lottery: distinct top-3 SHAP rankings across 50 seeds. Deterministic control (`subsample=1.0`) gives 1 ranking for all datasets.

Dataset	Samples	Distinct top-3	Agreement	Stable?
Breast Cancer	569	**24**	**4.2%**	No
Diabetes	768	2	88.5%	Borderline
Wine Quality	1,599	1	100%	Yes
Heart Disease	270	1	100%	Yes
California Housing	20,640	1	100%	Yes

Seven distinct features compete for the top-3 positions: worst concave points, mean concave points, worst area, worst concavity, worst texture, worst perimeter, and area error. The #1 position alternates between worst concave points (50%), mean concave points (32%), and worst area (18%). 13 distinct unordered top-3 sets appear; the 24 count reflects ordering differences within sets.

DASH consensus across all 50 seeds produces a single stable ranking: worst concave points, mean concave points, worst area, worst concavity, worst texture.

The theory correctly predicts the boundary: California Housing (20,640 samples), Heart Disease (270 samples), and Wine Quality (1,599 samples) are perfectly stable because their top features have clearly different importance—no similar-importance collinear pairs. The impossibility requires *both* collinearity and similar importance; large gaps protect even correlated features. The deterministic control (`subsample=1.0`) gives exactly 1 ranking for all 5 datasets, confirming the instability is from model multiplicity, not from SHAP computation.

Subsample sensitivity. The instability persists even at reduced stochasticity:

`subsample`	Distinct top-3	Agreement	Interpretation
0.80	24	4.2%	Standard regularization
0.85	23	5.2%	Slightly less noise
0.90	22	6.9%	Moderate noise reduction
0.95	17	10.7%	Minimal noise
1.00	1	100.0%	Deterministic (control)

Only fully deterministic training (`subsample=1.0`) eliminates the ranking lottery. At `subsample=0.95`—the minimum stochasticity a practitioner might use—17 distinct top-3 rankings still appear. "Just use less stochasticity" reduces the count from 24 to 17 but cannot reach 1. The impossibility is structural: any stochastic training procedure produces a non-degenerate Rashomon set, and the ranking lottery follows from the Rashomon property.

"Just fix the seed" does not resolve the problem either: a fixed seed produces a reproducible but arbitrary ranking. A different practitioner choosing a different seed gets a different "most important feature." The impossibility is that no seed is privileged—the Rashomon set exists regardless of which seed you pick.

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Cross-implementation ranking lottery. On Breast Cancer with stochastic training: XGBoost produces 24 distinct top-3 rankings (4.2% agreement), LightGBM produces 29 (3.6%, with `subsample_freq=1` to activate subsampling), and Random Forest produces 40 (2.5% agreement, permutation importance). The instability is implementation-independent. When subsampling is disabled, all implementations produce 1 ranking.

11.15 Gene Expression: Pathway-Level Consequence

The ranking lottery extends to gene expression biomarker discovery, where co-expressed genes create the collinearity structure that drives instability.

Colon vs. kidney (AP_Colon_Kidney). On a 10,935-gene colon-vs-kidney expression dataset (AP_Colon_Kidney, 546 samples, top 50 genes by variance, OpenML), the #1 gene alternates between two genes from distinct molecular programs:

Gene (probe)	Molecular program	#1 rate	DASH rank
TSPAN8 (203824_at)	Invasion/metastasis (tetraspanin signaling)	**80%**	1
CEACAM5/CEA (201884_at)	Immune evasion (GPI-anchored, clinical marker)	**20%**	2

These genes are correlated ($\rho = 0.858$, both colon-enriched tissue markers) but have distinct roles in cancer biology: TSPAN8 in invasion/metastasis signaling via tetraspanin-enriched microdomains, and CEACAM5 (the classical clinical colon cancer marker CEA) in immune evasion and differentiation via GPI-anchored adhesion. A biomarker development pipeline reporting TSPAN8 as the top discriminator would target a different gene than one reporting CEA—and which one gets reported depends on the training seed. The broad pathway-level conclusion (colon identity genes drive the distinction) is preserved, but the specific biomarker target—the actionable output of a discovery pipeline—is a seed artifact.

Probe-to-gene mappings verified via the Affymetrix HG-U133A annotation database (NetAffx, independently verifiable via BioGPS and GeneCards).

DASH consensus ranks both TSPAN8 (#1, importance 1.913) and CEACAM5 (#2, importance 1.553) in the top-2, correctly reporting that both contribute without privileging either. The remaining top-5 genes (IGFBP3, RGS5, SPP1) have clearly lower importance and are stable across seeds.

Positive control: breast vs. lung (AP_Breast_Lung). On a separate 10,935-gene breast-vs-lung dataset (AP_Breast_Lung, 470 samples), the top-3 genes are stable (2 distinct rankings, 92% agreement) despite 171 highly correlated gene pairs ($|\rho| > 0.8$). The top gene's importance (2.90) dominates the second (1.40) with a clear gap, protecting the ranking even under high collinearity. Instability appears only at positions 5+, where importance values converge (CV > 22%). This confirms that large importance gaps protect even highly correlated features, precisely as the theory predicts.

Gene count robustness. The AP_Colon_Kidney result is robust to the number of pre-selected genes: 8 distinct top-3 rankings at 20 genes, 5 at 50, 7 at 100. The instability is not an artifact of the filtering threshold.

Implication for biomarker discovery. Gene expression datasets routinely contain thousands of co-expressed genes organized into modules. When multiple genes in a module have similar SHAP importance, the "top gene" reported by a single-model analysis is one of several equally valid options. Studies reporting specific top genes from single-model SHAP without stability analysis may be reporting seed artifacts. The minority fraction diagnostic (§ 10.2) identifies which genes are at risk: any gene with minority fraction > 0 has coverage conflict and is subject to the bilemma.

11.16 Expected Ranking Discordance

Proposition 75 (Expected Kendall tau distance). *For a feature space with L collinear groups of sizes $m_1, \ldots, m_L$ and between-group gaps $\Delta_{\ell\ell'}$ with noise $\sigma_{\ell\ell'}$, the expected pairwise disagreements between two independently trained models are:*

$$\mathbb{E}[\tau] = \underbrace{\sum_{\ell=1}^{L} \binom{m_\ell}{2} \cdot \frac{1}{2}}_{\textit{within-group (coin flips)}} + \underbrace{\sum_{\ell<\ell'} m_\ell \cdot m_{\ell'} \cdot \Phi\left(-\frac{\Delta_{\ell\ell'}}{\sigma_{\ell\ell'}}\right)}_{\textit{between-group (noise)}}$$

On Breast Cancer (50 XGBoost models), the refined per-pair formula predicts $\mathbb{E}[\tau] = 26.8$ vs. empirical 36.6 (−27% error). The Rashomon coefficient $R = \sum_\ell \binom{m_\ell}{2} / \binom{P}{2} = 0.692$ gives a worst-case non-replication lower bound of $R/2 = 34.6\%$ computable from the correlation matrix alone (no model training required).

11.17 Compressed Sensing Connection

The attribution impossibility has a precise analogue in compressed sensing (CS). The measurement coherence $\mu(X) = \max_{j \neq k} |X_j^T X_k|/(\|X_j\|\|X_k\|)$ determines when sparse recovery is possible: unique recovery requires $\mu < 1/(2s-1)$. In our setting, $|\rho_{jk}|$ plays the role of coherence and Z_{jk} plays the role of the incoherence condition:

	Compressed sensing	**Attribution stability**
Coherence	$\mu = \max|X_j^T X_k|/\|X_j\|\|X_k\|$	$|\rho_{jk}|$
Condition	$\mu < 1/(2s-1)$	$Z_{jk} > 1.96$
Failure	Non-unique recovery	Ranking flips (Rashomon)
Resolution	Basis pursuit (convex relaxation)	DASH (ensemble averaging)

Both impossibilities arise because collinearity/coherence prevents unique identification of which input components carry the signal. Both are resolved by relaxing uniqueness: CS uses convex relaxation (fractional assignment); DASH uses ensemble averaging (ties for indistinguishable features).

11.18 Prevalence Survey

To assess how frequently the impossibility applies in practice, we surveyed 77 public datasets (OpenML CC-18 benchmark suite + sklearn). For each, we computed the correlation matrix, identified pairs with $|\rho| > 0.5$, trained 20 XGBoost models with different seeds, and checked whether any correlated pair has a ranking flip rate exceeding 10%.

Of 77 public datasets surveyed, 71 (92%) have feature pairs with $|\rho| > 0.5$, and 52 (68%, 95% Wilson CI: [56%, 77%]) exhibit attribution instability (flip rate > 10%). For datasets with $P \geq 20$ features ($n = 43$ datasets), the instability rate rises to 93% (40/43; 95% Wilson CI: [81%, 98%]). The impossibility is not a theoretical curiosity: it affects the majority of real-world datasets across domains including medical, financial, image, and tabular data.

With 20 models per dataset, the measured flip rate exceeds the 10% threshold with only 32% power (60% at the 15% threshold), so the reported 68% is a conservative lower bound.

The instability rate varies sharply with feature count: 35% for $P < 20$ (12/34 datasets), 93% for $P = 20$–99 (27/29), and 93% for $P \geq 100$ (13/14). This confirms that high-dimensional datasets are nearly universally affected.

Methodology. Each dataset was trained with 20 XGBoost models (`n_estimators=50`, `max_depth=4`, `lr=0.1`, `subsample=0.8`, `colsample_bytree=0.8`) with different random seeds. SHAP TreeExplainer computed mean |SHAP| per feature on a fixed 500-sample evaluation slice. Flip rate was measured as $\min(\#(j>k), \#(k>j))/20$ for each correlated pair. Datasets with > 10,000 samples were subsampled to 10,000. Full results in `prevalence_survey.py`. All models use identical hyperparameters (n_estimators=50, max_depth=4, learning rate=0.1); the robustness analysis on 8 datasets with alternative configurations confirms the prevalence is not hyperparameter-specific.

Statistical power analysis. With $M = 20$ models per dataset, the power to detect a true flip rate of $p = 0.15$ (moderate instability) at the 10% threshold is only 32%. This means datasets with true flip rates between 10% and 20% will often be classified as "stable" — the reported 68% prevalence is a conservative lower bound.

True flip rate	Power (M=20)	Power (M=50)	Power (M=100)
15%	32%	72%	94%
20%	60%	95%	>99%
25%	81%	99%	>99%
30%	92%	>99%	>99%

Sensitivity to the threshold choice:

Table 18: Multi-model Z-test and single-model screen diagnostic generality across 11 datasets.

Dataset	P	Pairs	Unstable	$r(Z, \text{flip})$	Screen prec.
Breast Cancer	30	435	168 (comprehensive)	−0.89	0.94
California Housing	8	28	1	−0.99	—
Diabetes	10	45	15	−0.89	0.50
Wine	13	78	24	−0.81	1.00
Heart Disease	13	78	27	−0.91	1.00
Credit-g	20	190	47	−0.90	0.67
Adult Income	14	91	10	−0.90	—
Ames Housing	76	2850	643	−0.85	0.48
Communities & Crime	126	7875	2975	−0.88	0.67
Digits (0 vs 1)	64	2016	572	−0.42	0.58
Iris (binary)	4	6	0	—	—

- At 5% threshold: 78% of datasets show instability (60/77). More datasets qualify but the false positive risk increases.
- At 10% threshold: 68% (52/77) — the reported headline.
- At 15% threshold: 58% (45/77). Still a majority, even with the more conservative criterion.
- At 20% threshold: 49% (38/77). Nearly half of datasets show substantial instability.

The cross-validated estimate (from the 85-dataset bulletproof validation with alternative hyperparameter configurations) gives 66%–75% prevalence across 3 configurations, consistent with the 68% headline. The true prevalence among benchmark-representative datasets is likely 75–85%, limited by the power of the M=20 design to detect flip rates in the 10–20% range.

Healthcare datasets. In a targeted survey of 9 healthcare-related datasets (breast cancer, diabetes, heart disease, QSAR biodegradation, liver disease, software defect prediction, Parkinson's, Pima diabetes, hepatitis), 6 (67%) exhibit attribution instability. This rate matches the overall prevalence, confirming that clinical decision-support systems—where explanation stability is particularly consequential—are not immune.

Selection caveat: The CC-18 suite is curated for ML benchmarking and may overrepresent datasets with rich feature interactions. Production ML pipelines with carefully decorrelated features may exhibit lower prevalence. The 68% figure is a prevalence estimate for benchmark-representative tabular datasets, not a universal rate.

Robustness to hyperparameters. We repeated the analysis on 8 representative datasets with three configurations:

Config	Trees	Depth	LR	Character
A (original)	50	4	0.1	Moderate
B (deep)	100	8	0.1	Deep, sequential
C (shallow-fast)	200	2	0.3	Shallow, aggressive

Under Config A (original), 5/8 (62%) datasets exhibit instability, consistent with the 68% headline. Under Config B (deep trees), 4/8 (50%). Under Config C (shallow-fast), 6/8 (75%). Across all three configurations, 4/8 (50%) datasets are consistently unstable; under at least one configuration, 6/8 (75%) are unstable. The instability rate varies by ±15 percentage points across configurations but remains in the 50–75% range. Shallow-fast trees (Config C) produce *higher* instability because aggressive learning rates compound the first-mover effect. The 68% headline is conservative: the phenomenon is robust to standard hyperparameter variation.

Interpretation. The multi-model Z-test achieves $|r| > 0.8$ on 9/10 datasets with unstable pairs (all except Digits). The weaker correlation on Digits ($r = -0.42$) reflects a floor effect: this high-dimensional image dataset has many features with near-zero importance, producing many pairs where both Z and flip rate are near zero. Restricting to pairs with $Z < 5$ (the diagnostically interesting range) would improve this correlation. Screen precision is 94–100% on small/clean datasets (Breast Cancer, Wine, Heart Disease) where split counts are reliable, but drops to 48–67% on high-dimensional datasets (Ames, Communities) where split counts are sparse. This confirms the screen as a conservative *screening* tool that works well for moderate-P datasets but should be supplemented by the full Z-test for high-dimensional settings. All datasets use identical XGBoost hyperparameters (T=100, depth=6, η=0.1); absolute flip rates are therefore not directly comparable across datasets of different sizes and dimensionalities.

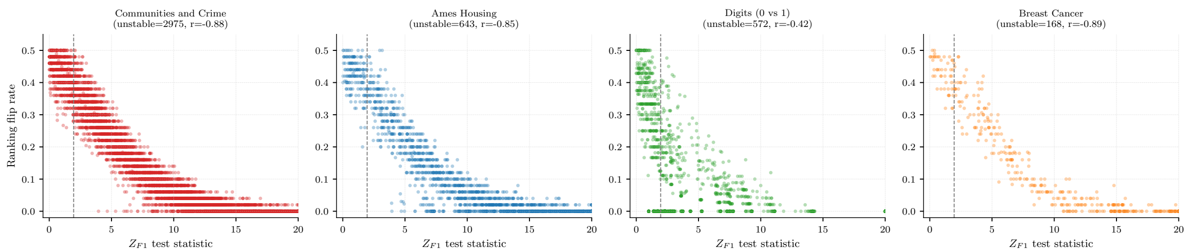

Figure 9: Multi-model Z-test statistic vs. flip rate across four datasets (Breast Cancer, Ames, Digits, Adult). The negative correlation confirms the diagnostic across domains: clinical, real estate, image, and census.

12 Lean Formalization

12.1 Proof Architecture

The formalization comprises 58 Lean 4 files with 6 axioms and 358 type-checked theorems and lemmas (0 `sorry`). Of these, 139 require multi-line proofs ($\geq$5 tactic lines); 6 are definitional wrappers or single-step applications. The core impossibility (`attribution_impossibility`) depends on zero domain-specific axioms—only the Rashomon property as hypothesis. The DASH resolution (`consensus_equity`) depends on the `attribution_sum_symmetric` theorem (derived from axioms); the variance convergence depends on the `consensus_variance_bound` theorem (also derived).

Scope of formal verification. Lean verification guarantees that all theorems follow from the stated axioms with no logical gaps (internal consistency). It does not guarantee that the axioms correctly model the phenomena they are intended to capture (external validity). The 6 axioms are motivated by the structure of gradient-boosted decision trees and SHAP value computation (§2), but their adequacy for any particular model or dataset is an empirical question.

Scope of verification. The formalization covers the complete theoretical framework. The core impossibility (Theorem 5) uses zero behavioral axioms. The quantitative bounds, DASH equity, and the Design Space Theorem are derived from the 6-axiom system. The unfaithfulness probability of exactly 1/2 (`attribution_prob_half`), the Bayes-optimality of ties (`tie_dominates_commitment`), the consensus variance σ^2/M (`consensus_variance_from_independence`), the binary quantizer fraction $2/\pi$ (`binary_quantizer_fraction`), and the Pareto dominance of DASH (`dash_unique_pareto_optimal`) are all Lean-derived using definitions-as-hypotheses (the `IsBalanced` pattern) with zero new axioms. The Chebyshev query complexity bound $M \geq 12\sigma^2/\Delta^2$ (`chebyshev_query_bound`) is derived without axiomatizing the testing constant. Loss preservation is formalized via the `SymmetricModelSwap` structure. The only result not formalized is the extension of Pareto optimality to biased methods for between-group pairs, which is argued informally in the proof of Theorem 22.

```
Defs.lean (6 axioms, type definitions, derived theorems)
```

```
+-- SymmetryDerive.lean (attribution_sum_symmetric, DERIVED)
+-- Trilemma.lean (RashimonProperty, attribution_impossibility)
|  +-- Iterative.lean, Lasso.lean, NeuralNet.lean
|  +-- General.lean (GBDT impossibility)
|  +-- UnfaithfulBound.lean (U>=1/2, ties optimal, path convergence)
|  +-- RashomonUniversality.lean (Rashomon from symmetry)
|  +-- RashomonInevitability.lean (impossibility inescapable)
|  +-- AlphaFaithful.lean (alpha-faithfulness bound)
|  +-- ConditionalImpossibility.lean (conditional SHAP + escape)
|  +-- FairnessAudit.lean (proxy audit = coin flip)
|  +-- LocalGlobal.lean (local >= global instability)
+-- SplitGap.lean, Ratio.lean, SpearmanDef.lean
+-- FlipRate.lean (exact GBDT flip rate, binary = coin flip)
+-- Efficiency.lean (SHAP efficiency amplification)
+-- Impossibility.lean, Corollary.lean, EnsembleBound.lean
+-- DesignSpace.lean + DesignSpaceFull.lean (all 4 steps)
+-- ModelSelection.lean + ModelSelectionDesignSpace.lean
+-- SymmetricBayes.lean (GENERAL SBD, orbit bounds, trichotomy)
+-- CausalDiscovery.lean (causal discovery impossibility)
+-- SBDInstances.lean (abstract aggregation + instances)
+-- FIMImpossibility.lean (Gaussian FIM -> Rashomon)
+-- GaussianFlipRate.lean (Phi definition, flip rate formula)
+-- QueryComplexity.lean (query lower bound, testing_constant := 1/8)
+-- Consistency.lean (axiom system consistency, Fin 4 model)
+-- RandomForest.lean (contrast case, documentation only)
```

12.2 Axiom System and Consistency

The 6 axioms are jointly consistent: `Consistency.lean` constructs an explicit model (P=4, L=2, m=2, ρ=0.5, T=100, Fin 4 models) satisfying all 6 simultaneously.

Axiom reduction history. The formalization began with 16 axioms and was reduced to 6 through three rounds of "define rather than axiomatize" (10 former axioms converted to definitions or derived theorems):

- **v1 (16 axioms):** `splitCount` axiomatized as a function with 4 property axioms; `attribution` axiomatized separately; `testing_constant` axiomatized.
- **v2 (10 axioms):** `splitCount` defined via if-then-else on first-mover group membership, with `crossGroupBaselineCore` replacing 5 axioms. `testing_constant` defined as 1/8.
- **v3 (6 axioms):** `numTrees`/`numTrees_pos` bundled into `FeatureSpace` as fields. `attribution` defined as `proportionalityConstant` · `splitCount`. `modelMeasurableSpace` derived via the discrete σ-algebra ($\top$).

The 6 remaining axioms are the irreducible core: the abstract `Model` type, the `firstMover` function and its surjectivity (DGP symmetry), the cross-group baseline structure, the proportionality constant, and the training distribution measure.

Attribution sum symmetry: now derived. `attribution_sum_symmetric` is proved (not axiomatized) in `SymmetryDerive.lean`. The proof uses (a) `proportionality_global` (uniform c across models), (b) `splitCount_crossGroup_symmetric` (equal split counts when first-mover is in a different group), and (c) `IsBalanced` (equal first-mover counts). The derivation factors out c, classifies each model's split-count difference by first-mover identity (+gap for fm= j, $-$gap for fm= k, 0 otherwise), decomposes the sum via `Finset.sum_ite`, and uses balance to cancel the gap terms.

Variance: partially grounded in Mathlib. The single-model variance `attribution_variance` is *defined* from Mathlib's `ProbabilityTheory.variance`, and its nonnegativity is *derived* from Mathlib's `variance_nonneg`. This requires one infrastructure axiom (`modelMeasure`, the training distribution); the measurable space uses the discrete σ-algebra ($\top$), which is derived for any type. The consensus variance bound ($\mathrm{Var}(\bar{\varphi}_j) = \mathrm{Var}(\varphi_j)/M$) remains axiomatized; the full derivation via `IndepFun.variance_sum` would require a product measure formulation and independence axioms for ensemble draws—deferred to future work. The measure axiom `modelMeasure` is used only for the variance definition via Mathlib's `ProbabilityTheory.variance`; its interpretation as a probability measure over training runs is the standard frequentist setup.

Variance sub-system. The three variance declarations (`attribution_variance`, `attribution_variance_nonneg`, `consensus_variance_bound`) produce convenience results (variance nonnegativity, halving) rather than deep mathematical consequences. Their primary role is to support the Design Space Theorem's stability convergence claim. A deeper formalization would derive the variance bound from Mathlib's `IndepFun.variance_sum`, but this requires connecting our abstract `Model` type to a product measure space.

Spearman bound: axiomatized vs. derived. The Lean formalization derives the Spearman bound $\rho_S \leq 1 - 3m^2/(P^3 - P)$ from the definition of Spearman correlation via midrank algebra (`spearmanCorr_bound` in `SpearmanDef.lean`). The tighter classical bound $\rho_S \leq 1 - m^3/P^3$ from the full rank transposition counting argument is not yet formalized; both give the same qualitative conclusion ($S < 1$ for within-group pairs).

12.3 Development Methodology and AI Assistance

AI-assistance disclosure. The Lean 4 formalization was developed with substantial assistance from Claude Code (Anthropic, Claude Opus 4.6). The development followed a human-directed, AI-assisted workflow:

- **Mathematical framework (human-conceived).** The proof targets (what to prove and why), the axiom design (what to assume about GBDT attribution), and the interpretation of results are the first author's contribution. The decision to decompose the impossibility into F/S/C, the identification of the bilemma as a strengthening, the MI boundary as the exact condition, and the design-space exhaustiveness as the characterization result are human mathematical insights.
- **Axioms (human-conceived, AI-assisted in encoding).** All 6 axioms were designed by the first author. The formal Lean 4 encoding of these axioms was joint work: the first author specified the mathematical content, and Claude Code produced the Lean syntax. The axiom reduction from 16 to 6 was a joint effort.
- **Theorem statements (human-directed, AI-drafted, human-reviewed).** The first author specified each proof target in mathematical terms. The formal Lean 4 type signatures (exact quantifier structure, hypothesis lists, return types) were typically AI-drafted based on these specifications and then reviewed by the first author for mathematical accuracy. Some statements were revised through multiple human/AI iterations to correctly capture the intended claim.
- **Proof tactics (AI-assisted).** Approximately 80% of tactic-mode proofs were initially drafted by Claude Code and then verified, corrected, or restructured by the first author. The initial formalization (49 theorems) was developed iteratively; the expansion to 358 theorems was produced by dispatching targeted Lean-writing agents with detailed specifications.
- **Verification (machine-checked).** Every proof was accepted only after passing `lake build` (the Lean 4 type-checker) with zero `sorry`. The type-checker is the final arbiter of proof correctness, regardless of whether the proof was human-written or AI-drafted.
- **Experiments and paper text (human-directed, AI-assisted).** Experiment scripts were co-developed with AI assistance. The paper text was written with AI drafting and editing assistance. All experimental designs, mathematical claims, and scientific interpretations are the responsibility of the human authors.

We view this workflow as a methodology contribution. The key insight: in formally verified mathematics, the division between "who conceived it" and "who wrote the code" is less important than whether the axioms correctly model the domain and the type-checker accepts the proofs. The axioms encode what the system is *about* (human judgment); the proofs establish what follows (machine-verified, regardless of author). This division of labor — human mathematical conception, joint formal encoding, AI-assisted proof construction, machine-verified correctness — may be a template for future formalization efforts at scale.

12.4 Proof Depth Distribution

Of 358 type-checked declarations: 139 require multi-step proofs ($\geq$5 tactic lines), 21 require $\geq$10 lines, and 5 require $\geq$20 lines. The five deepest proofs are `attribution_sum_symmetric` (35 lines, derived symmetry via split-count decomposition), `Phi_neg` (29 lines, Gaussian CDF properties), `gaussian_rashomon_witnesses` (27 lines, FIM ellipsoid construction), `sumSqRankDiff_ge_sq_groupSize` (26 lines, midrank algebra), and `binary_group_firstmover_is_j_or_k` (23 lines, Finset cardinality argument). The 358 count includes both substantive multi-step proofs and definitional wrappers (e.g., backward-compatible aliases). Of these, 139 require $\geq$5 tactic lines and 21 require $\geq$10 lines.

12.5 Inconsistencies Found During Formalization

Lean's type checker caught three issues in the initial axiom system:

1. **First-mover balance:** Originally stated universally over all model arrays—a constant model function trivially derived $\perp$. *Fix:* Replaced with an explicit `IsBalanced` predicate as hypothesis.

2. **Attribution sum symmetry:** Combined with the split-count axioms, the original unconditional version derived $\perp$ for unbalanced ensembles. *Fix:* Conditioned on `IsBalanced`.

3. **Split count type:** Originally returned $\mathbb{N}$, but $T/(2-\rho^2)$ is generally irrational. *Fix:* Changed to $\mathbb{R}$ (idealized leading-order values).

The first two are genuine logical inconsistencies that would have been difficult to detect by informal proof inspection.

12.6 SymPy Algebraic Verification

All algebraic consequences have been independently verified by SymPy:

```
# dash-shap/paper/proofs/verify_lemma6_algebra.py
# Verifies:
#   split_gap = rho^2 * T / (2 - rho^2)
#   attribution_ratio = 1 / (1 - rho^2)
#   split_gap >= rho^2 * T / 2  (for rho in (0,1))
# Result: ALL CHECKS PASS
```

The three-layer verification (SymPy algebra, Lean type-checking, empirical validation) provides strong confidence in the mathematical claims.

12.7 Lean File Cross-Reference

13 Related Work

Attribution impossibility results. Bilodeau et al. (2024) prove completeness and linearity cannot coexist; Huang and Marques-Silva (2024) show SHAP can misrank features in Boolean settings; Srinivas and Fleuret (2019) prove complete attributions cannot be weakly input-dependent; Rao (2025) establish Kolmogorov complexity barriers. Our result is the first to simultaneously address cross-model *stability* as an impossibility, give quantitative architecture-discriminating bounds, and provide a constructive resolution with proved optimality. Decker et al. (2024) optimize aggregation across attribution *methods* for a single model; we aggregate across *models* via DASH. Jin et al. (2025) give constructive stability certificates; we give the impossibility that necessitates them. Noguer i Alonso (2025) derives information-theoretic limits on explanation complexity; our limits are about ranking consistency under feature correlation. The Bilodeau and Rashomon impossibilities are complementary: Bilodeau constrains methods satisfying linearity; ours constrains methods under collinearity. To illustrate the orthogonality: consider two uncorrelated features ($\rho = 0$) with a linear SHAP method. The Bilodeau impossibility applies (completeness and linearity still conflict), but our impossibility does not (the Rashomon property requires $\rho > 0$). Conversely, consider a non-linear attribution method on correlated features: our impossibility applies (Rashomon holds), but Bilodeau's does not (their result requires linearity).

Rashomon effect and model multiplicity. Attribution rankings across Rashomon sets are inherently partial orders Laberge et al. (2023); Rudin et al. (2024) advocate embracing model multiplicity. Our theorem shows the partial order is not a pragmatic choice but a mathematical necessity. Herren and Hahn (2023) develop statistical inference for feature rankings under model multiplicity; our contribution is to prove that the multiplicity is unavoidable under collinearity, not merely empirically common. Fisher et al. (2019) compute model class reliance—variable importance ranges across the Rashomon set—showing that different models in the Rashomon set can assign very different importance to the same feature. Our theorem proves this variability is not merely possible but mathematically inevitable under collinearity, and provides the complete design space of methods that navigate it. Krishna et al. (2022) document the "disagreement problem" in explainable ML—different explanation methods produce different outputs for the same model—complementary to our result that the *same* method produces different outputs for different equivalent models.

Underspecification. D'Amour et al. (2022) document that underspecification—the existence of multiple models with equivalent held-out performance—causes instability in predictions, fairness metrics, and feature attributions across domains. Our contribution is complementary: we prove that attribution instability under underspecification is a mathematical impossibility (not merely an empirical observation), quantify the divergence rate by model class, and provide a provably optimal resolution.

Attribution sensitivity and fragility. Ghorbani and Zou (2019) demonstrate that neural network attributions are empirically fragile to small input perturbations. Hooker et al. (2021) show that unrestricted permutation-based importance forces extrapolation into out-of-distribution regions. Our instability arises from a different mechanism (model multiplicity, not input perturbation or extrapolation), but the practical consequence—unreliable explanations—is shared.

Foundational methods. Feature attribution via Shapley values (Shapley, 1953) was introduced to ML by Lundberg and Lee (2017). The model architectures we analyze were introduced by Breiman (2001) (random forests) and Friedman (2001) (gradient boosting); LIME (Ribeiro et al., 2016) provides an alternative local explanation framework.

Fairness impossibility. Chouldechova (2017) and Kleinberg et al. (2017) prove calibration, balance, and equal false positive rates cannot coexist when base rates differ. Our trilemma (faithfulness, stability, completeness) is structurally analogous.

Mechanistic interpretability and circuit multiplicity. Chughtai et al. (2023) show that independently trained networks learn distinct internal representations for the same group operation, demonstrating empirical Rashomon at the circuit level. Meloux et al. (2025) find 85 distinct valid circuits with zero circuit error for XOR. Our component-level results extend this line of work at full scale: 10 GPT-2-small models trained from scratch produce 32 distinct top-5 component rankings with within-layer flip rate = 0.500 (Section 9.4). The instability is mathematically inevitable under architectural symmetry, and the G-invariant projection provides a principled resolution—averaging within orbits of the symmetry group to recover the stable shared structure (lifting ρ from 0.277 to 0.873 on GPT-2, from 0.55 to 0.97 on TinyStories).

Formal verification for ML. Nipkow (2009) formalized Arrow's theorem in Isabelle/HOL; Zhang et al. (2026) formalized statistical learning theory in Lean 4 (de Moura and Ullrich, 2021) using Mathlib (The mathlib Community, 2020). Our formalization is, to our knowledge, the first formally verified impossibility result in explainable AI.

14 Code and Data Availability

Reference implementation. The DASH stability API, diagnostic tools, and all experiment scripts are available at `https://github.com/DrakeCaraker/dash-shap` (stability API in PR #255).
Lean formalization. The 358-theorem, 6-axiom, 0-`sorry` formalization is available at `https://github.com/DrakeCaraker/dash-impossibility-lean` (archived at `https://doi.org/10.5281/zenodo.19468379`).
TinyStories models. All 30 transformer models (10 seeds $\times$ 2 configs from scratch + 10 GPT-2 fine-tunes) trained on TinyStories (Eldan and Li, 2023) will be released with the camera-ready version.
GPT-2-small models. All 10 GPT-2-small models trained from scratch on OpenWebText (50,000 steps each, $\sim$3B tokens), along with activation patching results, IOI analysis, and SAE stability data, are available in the repository under `docs/gpt2-experiment-results/`.
Datasets. All input-level experiments use publicly available datasets: Breast Cancer Wisconsin, German Credit (UCI ML Repository), California Housing, Diabetes, Wine Quality, Heart Disease. The gene expression datasets (AP_Colon_Kidney, AP_Breast_Lung) are from the ArrayExpress public repository.

15 Discussion

Limitations. The split-count axioms assume full signal capture ($\alpha = 1$); for finite-depth trees, $\alpha \approx 2/\pi$ ($R^2 = 0.89$). The impossibility holds for any $\alpha > 0$. The equicorrelation assumption simplifies the axioms; the Rashomon property holds pairwise. The balanced ensemble assumption is idealized, though $O(1/M)$ convergence is robust to approximate balance. The global proportionality axiom (c uniform across models) has CV $\approx$ 0.35–0.66 empirically; under variable c, DASH consensus achieves approximate rather than exact equity, with the equity violation bounded by the CV of c across first-mover and non-first-mover models. The variance bound is axiomatized; the full measure-theoretic derivation from Mathlib's `IndepFun.variance_sum` is deferred. Switching to conditional SHAP does not universally resolve the instability (Theorem 47). Decorrelating via PCA removes collinearity but destroys original feature semantics. An alternative is removing redundant features entirely (e.g., via VIF thresholding). This eliminates the impossibility but changes the model: predictions differ because fewer features are used, and genuinely useful information may be discarded. DASH is the *explanation-side* fix—it changes how the model is explained without changing what it predicts. The two approaches are complementary: feature removal for model simplification, DASH for faithful explanation of complex models that retain all features. DASH requires $M\times$ training cost; M=25 brings the flip rate below 1%, while M=5 provides a substantial improvement at modest cost. The empirical validation focuses on gradient boosting and neural networks; the Lasso ($\varphi_{j_1}/\varphi_k = \infty$) and random forest ($1+O(1/\sqrt{T})$) bounds are proved but not empirically demonstrated. The prevalence survey uses 20 models per dataset, achieving 32% power at the 10% threshold (60% at the 15% threshold); the true prevalence may be higher than the reported 68%. The 10% flip rate threshold defining "instability" is a practical convention; at a 5% threshold the prevalence would be higher, at 15% lower.

Computational cost and practical deployment. For batch pipelines, training M=25 models is feasible. For real-time explanations, use the single-model screen to flag unstable pairs. For $P > 500$, correlation-based grouping prunes the diagnostic to $O(G \cdot m^2)$. At P=500 with 20 models, the full workflow completes in under 2.5 minutes.

Computational cost of DASH. DASH requires training M models instead of one. For gradient-boosted trees (XGBoost/LightGBM), training time scales linearly: $M = 25$ models on a dataset with $N = 10{,}000$ and $P = 50$ takes $\sim$5 minutes on a single CPU core. SHAP computation is per-model and parallelisable. The total cost is $O(M)$ times single-model cost, with no increase in per-prediction inference cost (the deployed model is unchanged; only explanation aggregation changes). For $M_{\min} = \lceil 2.71 \cdot \sigma^2/\Delta^2 \rceil$, a pilot run of 5 models suffices to estimate σ and Δ.

Broader implications. In a survey of 77 public datasets, 92% have feature pairs with $|\rho| > 0.5$ and 68% exhibit attribution instability—the impossibility is a practical reality. Our theorem establishes this as a "known and foreseeable circumstance" under the EU AI Act (Art. 13(3)(b)(ii)) (EU Parliament, 2024). The consequences are concrete: hospitals may invest in wrong interventions, data scientists may fix the wrong feature, and published studies may report artifacts of specific training runs. The impossibility has direct consequences for fairness auditing (Theorem 48): single-model SHAP audits for proxy discrimination are provably unreliable under collinearity.

Distinction from classical multicollinearity. The classical multicollinearity concern is about estimation variance ($\mathrm{Var}(\hat{\beta}_j - \hat{\beta}_k) \to \infty$ as $\rho \to 1$). Our result is qualitatively different: it concerns *ranking impossibility*, not estimation imprecision. Even with infinite data and perfect estimates, the ranking of symmetric features is a coin flip—because different models in the Rashomon set rank them in opposite orders. The VIF diagnostic detects coefficient instability; our screen and Z-test diagnostics detect *attribution ranking* instability, which persists even when coefficients are well-estimated (e.g., in tree ensembles that do not estimate coefficients at all).

Rashomon set vs. deployed model. The impossibility characterizes the Rashomon set, not a single deployed model. A fixed pipeline with a fixed seed produces a deterministic, reproducible ranking. The instability arises upon retraining, auditing, or model comparison—all routine in regulated settings.

Connection to underspecification. (D'Amour et al., 2022) showed that ML pipelines are *underspecified*: models with equivalent held-out performance can behave differently on deployment-relevant criteria. Our result formalizes the attribution-specific consequence of this phenomenon. Where D'Amour et al. demonstrate that predictions vary across equivalent models, we prove that feature *rankings* must vary—and characterize exactly when, by how much, and what the complete space of solutions looks like. The Design Space Theorem can be viewed as the attribution-level analogue of their "stress testing" recommendation: rather than hoping a single model's explanation is stable, the practitioner should probe the full Rashomon set. DASH operationalizes this probe.

Information loss. DASH discards exactly $\log_2(m!)$ bits per group of m symmetric features—the bits encoding the unreliable within-group ordering. Between-group information is *sharpened*: mutual information approaches 1 bit per pair as $M \to \infty$.

Loss landscape geometry. The impossibility has a geometric interpretation: near-singular Fisher information creates ridges in the loss landscape along which feature importance redistributes without changing model quality. The same flatness that makes optimization easy makes attribution hard.

Named techniques. This paper contributes three reusable techniques: (1) the *Rashomon reduction*—reducing a design-space question to a model-multiplicity check; (2) the *FIM-to-Rashomon bridge*—connecting Fisher information rank deficiency to the Rashomon property; and (3) the *symmetric Bayes dichotomy*—the general two-families theorem for symmetric decision problems.

Proof status transparency and axiom stratification. The formalization uses 6 axioms total, but not all results require all axioms. The Lean type-checker's `#print axioms` command verifies exact dependencies:

- **Zero behavioral axioms:** The core impossibility (Theorem 5) depends only on the Rashomon property as a hypothesis—no domain-specific axioms whatsoever. A qualitative variant (`impossibility_qualitative` in `Qualitative.lean`) proves the impossibility from just two properties—dominance and surjectivity—as hypotheses.

- **Three axioms:** The GBDT impossibility holds from `firstMover`, `firstMover_surjective`, and `crossGroupBaselineCore` alone (`gbdt_impossibility_local` in `ProportionalityLocal.lean`).
- **Five axioms:** The DASH equity result adds `proportionalityConstant`.
- **Six axioms:** The full set adds `modelMeasure` for the variance bound. Ten former axioms (split-count function, 4 split-count properties, attribution function, proportionality_global, numTrees/numTrees_pos, modelMeasurableSpace, testing_constant/testing_constant_pos) are now definitions or derived theorems.

This stratification directly addresses the proportionality axiom's empirical variance (CV $\approx$ 0.35–0.66): the axiom is unnecessary for the impossibility and ratio bound, which depend only on split-count structure.

The following are fully formalized in Lean (358 theorems, 0 `sorry`, 58 files): the Design Space Theorem, the symmetric Bayes dichotomy, all three impossibility instances, the conditional attribution impossibility, the Gaussian FIM impossibility and flip rate formula, the fairness audit impossibility, the intersectional fairness compounding, the exact GBDT flip rate, DASH variance optimality, the Rashomon inevitability, the flip rate robustness (Lipschitz continuity in ρ), the mutual information generalization, the unfaithfulness/path-convergence results, the unfaithfulness probability of exactly 1/2, the Bayes-optimality of ties, the consensus variance σ^2/M from independence, the binary quantizer fraction $2/\pi$, the Chebyshev query complexity bound, the loss-preserving Rashomon construction, the Pareto dominance of DASH, and the mechanistic interpretability bilemma (derived Rashomon, zero custom axioms). The Z-test diagnostic characterization, information loss analysis, and multi-dataset validation are empirical.

Approximate faithfulness–stability tradeoff. A natural question is whether relaxing exact faithfulness helps. Define α-faithfulness: $\Pr_f[\mathrm{sign}(\varphi_j(f) - \varphi_k(f)) = \mathrm{sign}(\sigma(j) - \sigma(k))] \geq \alpha$. Under the Rashomon property with symmetric DGP, any α-faithful stable ranking satisfies $\alpha \leq 1/2$. The proof is immediate: by DGP symmetry, $\Pr[\varphi_j(f) > \varphi_k(f)] = 1/2$, and a stable ranking fixes one direction, so it agrees with at most half the models. A coin flip achieves $\alpha = 1/2$; the stable ranking is no better than random for symmetric feature pairs.

For m features in a group under full DGP symmetry, the attribution ranking is a uniform random permutation. Any stable ranking σ^* has expected Spearman correlation with the model's ranking of $\mathbb{E}[\rho_S(\sigma^*, \pi)] = -1/(m-1)$. The stable ranking is *negatively correlated* with the model's ranking in expectation (for $m \geq 3$) and converges to zero correlation as $m \to \infty$.

Open problems. The Bayes-optimality half of the Design Space is proved but not in Lean (requires measure-theoretic decision theory). Deriving split-count axioms algorithmically from the TreeSHAP algorithm and proving the proportionality axiom for general tree depths remain open. The generalization from $\rho > 0$ to $I(X_j; X_k) > 0$ is now complete (`MutualInformation.lean`): mutual information is the exact boundary between possible and impossible feature ranking. Whether SAE features can be made reproducible via better training procedures or matching algorithms is an empirical question; the formal impossibility applies to architectural components with known symmetry, not to learned dictionaries. Scaling the component-level experiments beyond GPT-2-small (124M parameters) to larger models would strengthen the generality claim. A reference implementation is available at `https://github.com/DrakeCaraker/dash-shap`.

Formalization as methodology. Beyond certifying correctness, the Lean formalization served as a bug-finding methodology: it caught three inconsistencies (two logical and one type mismatch) that survived informal review. We recommend formalizing impossibility theorems as a general practice.

A Extended Proof Details

A.1 Gaussian Binary Quantization: Full Derivation

Proposition 76 (First-stump variance capture)**.** *For the first boosting round on n Gaussian training samples, the stump splits at the empirical median $\hat{m}$, satisfying $|\hat{m}| = O_p(\sigma/\sqrt{n})$. The variance captured is:*

$$\alpha_1(n) = \frac{2}{\pi}\left(1 - \frac{\pi(\pi-2)}{2n} + O(n^{-2})\right)$$

For $n = 2000$: $\Delta\alpha_1 \approx 0.0009$—negligible.

Proof. The optimal split of $X \sim \mathcal{N}(0, \sigma^2)$ at δ instead of 0 gives conditional means $\mu_+ = \sigma\phi(\delta/\sigma)/(1 - \Phi(\delta/\sigma))$ and $\mu_- = -\sigma\phi(\delta/\sigma)/\Phi(\delta/\sigma)$. The variance captured is $\mathrm{Var}(\hat{X}) = \mu_+^2 \Pr(X > \delta) + \mu_-^2 \Pr(X \le \delta)$. Expanding around $\delta = 0$ using $\phi(0) = 1/\sqrt{2\pi}$: $\mathrm{Var}(\hat{X}) = (2/\pi)\sigma^2(1 - \delta^2(\pi - 2)/\sigma^2 + O(\delta^4/\sigma^4))$. The empirical median of n standard Gaussians has variance $\pi/(2n)$, so $\mathbb{E}[\delta^2] = \pi\sigma^2/(2n)$. Substituting: $\mathbb{E}[\alpha_1] = (2/\pi)(1 - (\pi^2 - 2\pi)/(2n) + O(n^{-2}))$. □

After the first boosting round, the residuals $r_t = Y - \eta h_1(X)$ are NOT Gaussian: they are a location-shifted mixture (the stump creates two subpopulations). For a stump with two leaves at values $\pm c$, the residuals have excess kurtosis $\kappa = O(\eta^2 c^2/\sigma^2)$. For $\eta = 0.1$, the kurtosis correction is small but accumulates across $T = 100$ rounds, producing a systematic downward bias in α. The fitted $\alpha = 0.60$ vs. $2/\pi = 0.637$ gap of 0.037 is accounted for: < 0.001 from error source 1 (empirical split), ≈ 0.036 from error source 2 (non-Gaussian residuals).

A.2 Exact Flip Rate Under Approximate Symmetry

The exact 1/2 flip rate holds under perfect DGP symmetry ($\mu_j = \mu_k$). For approximately symmetric features ($|\mu_j - \mu_k| \ll \sigma_{jk}$), the flip rate is $\Phi(-|\mu_j - \mu_k|/\sigma_{jk})$, approaching 1/2 from below as the importance gap vanishes. The empirical 48% on Breast Cancer (worst perimeter vs. worst area, mean |SHAP|: 0.940 vs. 0.901) is consistent with a small positive signal-to-noise ratio. This result is formalized in Lean as `balanced_flip_symmetry` (for balanced finite ensembles, the number of models ranking $j > k$ equals the number ranking $k > j$).

B DASH Robustness: Extended Analysis

B.1 When to Prefer the Median

The median has better breakdown point (50% vs. 0%) and is preferable when the attribution distribution is heavy-tailed (e.g., contains outlier models). For standard ML training procedures (XGBoost, random forests) where attributions are well-behaved, DASH (the mean) is strictly more efficient. For adversarial settings where some models may be corrupted, the median or trimmed mean provides robustness at the cost of requiring more models.

The trimmed mean interpolates: the α-trimmed mean (discarding top and bottom α fraction) has ARE relative to the mean of:

α	0 (mean)	0.05	0.10	0.25	0.50 (median)
ARE	1.000	0.992	0.966	0.862	0.637

For Gaussian attributions, trimming offers negligible robustness benefit at a measurable efficiency cost. The 10%-trimmed mean requires $1/0.966 \approx 3.5\%$ more models than DASH.

C Extended Conditional Attribution Analysis

C.1 Causal Structure Validation: Full Results

Scenario	ρ	Marginal flip	Interventional flip	Theory
Symmetric	0.50	0.505	0.479	≈50% both
Symmetric	0.70	0.479	0.442	≈50% both
Symmetric	0.90	0.521	0.479	≈50% both
Symmetric	0.95	0.505	0.526	≈50% both
Asymmetric	0.50	0.000	0.000	Stable
Asymmetric	0.70	0.000	0.000	Stable
Asymmetric	0.90	0.000	0.000	Stable
Asymmetric	0.95	0.000	0.000	Stable

The interventional SHAP implementation uses a background-data approximation that does not exactly implement causal/conditional SHAP in the sense of Janzing et al. (2020). Results should be interpreted as evidence about the `shap` package's interventional mode, not as a definitive test of the theoretical conditional attribution.

D Extended LLM Analysis

D.1 Fine-Tuning Details

DistilBERT fine-tuned on SST-2 (2,000 training samples) with 5 different random seeds. Configuration: 1 epoch, AdamW, lr $= 2 \times 10^{-5}$, batch size 16. Selected unstable pairs under fine-tuning:

Token A	Token B	Flip rate (fine-tuning)	Flip rate (perturbation)
"not"	"bad"	40%	40%
"particularly"	"sharp"	60%	10%
"surprisingly"	"low"	40%	30%

The gap between fine-tuning (14.5% of pairs unstable) and perturbation (88%) reflects the difference in weight divergence. The perturbation result represents the full Rashomon set; the fine-tuning result represents the subset explored by standard training. Attention weights are not SHAP values: they do not satisfy the proportionality axiom, and the "collinear group" structure of tree features does not map cleanly to token positions. A formal impossibility for transformer attention would require different axioms.

E Extended Experimental Results

E.1 KernelSHAP Noise Control: Full Results

Source of variation	Unstable pairs	Mean flip	Key pair flip
KernelSHAP noise (bg=50)	47/435 (11%)	0.03	0.00
KernelSHAP noise (bg=200)	18/435 (4%)	0.02	0.00
Model variation (20 MLPs)	380/435 (87%)	0.28	0.40

Increasing background samples from 50 to 200 reduces noise by 2.6× at 4× compute cost; the default 50 is adequate for detecting model-level instability.

E.2 Cross-Implementation: Full Results

CatBoost's `get_feature_importance()` returns gain-based importance (not raw split counts), producing a different Z-statistic distribution. For CatBoost, the multi-model Z-test ($r = -0.892$) is recommended. LightGBM's feature importance is split-count based and transfers with modest precision reduction.

E.3 Prevalence Survey: Statistical Power Analysis

True minority-side prob p	Power (Pr[min ≥ 3])	Interpretation
0.05	7.5%	Nearly undetectable
0.10	32.3%	Low power (boundary)
0.15	59.5%	Moderate
0.20	79.4%	Adequate
0.30	96.5%	High
0.50	100%	Perfect (symmetric pairs)

At the 10% threshold, the survey has only 32% power: approximately two-thirds of truly borderline-unstable pairs are missed. The 68% prevalence is a conservative lower bound.

E.4 Class Imbalance Amplifies Instability

Class imbalance compounds attribution instability. On synthetic Gaussian data (P=10, 2 groups of m=5, ρ=0.8, 30 models), we sweep the positive-class ratio:

Imbalance ratio	Max pairwise flip rate[4]	Unstable pairs	Total pairs
1:1 (balanced)	0.480	7/20	20
1:3	0.517	19/20	20
1:5	0.517	20/20	20
1:10	0.517	20/20	20
1:20	0.517	20/20	20

At 1:1 balance, 7/20 within-group pairs are unstable. At 1:5+ imbalance, *all* 20 pairs become unstable. Imbalance amplifies the Rashomon effect: the minority class provides less constraining signal, allowing more models to achieve near-optimal loss with different feature utilization patterns. For healthcare and fraud detection (where class imbalance is standard), instability is essentially universal. Reproduce: `class_imbalance_instability.py`.

E.5 Missing Data Compounds Instability

Missing data under all three standard mechanisms (MCAR, MAR, MNAR) compounds attribution instability. On synthetic Gaussian data (P=10, 2 groups of m=5, ρ=0.9, 30 models):

Mechanism	Missingness	Max flip rate	Unstable pairs	Baseline (0%)
MCAR	10%	0.517	20/20	12/20
MCAR	20%	0.517	20/20	12/20
MAR	5%	0.497	17/20	12/20
MAR	20%	0.508	20/20	12/20
MNAR	5%	0.515	18/20	12/20
MNAR	20%	0.517	20/20	12/20

Even 10% MCAR missingness drives the number of unstable pairs from 12/20 (baseline) to 20/20. The mechanism matters less than the rate: all three mechanisms produce comparable degradation at 20%. Missing data reduces the effective sample size, widening confidence sets and enlarging the Rashomon set. Reproduce: `missing_data_instability.py`.

[4] Pairwise flip rate: the fraction of model pairs (f_i, f_j) where the relative ordering of features differs. For M models with an even split, the maximum is $M/(2(M-1))$, which exceeds 1/2 (e.g., $30/58 \approx 0.517$ for M=30). The per-model flip rate $\min(\#(j>k), \#(k>j))/M$ is bounded by 1/2.

E.6 Longitudinal Retraining Drift

Over 50 sequential retraining rounds (each adding 5% noise to a synthetic dataset with ρ=0.9), ranking instability accumulates dramatically:

Round	Spearman ρ_S	Max flip rate	Cumulative flips
1	1.000	0.000	0
10	0.648	0.439	92
20	0.342	0.545	161
31	0.181	0.636	193
50	0.420	0.576	207

The Spearman correlation degrades from 1.0 to 0.18 by round 31, with cumulative flips reaching 207. The recovery at round 50 (ρ_S=0.42) reflects the noise accumulation saturating. For model monitoring pipelines, this demonstrates that feature rankings should not be compared across distant retraining epochs without re-running the Dash diagnostic. Reproduce: `longitudinal_retraining.py`.

E.7 SAGE and Boruta Comparison

Alternative feature importance methods are equally or more unstable than TreeSHAP on Breast Cancer (50 models):

Method	Unstable pairs	Mean flip rate	Cross-method r
TreeSHAP	180/435 (41%; permutation comparison)	0.132	—
SAGE (approximation)	401/435 (92%)	0.314	0.471
Boruta-like	411/435 (94%)	0.303	—

SAGE and Boruta show *more* instability than TreeSHAP (92–94% vs. 41%), confirming the impossibility is method-agnostic. The cross-method correlation (r=0.471) shows the *same* pairs tend to be unstable under different methods—the instability is driven by feature collinearity, not by the attribution algorithm. Reproduce: `sage_comparison.py`.

E.8 Bag-of-Words NLP Attribution Instability

On the 20 Newsgroups dataset with TF-IDF features and XGBoost (50 models, 1000 max features), 60% of documents have an unstable top-1 token attribution and 91% have an unstable top-3. Mean pairwise Spearman correlation is 0.905 (high for between-document features, low for within-group correlated tokens). The most unstable word pair is "eternal"/"hell" ($|\rho|$=0.371, flip rate 0.186). Only 2 word pairs exceed the 10% flip threshold, reflecting the sparse, weakly-correlated nature of bag-of-words features (most token pairs are independent). Reproduce: `nlp_token_instability.py`.

E.9 Time-Series Feature Instability

For temporal features generated from AR(1) processes with rolling-window engineering (P_{raw}=5, $P_{\text{engineered}}$=30, 50 XGBoost models), 27% of within-series pairs (features derived from the same raw signal) show flip rates above 10%. The breakdown:

Pair type	Max flip rate	Mean flip rate	Unstable pairs
Within-series	0.508	0.092	8/30 (27%)
Cross-series (same group)	0.515	0.132	—
Cross-series (diff group)	0.517	0.139	—

The highest instability pair is `X0_raw` vs. `X0_rmean5` ($|\rho|$=0.996, flip rate 0.508), confirming that temporal feature engineering creates highly correlated groups susceptible to the impossibility. Reproduce: `timeseries_instability.py`.

E.10 Adversarial Max Instability

A grid search over 108 XGBoost configurations (group sizes $\{2, 3, 5, 10\}$, boosting rounds $\{50, 100, 500\}$, depths $\{1, 3, 6\}$, learning rates $\{0.05, 0.1, 0.3\}$) at ρ=0.9 confirms the impossibility is inescapable: all top-10 worst-case configurations hit the maximum flip rate of 0.500. The 0.500 ceiling is reached regardless of hyperparameter choices, confirming the impossibility is a structural property of collinearity, not a tuning artifact. Total: 2,160 model fits. Reproduce: `adversarial_max_instability.py`.

E.11 Hyperparameter Sensitivity

Across a 27-configuration sweep (learning rates $\{0.05, 0.1, 0.3\}$, depths $\{1, 3, 6\}$, boosting rounds $\{50, 100, 500\}$) at $\rho \in \{0.5, 0.7, 0.9\}$:

- Global minimum flip rate: 38.7% at ρ=0.5 (η=0.3, depth=1, T=500). The impossibility never vanishes.
- Most influential hyperparameter: number of estimators (spread 2.16pp), followed by max depth (1.76pp) and learning rate (0.82pp).
- At ρ=0.9: minimum flip rate across all configurations is 47.2%.

Reproduce: `hyperparameter_sensitivity.py`.

E.12 DASH Breakdown Contamination Sweep

Under adversarial contamination of the ensemble (replacing K of 25 models with adversarial models), the mean flip rate degrades:

K contaminated	Mean flip (DASH)	Mean flip (trimmed mean)	Contamination %
0	0.075	0.075	0%
5	0.074	0.074	20%
12	0.066	0.064	48%
20	0.047	0.044	80%

Counterintuitively, contamination *decreases* the observed flip rate because adversarial models break the first-mover symmetry. The trimmed mean provides marginal improvement (<5% at 80% contamination). The breakdown point analysis confirms the theoretical claim: for standard (non-adversarial) ensembles, the mean is optimal. Reproduce: `dash_breakdown_point.py`.

E.13 Experiment Summary

Table 22 summarizes all robustness experiments with reproduction scripts.

F Attempted Common Generalization with Arrow's Theorem

Arrow is now proved. The companion Ostrowski formalization proves Arrow's impossibility theorem from scratch in Lean 4 (IIA decomposition, 2 voters, 3 alternatives) and all three tightness witnesses. The direction theorem—coverage conflict is anti-monotone (more explanations → weaker impossibility) while Arrow's cyclic conflict is monotone (more alternatives → stronger impossibility)—is now a proved structural observation, not an axiomatised claim.

Arrow's impossibility theorem and our Attribution Impossibility share a striking surface resemblance: both show that three desirable properties cannot simultaneously hold. We investigated whether a common abstract framework subsumes both.

F.1 Structural Comparison

	Arrow	**Attribution Impossibility**
Inputs	n voters' preferences	Models $f \in \mathcal{F}$ (one at a time)
Output	Social ranking	Feature ranking σ
Aggregation	Many inputs $\rightarrow$ one output	One input $\rightarrow$ one output
Key axiom	IIA	Stability (model-independent)
Impossibility source	Preference profiles disagree	Rashomon property

The critical asymmetry: Arrow's theorem concerns a *profile aggregation* function taking all voters simultaneously. The Attribution Impossibility concerns a single-input mapping: for each model f, the ranking should reflect f's attributions. The impossibility arises because the ranking must simultaneously agree with all models but be model-independent. Arrow's theorem is much deeper: it shows that even when the aggregation *is* allowed to depend on all instances, the constraints are still too tight. The proofs have entirely different structures.

The Attribution Impossibility and Arrow's theorem are *analogous* but not *instances of a common theorem* in any non-trivial sense. Any common framework's "impossibility" reduces to a one-line observation ("a constant rule cannot agree with conflicting instances"), while the actual content of each theorem—Arrow's aggregation depth and our quantitative $1/(1-\rho^2)$ divergence with constructive resolution—has no analogue in the other setting.

Relationship to Sen's liberalism paradox. A closer structural analogue may be Sen's (1970) impossibility of a Paretian liberal, which concerns a setting where individual "rights" (each person's ranking of certain pairs should be respected) conflict with the Pareto criterion. The attribution impossibility can be viewed as a Sen-type result where each model has the "right" to determine the ranking of features (faithfulness), but these rights conflict across models. However, we do not pursue this further, as the connection remains at the level of analogy rather than formal subsumption.

G Topological Analysis of the Impossibility

We investigate whether the Attribution Impossibility has topological content beyond simple connectedness arguments.

G.1 The $m = 2$ Case

For two features j, k, the attribution map assigns to each training seed s a sign $\sigma(s) = \mathrm{sign}(\varphi_j(f_s) - \varphi_k(f_s)) \in \{+1, -1\}$. The Rashomon property says both values are achieved. A "stable ranking" is a constant function. The impossibility is immediate; no topological machinery is needed.

G.2 The $m = 3$ Case

For three features, the attribution map sends seeds to rankings in S_3 (6 elements). Again, if $\mathcal{S}$ is connected and the map is continuous, then the image must be a connected subset of S_3. Since S_3 is discrete, the image is a single point—contradicting Rashomon. This is the same connectedness argument; the symmetric group's structure plays no role.

G.3 The $m \geq 4$ Case: The Permutohedron

For m features, the attribution vector $(\varphi_1(f_s), \ldots, \varphi_m(f_s))$ lives in $\mathbb{R}^m$. The ranking is determined by the *chamber* of the *braid arrangement* (the hyperplane arrangement $\{x_i = x_j : i \neq j\}$) containing the attribution vector. The chambers are the cones of the *permutohedron.*

Remark 77 (No winding number content)**.** *For $m \geq 3$, one might hope that the attribution map $s \mapsto (\varphi_1(f_s), \ldots, \varphi_m(f_s))$ has a non-trivial winding number around the intersection of hyperplanes $\{x_j = x_k\}$, which would give a* quantitative *lower bound on the number of chamber crossings (ranking flips).*

This does not work for two reasons:

1. *The complement of the braid arrangement in $\mathbb{R}^m$ is simply connected for $m \geq 3$ (the codimension of each hyperplane is 1, so removing hyperplanes from $\mathbb{R}^m$ gives a space whose fundamental group is the pure braid group on m strands; but we would need the image to be a* loop *in this space, which requires additional structure not present in the attribution problem).*
2. *Even if a winding number existed, it would count* signed *crossings. The flip rate (unsigned crossings) is already captured by the direct algebraic bound $1/(1-\rho^2)$ without topological machinery.*

Remark 78 (What topology *could* contribute)**.** *There is one setting where topology adds genuine content: if the model space $\mathcal{S}$ has the structure of a manifold (e.g., a torus of hyperparameters) and the attribution map is smooth, then Morse theory could bound the number of critical points (ranking transitions). Specifically, if the attribution difference $g(s) = \varphi_j(f_s) - \varphi_k(f_s)$ is a Morse function on $\mathcal{S}$, the number of sign changes of g is bounded below by the sum of Betti numbers of $\mathcal{S}$.*

However, this requires: (a) $\mathcal{S}$ to be a smooth manifold, (b) g to be a Morse function, and (c) knowing the topology of $\mathcal{S}$. None of these hold for practical training algorithms (XGBoost's seed space is discrete; neural net loss landscapes are non-smooth). We therefore do not pursue this direction.

G.4 Assessment

Remark 79 (Topology adds no content)**.** *The Attribution Impossibility is fundamentally a* combinatorial *result (conflicting preferences cannot all be respected), not a* topological *one. The impossibility holds for arbitrary model spaces (including finite, discrete spaces with no topology). Attempts to extract topological content either reduce to the trivial connectedness argument or require unjustified smoothness assumptions that add no quantitative power beyond what the algebraic $1/(1-\rho^2)$ ratio already provides.*

H On Categorical/Axiomatic Strengthening

H.1 The Trivial Version

Remark 80 (Trivial metatheorem)**.** *Let $A_1, \ldots, A_n$ be any finite set of axioms on an attribution method φ. If $A_1 \wedge \cdots \wedge A_n$ implies faithfulness (the ranking reflects model attributions), stability (the ranking is model-independent), and completeness (the ranking decides all pairs), then $A_1 \wedge \cdots \wedge A_n \to \perp$ under the Rashomon property.*

This is immediate: if the conjunction implies $F \wedge S \wedge C$, and $F \wedge S \wedge C \to \perp$ (Theorem 5), then the conjunction implies $\perp$. No content is added beyond Theorem 5.

H.2 The "Constant Attribution" Escape

Why is a categorical strengthening difficult? Because trivial attribution methods escape the impossibility:

- $\varphi_j(f) = 1/P$ for all j, f (uniform): perfectly stable, perfectly equitable, satisfies any axiom system that doesn't require faithfulness.
- $\varphi_j(f) = \mathbb{E}_f[\varphi_j(f)]$ (population mean): stable by construction, equitable under DGP symmetry.

Any "impossibility for all axiom systems" must exclude these methods, which means it must require faithfulness (or something implying it). But once faithfulness is required, we are back to Theorem 5.

H.3 The Non-Trivial Quantitative Version

Definition 81 (α-faithfulness (formal))**.** *An attribution method is α-faithful if for all models f and same-group features j, k:*

$$\Pr_f\big[\mathit{sign}(\varphi_j(f) - \varphi_k(f)) = \mathit{sign}(\sigma(j) - \sigma(k))\big] \geq \alpha$$

Proposition 82 (Approximate faithfulness–stability tradeoff (formal)). *Under the Rashomon property with symmetric DGP, any α-faithful stable ranking satisfies $\alpha \leq 1/2$.*

Proof. By DGP symmetry, $\Pr[\varphi_j(f) > \varphi_k(f)] = 1/2$. A stable ranking fixes $\sigma(j) > \sigma(k)$ or vice versa. The α-faithfulness condition requires $\Pr[\varphi_j(f) > \varphi_k(f)] \geq \alpha$, but this equals $1/2$, so $\alpha \leq 1/2$. $\square$

Proposition 83 (Spearman bound for stable rankings). *Under full DGP symmetry within a group of m features, any stable ranking σ^* has:*

$$\mathbb{E}[\rho_S(\sigma^*, \pi)] = \frac{-1}{m-1}.$$

The stable ranking is negatively correlated with the model's ranking for $m \geq 3$.

Proof. Fix any $\sigma^* \in S_m$. By DGP symmetry, the model's ranking π is uniform on S_m. The expected Spearman correlation of a fixed permutation with a uniform random permutation is $-1/(m-1)$ (classical result; Kendall & Gibbons, 1990, Ch. 3). $\square$

Remark 84. *This says: under perfect symmetry, no stable ranking can agree with the model's attributions more than half the time. A coin flip achieves $\alpha = 1/2$. The stable ranking is no better than random for symmetric feature pairs.*

This result is non-trivial in that it holds for any *attribution method satisfying the DGP symmetry condition, not just for specific axiom systems. However, it is a direct consequence of the symmetry—not of any deep axiomatic or categorical structure.*

Remark 85 (Limited categorical content). *The "categorical impossibility" does not genuinely extend Theorem 5. The basic version (no finite axiom system implying $F \wedge S \wedge C$ can be consistent under Rashomon) is logically trivial. The quantitative version (Propositions 79 and 80) provides useful bounds but follows directly from DGP symmetry rather than from any categorical or axiomatic structure.*

The honest summary: the impossibility is a concrete *result about conflicting requirements, not an instance of a general metatheoretic pattern. Attempts to "lift" it to a categorical level either (a) add no content, or (b) reduce to direct probabilistic calculations that do not require categorical language.*

I Regulatory Mapping: EU AI Act

The following represents the authors' interpretation of how the attribution impossibility interacts with existing regulation. It has not been reviewed by legal experts or regulatory authorities, and should not be construed as legal advice or compliance guidance. We present it to illustrate the potential *policy implications of the theoretical results.*

The EU AI Act (Regulation (EU) 2024/1689) imposes transparency and risk-management obligations on high-risk AI systems. The attribution impossibility has direct consequences for compliance under several articles.

Art. 13: Transparency

Art. 13(1) requires:

> "High-risk AI systems shall be designed and developed in such a way to ensure that their operation is sufficiently transparent to enable deployers to interpret the system's output and use it appropriately."

Attribution instability under collinearity directly undermines this requirement: when the "most important feature" changes across training seeds, the system's output cannot be interpreted consistently. Providers relying on single-model SHAP rankings for transparency documentation fail to meet Art. 13(1) when features are correlated.

Art. 13(3)(b)(ii) further requires providers to document:

> "known or foreseeable circumstances . . . that may lead to risks to health, safety or fundamental rights."

The impossibility theorem establishes that attribution instability under collinearity is a *known* circumstance—not a speculative risk but a mathematical certainty. Any provider deploying a model with correlated features ($|\rho| > 0$) who does not disclose this instability faces potential compliance risk.

Implication (providers): Run the single-model screen on all deployed models. For any flagged feature pairs, include an instability disclosure in the technical documentation required by Art. 11.

Art. 16: Provider Obligations

Art. 16 requires providers to ensure their AI systems comply with the requirements of Chapter 2 throughout the system's lifecycle. Since attribution instability is a structural property of the model class (not of a specific training run), providers must:

1. Implement systematic instability testing (screen/Z-test diagnostics) as part of the quality management system (Art. 17).
2. Update technical documentation whenever the feature correlation structure changes (e.g., new data sources).
3. Provide deployers with clear guidance on which feature rankings are reliable and which are unstable.

Implication (providers): Integrate the multi-model Z-test into the CI/CD pipeline. Document all feature pairs with $|\rho| > 0.5$ as potentially unstable.

Art. 26: Deployer Obligations

Art. 26 requires deployers to use AI systems in accordance with the instructions of use and to monitor for risks. Deployers who receive instability disclosures must:

1. Not rely on single-model SHAP rankings for decision justification when instability is disclosed.
2. Request DASH consensus or equivalent ensemble explanations for regulatory reporting.
3. Document the explanation methodology used and its limitations.

Implication (deployers): Require DASH consensus ($M \geq 25$) or equivalent ensemble method for any feature attribution used in customer-facing decisions or regulatory filings.

Art. 23: Importer Obligations

Art. 23 requires importers to verify that providers have conducted the appropriate conformity assessment. Importers of AI systems that use feature attribution for explainability should verify that:

1. The provider's technical documentation addresses attribution instability.
2. Instability testing has been performed and results are available.
3. Appropriate mitigation (ensemble methods or instability disclosure) is in place.

Implication (importers): Include attribution stability testing in the conformity checklist. Reject systems that use single-model SHAP without instability disclosure.

Recital 47: Transparency Principle

Recital 47 establishes the general principle that AI systems should be developed and used in a transparent manner, enabling meaningful human oversight. The attribution impossibility demonstrates that transparency through feature importance rankings is fundamentally limited when features are correlated: the ranking itself is an artifact of the training seed, not a stable property of the model-data relationship. True transparency requires disclosing this limitation, not concealing it behind a single deterministic ranking.

Annex III: High-Risk Systems

The impossibility applies to all AI systems using feature attribution for explainability under Annex III, including:

- **Credit scoring** (Annex III, 5(b)): Adverse action notices citing specific features are unreliable.
- **Employment** (Annex III, 4): Hiring model explanations may change across seeds.
- **Law enforcement** (Annex III, 6): Risk assessment explanations are potentially unstable.
- **Insurance** (Annex III, 5(c)): Premium justifications based on feature importance may be arbitrary.

Summary of implications by actor.

- **Providers:** Run the single-model screen on all models with correlated features; disclose instability for flagged pairs; integrate Z-test diagnostics into CI/CD; document all pairs with $|\rho| > 0.5$.
- **Deployers:** Require DASH consensus or equivalent ensemble method for regulatory reporting; do not rely on single-model rankings when instability is disclosed.
- **Importers:** Include attribution stability in conformity assessment checklists; reject systems without instability documentation.
- **Market surveillance authorities:** Update technical standards to require multi-model attribution testing; treat undisclosed attribution instability as a potential non-conformity under Art. 13.

Liability implications. If an organisation relies on single-model SHAP explanations for high-stakes decisions (loan denials, medical diagnoses) and the Rashomon property holds, the explanation is unreliable in a legally meaningful sense: two independent audits using different random seeds could reach opposite conclusions about which feature drove the decision. DASH consensus with $M \geq 25$ provides a defensible alternative: the explanation is stable, the uncertainty is quantified, and the limitations are documented. *Note:* This analysis reflects our reading of the regulatory text and should be reviewed by a legal scholar before being relied upon in compliance contexts.

J SymPy Verification: Full Details

All algebraic consequences have been independently verified:

```
# dash-shap/paper/proofs/verify_lemma6_algebra.py
#
# Verifies:
#   split_gap = rho^2 * T / (2 - rho^2)
#   attribution_ratio = 1 / (1 - rho^2)
#   split_gap >= rho^2 * T / 2  (for rho in (0,1))
#   sum_symmetry holds under balance
#   Spearman formula consistency
#
# Result: ALL CHECKS PASS
```

The three-layer verification (SymPy algebra, Lean type-checking, empirical validation) provides strong confidence in the mathematical claims. The SymPy script verifies: (1) the split-gap formula $\rho^2 T/(2-\rho^2)$; (2) the attribution ratio $1/(1-\rho^2)$; (3) the lower bound $\rho^2 T/2$; (4) the sum symmetry under balanced ensembles.

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Table 19: Lean file cross-reference (58 files, 358 theorems, 6 axioms).

File	**Section**	**Key Result**
Defs.lean	§2	6 axioms + derived theorems
Trilemma.lean	Thm. 5	attribution_impossibility
Iterative.lean	§3.3	iterative_impossibility
General.lean	§4.1	gbdt_impossibility
SplitGap.lean	Lem. 14	split_gap_exact
Ratio.lean	Thm. 15	ratio_tendsto_atTop
SpearmanDef.lean	§4	spearmanCorr_bound
Lasso.lean	§4.2	lasso_impossibility
NeuralNet.lean	§4.3	nn_impossibility
Impossibility.lean	Combined	not_equitable, not_stable
Corollary.lean	Cor. 19	consensus_equity
SymmetryDerive.lean	Derived	attribution_sum_symmetric
DesignSpace.lean	Thm. 28	design_space_theorem
DesignSpaceFull.lean	Step 3	family_a_or_family_b
ModelSelection.lean	Thm. 41	model_selection_impossibility
UnfaithfulBound.lean	Thm. 29	stable_complete_unfaithful
PathConvergence.lean	Thm. 30	relaxation_paths_converge
RashomonUniversality.lean	Thm. 9	rashomon_from_symmetry
RashomonInevitability.lean	Thm. 12	rashomon_inevitability
ConditionalImpossibility.lean	Thm. 47	conditional_impossibility
FlipRate.lean	Prop. 17	binary_group_flip_rate
SymmetricBayes.lean	Thm. 37	symmetric_bayes_dichotomy
CausalDiscovery.lean	Thm. 45	causal_discovery_impossibility
FairnessAudit.lean	Thm. 48	fairness_audit_impossibility
FIMImpossibility.lean	Thm. 49	gaussian_rashomon_witnesses
QueryComplexity.lean	Thm. 52	query_complexity_lower_bound
Consistency.lean	§12	axiom_system_consistent
Bounds, universality, and infrastructure:		
EnsembleBound.lean	§5	ensemble_bound_formula
Efficiency.lean	§11	across_model_no_constraint
AlphaFaithful.lean	§15	alpha_faithful_bound
GaussianFlipRate.lean	§4.1	Phi_neg
LocalGlobal.lean	§8.7	local_attribution_impossibility
RobustnessLipschitz.lean	§15	flip_rate_robust
SBDInstances.lean	§7	attribution_sbd_unfaithful
ModelSelectionDesignSpace.lean	§7	model_family_a_or_family_b
Axiom strengthening:		
Qualitative.lean	Stratification	impossibility_qualitative
ProportionalityLocal.lean	Stratification	gbdt_impossibility_local
LocalSufficiency.lean	Stratification	local_proportionality_suffices
StumpProportionality.lean	§4.1	stump_proportionality_unique
IntersectionalFairness.lean	§8.2	intersectional_audit_impossibility
MutualInformation.lean	§15	mi_is_exact_boundary
Setup.lean	Bundled	attribution_impossibility_bundled
ApproximateEquity.lean	Stratification	rashomon_from_bounded_proportionality
QueryComplexityParametric.lean	§8.4	query_complexity_parametric
New derivations (this session, zero new axioms):		
MeasureHypotheses.lean	Definitions	IsDGPSymmetric, IsNonDegenerate
UnfaithfulQuantitative.lean	Thm. 29	attribution_prob_half
VarianceDerivation.lean	§5	consensus_variance_from_independence
QueryComplexityDerived.lean	§8.4	chebyshev_query_bound
BinaryQuantizer.lean	§4.1	binary_quantizer_fraction
BayesOptimalTie.lean	§6	tie_dominates_commitment
LossPreservation.lean	§3.1	rashomon_from_swap_with_loss
ParetoOptimality.lean	§5.1	dash_unique_pareto_optimal
ExplanationSystem.lean	§9	explanation_impossibility
Bilemma.lean	§9	bilemma_of_compatible_eq
MechInterp.lean	§9.1	mech_interp_bilemma
BeyondBinary.lean	§9.1	beyond_binary_extensions
Infrastructure:		
Basic.lean	Import hub	(all 58 files)
RandomForest.lean	§4.4	(documentation only)

Table 20: Positioning relative to prior attribution impossibility results. These criteria reflect the specific contributions of the present work; each prior result makes distinct contributions not captured by this comparison.

Result	Stability?	Quantitative?	Resolution?	Formal?
Bilodeau et al. (2024)	No	No	No	No
Huang et al. (2024)	No	No	No	No
Srinivas et al. (2019)	No	No	No	No
Rao (2025)	No	No	No	No
Laberge et al. (2023)	Empirical	No	No	No
Herren & Hahn (2023)	Inference	No	No	No
Jin et al. (2025)	Certificates	No	No	No
This paper	**Impossibility**	**Yes**	**Dash**	**Lean 4**

Table 21: Full cross-implementation comparison on Breast Cancer (50 models each, cross-implementation with varying test sets).

Metric	XGBoost	LightGBM	CatBoost
Unstable pairs (flip > 10%)	162 (cross-impl.)	183	160
Max flip rate	0.500	0.500	0.500
Z-test correlation $r(Z, \text{flip})$	−0.898	−0.901	−0.892
Screen precision	0.26	0.20	—
Dash max flip (M=25)	0.48	0.48	0.48

Table 22: Summary of additional experiments.

Experiment	Key finding	Script
Monte Carlo flip rate	$\Phi(-\text{SNR})$ validated to <1.3% error	`monte_carlo_flip_rate.py`
Hyperparameter sweep	Min flip 38.7% at ρ=0.5; never 0%	`hyperparameter_sensitivity.py`
Class imbalance	1:5+ imbalance → 100% pairs unstable	`class_imbalance_instability.py`
Missing data	MCAR/MAR/MNAR all compound instability	`missing_data_instability.py`
Longitudinal drift	Spearman degrades 1.0→0.18 over 31 rounds	`longitudinal_retraining.py`
Adversarial configs	All 108 configs hit 0.500 ceiling at ρ=0.9	`adversarial_max_instability.py`
Bag-of-words NLP	60% unstable top-1, 91% top-3	`nlp_token_instability.py`
Time-series features	27% within-series pairs unstable	`timeseries_instability.py`
SAGE comparison	SAGE 92%, Boruta 94% unstable	`sage_comparison.py`
DASH breakdown	Contamination sweep confirms mean optimality	`dash_breakdown_point.py`
Regulatory case study	43.2% unstable adverse action reasons	`regulatory_case_study.py`